

Service Troubleshooting Guide

Technical Instructions

LIAISON® XS



Chemiluminescence Analyzer

LIAISON® XS (Ref. I0090)

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IMPORTANT

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Description of Changes

4	Chapter 4.2 Added event 000.006.00096 according to SW version 1.4.7 and above.
7	Chapter 7.2 Modified event 018.000.00040 and 018.000.00041 according to SW version 1.4.7 and above. Added event 018.000.00042 according to SW version 1.4.7 and above.
9	Chapter 9.2 Modified event 020.005.00144 and 020.005.00145 and 020.005.00147 according to SW version 1.4.7 and above.
12	Chapter 12.2 Modified event 022.001.00083 according to SW version 1.4.7 and above.
13	Chapter 13.2 Modified event 024.005.00133, 024.003.00167, 024.004.00167 according to SW version 1.4.7 and above. Added event 024.005.00113 according to SW version 1.4.7 and above.
18	Chapter 18.2 Added event 209.000.00016, 209.000.00190 and 209.000.00207 according to SW version 1.4.7 and above.
29	Chapter 29.2 Added event 227.000.00042 according to SW version 1.4.7 and above.
31	Chapter 31.2 Added event 229.001.00011 according to SW version 1.4.7 and above.
35	Chapter 35.2 Added event 239.002.00034, 239.002.00035, 239.002.00151 according to SW version 1.4.7 and above. Modified event 239.002.00027, 239.002.00150 according to SW version 1.4.7 and above.
36	Chapter 36.2 Added event 253.014.00192 and 253.014.00193 according to SW version 1.4.7 and above. Modified event 253.014.00006, 253.014.00007, 253.014.00016, 253.015.00016, 253.016.00016, 253.014.00031, 253.014.00164, 253.014.00168 according to SW version 1.4.7 and above. Deleted event 253.014.00015, 253.015.00015, 253.016.00015, 253.014.00112, 253.015.00112, 253.016.00112, 253.014.00113, 253.015.00113, 253.016.00113, 253.014.00166 according to SW version 1.4.7 and above.

38	Chapter 38.2 Deleted event 253.009.00176, 253.009.00177, 253.009.00178, 253.009.00179, 253.009.00180, 253.009.00181 according to SW version 1.4.7 and above.
41	Chapter 41.2 Added event 253.000.00069, 253.002.00096, 253.000.00209 according to SW version 1.4.7 and above. Modified event 253.001.00059, 253.002.00063 (now 253.000.00069), according to SW version 1.4.7 and above. Deleted event 253.001.00060, 253.001.00060, 253.002.00060, 253.003.00060, 253.004.00060, 253.000.00061, 253.001.00061, 253.003.00061, 253.004.00061, 253.005.00061, 253.006.00061, 253.007.00061, 253.008.00061, 253.009.00061, 253.010.00061, 253.011.00061, 253.012.00061, 253.013.00061, 253.014.00061, 253.015.00061, 253.016.00061, 253.017.00061, 253.018.00061, 253.019.00061, 253.020.00061, 253.021.00061, 253.000.00062, 253.002.00062, 253.003.00062, 253.004.00062, 253.005.00062, 253.006.00062, 253.007.00062, 253.008.00062, 253.009.00062, 253.010.00062, 253.011.00062, 253.012.00062, 253.013.00062, 253.014.00062, 253.015.00062, 253.016.00062, 253.017.00062, 253.018.00062, 253.019.00062, 253.020.00062, 253.021.00062, 253.000.00063, 253.003.00063, 253.004.00063, 253.005.00063, 253.006.00063, 253.007.00063, 253.008.00063, 253.009.00063, 253.010.00063, 253.011.00063, 253.012.00063, 253.013.00063, 253.014.00063, 253.015.00063, 253.016.00063, 253.017.00063, 253.018.00063, 253.019.00063, 253.020.00063, 253.021.00063 according to SW version 1.4.7 and above.
43	Chapter 43.2 Deleted event 253.012.00176, 253.012.00177, 253.012.00178, 253.012.00179, 253.012.00180, 253.012.00181 according to SW version 1.4.7 and above.
44	Chapter 44.2 Added event 253.017.00130, 253.017.00131, 253.017.00192, 253.017.00193 according to SW version 1.4.7 and above. Modified event 253.017.00054, 253.017.00064, 253.017.00068, 253.017.00160, 253.017.00161, 253.017.00164, 253.017.00169, 253.017.00170 according to SW version 1.4.7 and above. Removed event 253.017.00015, 253.018.00015, 253.019.00015, 253.017.00036, 253.017.00038, 253.017.00067, 253.017.00112, 253.018.00112, 253.019.00112, 253.017.00113, 253.018.00113, 253.019.00113, 253.017.00162, 253.017.00176 and 253.017.00166 according to SW version 1.4.7 and above.

45	Chapter 45.2 Added events 500.000.00059, 500.000.00060, 500.000.00061, 500.000.00062, 500.000.00063 according to SW version 1.4.7 and above.
47	Chapter 47.2 Added event 501.000.00012 according to SW version 1.4.7 and above.
48	Chapter 48.2 Added event 502.000.00018, 502.000.00019, 502.000.00100 and 502.000.00101 according to SW version 1.4.7 and above.
49	Chapter 49.2 Added event 503.001.00011 and 503.001.00050 according to SW version 1.4.7 and above. Chapter 49.3 Added event 503.001.00011 and 503.001.00050 according to SW version 1.4.7 and above. Modified event 503.001.00029 according to SW version 1.4.7 and above.
50	Chapter 50.2 Modified event 504.000.00000, 504.000.00001, 504.000.00002, 504.000.00003, 504.000.00004, 504.000.00005, 504.000.00006 and 504.000.00007 according to SW version 1.4.7 and above. Added event 504.000.00008, 504.000.00009, 504.000.00010, 504.000.00011, 504.000.00012, 504.000.00013, 504.000.00014, 504.000.00015, 504.000.00016, 504.000.00017, 504.000.00018, 504.000.00019, 504.000.00020, 504.000.00021, 504.000.00022, 504.000.00030, 504.000.00031, 504.004.00000 and 504.004.00001 according to SW version 1.4.7 and above.
53	Chapter 53.3 Modified event 505.006.00002 and 505.006.00003 and 505.006.00008 (now 505.006.00004) according to SW version 1.4.7 and above.
55	Chapter 55.2 Added new events 505.000.00009, 505.000.00012 according to SW version 1.4.7 and above.
57	Chapter 57.2 Modified event 505.007.00000 according to SW version 1.4.7 and above.

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1 Introduction

This Manual was created to guide through the Troubleshooting on the **LIAISON® XS** Analyzer. This document is intended for Field Service Engineers (FSEs) for training and performing field service. FSEs shall follow all the instructions included in this Troubleshooting Guide together with those provided in the Service and in the User Manuals.

1.1 Safety

WARNING**Non-observance of safety instructions**

The non-observance of safety instructions may result in serious personal injury and material damage.

- Follow all safety instructions written in the Service and User Manuals.
 - Follow all safety instructions included in this guide.
 - Follow all warnings marked on the instrument.
-

WARNING**Handling of the Troubleshooting guide**

The Troubleshooting guide must be used together with all safety informations and service instructions provided in the Service manual. Service manual is provided for your safety and gives important instructions for the handling of the instrument described.

- Read all instructions!
 - Keep the Service manual nearby the instrument.
-

WARNING**Non-observance of rules and regulations**

Non-observance of rules and regulations will cause serious personal injury with deadly consequences and material damage.

- National rules and legal regulations for the safe electrical operation of the instrument shall be observed.
-

2 Event ID and category description

In this chapter it is assumed that the Service Engineer is generally familiar with the Software tools and Hardware components to describe the diagnostic approaches in a short form. In case more detailed background information is needed, please refer to the individual Module chapters.

2.1 User Software Event ID

User Software Event ID is always in decimal (dez):

Module ID	SubUnit ID	Event Number
22	001	0070

Table 2–1: User Software Event ID

- Model ID is in decimal
- SubUnit ID is the sub unit of a module in decimal.
- Event Number is the detectable error of the module in decimal.

2.2 Service Software Error Code

Service Software Error Code is always in hexadecimal (hex):

Device ID	SubUnit ID	Event Number	Severity
16	001	0070	Warning

Table 2–2: Service Software Error Code

- Device ID is in hexadecimal
- SubUnit ID is the sub unit of a module in decimal.
- Event Number is the detectable error of the module in decimal.
- Severity is the Error level.

2.3 Error codes

The **User Software** translates most of the error codes. However, when using the **Service Software**, only the raw error codes will be displayed in the status line of the command line display at the bottom of the service software dialog window. All error codes are prefixed with 5 Bytes:

1. Device ID (e.g. 20)
2. Command ID (e.g. 30)
3. Error level:

- 00 = Message
- 01 = Warning

A warning described an error message which has no effect to the result. This means for example an malfunction of the module which can be detect by the module and the module is able to solve the malfunction itself.

- 02 = Critical error

A critical error describes an error message which has a possible effect to the result. The running process can be continued without any effect.

- 03 = Fatal error

A fatal error is occurred if a malfunction of the module has effect to the whole process. The whole process has to be stopped.

4. Sub assembly (e.g. 01)
5. Error code (e.g. 35)

CAUTION

All byte values are hexadecimal!

How to find an error

All byte values are hexadecimal!

1. Identify the affected module (device ID), see below.
2. Identify the affected sub assembly ID (note the preceding table in the respective chapters).
3. Identify the error code.

2.4 Categories

Category	Example	Corrective action
Configuration Issue	Internal error Command parameter out of range	Check Instrument configuration - Run Instrument setup Call manufacturer
Electronics Issue	Command takes too long Device registered to CAN bus failed Sensor plausibility check failed High side switch failure Temperature is out of range for too long	Check cable and connection Check instrument configuration - Run instrument setup Exchange PCB Exchange module
Mechanics Issue	Target not reached LB not reached	Clean module Check teaching of instrument Check calibration Check mechanic Exchange module
Consumable Issue	Wrong barcode detected RFID access not possible	Exchange consumable Exchange module
Environmental Issue	Temperature out of range	Check environment (temperature, humidity) Exchange module
Process control Issue	OLV error No liquid found Aspiration failed Bubble detected	Individual corrective action
Calibration Issue	Bottom touch not reach Sensor not calibrated	Calibrate module via service software Exchange module
Subsequent Issue	Error from submodule	
Instrument handling Issue	Tip not found Resource empty Module not initialized	Check consumable state Initialize module

Table 2-3: Categories

2. Event ID and category description

3 Reference list for device driver

Module ID. Hex code	Device ID. Dec code	Device chapter and link to the message list	No of Sub Assembly Hex code	Sub Assembly
00	00	COP	00	General
			01	COP Main
			02	COP Com
			03	Work list
			05	App
			06	App Config
			0D	Notify Handler
			1E	COP Drv Frw
10	16	Pwr Can Extension	00	General
11	17	User Access Panel	00	General
12	18	Incubator	00	General
13	19	Washer	00	General
			01	Probe 1
			02	Probe 2
			03	Probe 3
			04	Ring
			05	Asp Monitoring 1
			06	Asp Monitoring 2
			07	Asp Monitoring 3
14	20	Reader / Measurement	00	General
			01	Measurement (Reader)
			02	Rotor
			03	Lift
			04	Data Transmission
			05	Bubble sensor
			06	Pump starter 1
			07	Pump starter 2
16	22	Loading	00	General

3. Reference list for device driver

Module ID. Hex code	Device ID. Dec code	Device chapter and link to the message list	No of Sub Assembly Hex code	Sub Assembly
			01	Loading
			02	RFID
			03	Stirrer
			04	Cooling
			05	Scanner
18	24	Liquid Support	00	General
			01	Bubble sensor 1
			02	Bubble sensor 2
			03	Heat Tube 1
			04	Heat Tube 2
			05	RFID Starter
32	50	Washer Dispense Pump	00	General
35	53	WAP	00	General
40	64	Front Cover	00	General
D0	208	Can Bus Support	00	General
D1	209	Application Manager	00	General
D2	210	Job Manager	00	General
D3	211	Assay Manager	00	General
D4	212	Dispatcher	00	General
			01	Left arm
			02	Right arm
D5	213	Cuvette Handler	00	General
D6	214	Loading Manager	00	General
D7	215	Diti Manager	00	General
DF	223	System Monitor Driver	00	General
E0	224	Driver Framework Support Driver	00	General
E1	225	User Access Driver	00	General
E2	226	Service Driver	00	General
E3	227	Incubator Driver	00	General
E4	228	Washer Driver	00	General

3. Reference list for device driver

Module ID. Hex code	Device ID. Dec code	Device chapter and link to the message list	No of Sub Assembly Hex code	Sub Assembly
			01	Probe 1
			02	Probe 2
			03	Probe 3
E5	229	Reader / Measurement Driver	00	General
			01	Bubble Sensor
			00	General
			01	Reagent
			02	Sample
			00	General
E8	232	Liquid Support Driver	00	General
			01	Bubble Sensor 1
			02	Bubble Sensor 2
			03	Heat Tube 1
			04	Heat Tube 2
			05	RFID Starter
E9	233	Front Cover Driver	00	General
EF	239	Com Gen Driver	00	General
			01	Left Arm
			02	Right Arm
FD	253	Air Pipettor Left Arm	0E	General
			0F	Sctl CU
			10	AP Head CU
FD	253	Dual Sledge Y	05	General
			06	Y_Drive
			09	General
			0A	ZDrive1
			0B	ZDrive2
			14	General
FD	253	Gripper	15	General
FD	253	LWP	01	Left Arm
FD	253	Pipettor		

3. Reference list for device driver

Module ID. Hex code	Device ID. Dec code	Device chapter and link to the message list	No of Sub Assembly Hex code	Sub Assembly
			02	Right Arm
			03	Left Y Motor
			04	Right Y Motor
			00	General
FD	253	Single Sledge Y	07	General
FD	253	Single Sledge Z	08	Y Drive
			0C	General
FD	253	SPOLV Right Arm	0D	Z Drive 1
			11	General
			12	Sctl CU
			13	AP Head CU
1F4	500	Application	00	Started
1F4	500	COM Protocol	3E7	Research and development exception (Fake Event)
			00	Com Protocol Exception
			00	Repository Exception
			00	Coordinate File Not Available
1F5	501	Database	00	Job Scheduled
1F6	502	Instrument Control	00	Abort Ongoing Job On SW Start Or Initialization
1F7	503	Process Control	00	Vigilant Online Field Gateway Service Connected
1F8	504	Infrastructure	01	Event Creation Exception
			02	Desk Shield Language File Not Found Exception
			03	Windows time changed
1F9	505	Ancillary Loading	02	Ancillary Shelf Expired
1F9	505	Consumables	04	Drawer Withdraw Request

Module ID. Hex code	Device ID. Dec code	Device chapter and link to the message list	No of Sub Assembly Hex code	Sub Assembly
1F9	505	Liquid Containers	06	Wash Buffer Removed While In Use
1F9	505	Reagent Loading	01	Integral Shelf Expired
1F9	505	Sample Loading	00	Sample Rack Inserted
1F9	505	Starter Loading	03	Starter RFID Data Block Error
1F9	505	Top Cover	07	Top cover open for priming
1F9	505	Waste	05	Solid Waste Emptied
1FA	506	Maintenance	00	Import Successful
1FB	507	UI	00	Importing Assay File
1FC	508	Reporting	00	Print Error
1FD	509	LIS	00	No Response To Query
1FE	510	ICoL	00	Unknown Message
1FF	511	TDef	00	Process Input Description Exception
200	512	Data Reduction	00	Calibration Set To Invalid Due To Kulovic
201	513	Auto Backup	01	Second Path Does Not Exist
201	513	Backup	00	Definition File Not Found
202	514	Q CSTAT	00	Rule passed

Table 3–1: Reference list for device driver

4 COP

Device ID: 0x00

4.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General
1	0x01	CopMain
2	0x02	CopCom
3	0x03	Worklist
5	0x05	App
6	0x06	AppConfig
13	0x0D	NotifyHandler
30	0x1E	CopDrvFrw

Table 4–1: Sub assemblies, COP

4.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.000.00002	00 00 02	Critical	This is the Custom Text for 000.000.00002	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00002	00 01 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.002.00002	00 02 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

4. COP

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.003.00002	00 03 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.005.00002	00 05 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00002	00 06 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.013.00002	00 0D 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.030.00002	00 1E 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.000.00003	00 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.000.00004	00 00 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
000.000.00005	00 00 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
000.000.00012	00 00 0C	Fatal	An Internal error occurred during self test	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
000.000.00014	00 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00014	00 01 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.002.00014	00 02 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.003.00014	00 03 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

4. COP

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.005.00014	00 05 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00014	00 06 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.013.00014	00 0D 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.030.00014	00 1E 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.000.00015	00 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00015	00 01 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.002.00015	00 02 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.003.00015	00 03 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.005.00015	00 05 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00015	00 06 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.013.00015	00 0D 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.030.00015	00 1E 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.000.00016	00 00 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00016	00 01 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

4. COP

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.002.00016	00 02 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.003.00016	00 03 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.005.00016	00 05 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00016	00 06 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.013.00016	00 0D 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.030.00016	00 1E 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.000.00017	00 00 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.000.00018	00 00 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.000.00019	00 00 13	Critical	The received parameter number needs an authorisation which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
000.000.00020	00 00 14	Critical	Received parameter authorisation code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
000.000.00021	00 00 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

4. COP

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.000.00022	00 00 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00032	00 06 20	Fatal	The device cannot be addressed or is not responsive anymore.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
000.006.00033	00 06 21	Fatal	Internal CAN-bus error while command transmission to the device.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00034	00 06 22	Fatal	A device failed registering to CAN-bus.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
000.006.00035	00 06 23	Fatal	Configuration for connection definition has failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00036	00 06 24	Fatal	A routing definition for device has failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00037	00 06 25	Fatal	Configuration for set headless communication by device at port has failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00038	00 06 26	Fatal	Configuration for set exclusive read mode at port has failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.006.00039	00 06 27	Fatal	Communication to the device failed.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00040	00 06 28	Fatal	Configuration for link definition has failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00041	00 06 29	Fatal	CAN communication system failed.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00042	00 06 2A	Fatal	Configuration for unused port definition has failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00046	00 06 2E	Fatal	Set plug-and-play configuration for device has failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.000.00043	00 00 2B	Fatal	Driver Framework Error at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00043	00 01 2B	Fatal	Driver Framework Error at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.002.00043	00 02 2B	Fatal	Driver Framework Error at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

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Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.003.00043	00 03 2B	Fatal	Driver Framework Error at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.005.00043	00 05 2B	Fatal	Driver Framework Error at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00043	00 06 2B	Fatal	Driver Framework Error at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.013.00043	00 0D 2B	Fatal	Driver Framework Error at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.030.00043	00 1E 2B	Fatal	Driver Framework Error at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.000.00044	00 00 2C	Warning	Driver Framework Warning at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00044	00 01 2C	Warning	Driver Framework Warning at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.002.00044	00 02 2C	Warning	Driver Framework Warning at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.003.00044	00 03 2C	Warning	Driver Framework Warning at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.005.00044	00 05 2C	Warning	Driver Framework Warning at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.006.00044	00 06 2C	Warning	Driver Framework Warning at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.013.00044	00 0D 2C	Warning	Driver Framework Warning at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.030.00044	00 1E 2C	Warning	Driver Framework Warning at device occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00045	00 06 2D	Fatal	exception message(s) destined for work list lost due to buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.002.00049	00 02 31	Warning	Error message(s) lost due to buffer overflow.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.002.00050	00 02 32	Fatal	No communication from PC for too long time.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.002.00051	00 02 33	Fatal	No communication from peer for too long time.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.003.00064	00 03 40	Message	time stamp of start command sequence.	Process Control
000.003.00065	00 03 41	Message	time stamp of end command sequence.	Process Control

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Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.003.00066	00 03 42	Fatal	command cannot be scheduled to worklist.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.000.00072	00 00 48	Fatal	Driver Framework Work list: unregistered queue requested.	Process Control
000.001.00072	00 01 48	Fatal	Driver Framework Work list: unregistered queue requested.	Process Control
000.002.00072	00 02 48	Fatal	Driver Framework Work list: unregistered queue requested.	Process Control
000.003.00072	00 03 48	Fatal	Driver Framework Work list: unregistered queue requested.	Process Control
000.005.00072	00 05 48	Fatal	Driver Framework Work list: unregistered queue requested.	Process Control
000.006.00072	00 06 48	Fatal	Driver Framework Work list: unregistered queue requested.	Process Control
000.013.00072	00 0D 48	Fatal	Driver Framework Work list: unregistered queue requested.	Process Control
000.030.00072	00 1E 48	Fatal	Driver Framework Work list: unregistered queue requested.	Process Control
000.013.00077	00 0D 4D	Message	Notification handler storage memory dump received.	Process Control
000.013.00078	00 0D 4E	Fatal	Notification handler storage overflow occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.013.00079	00 0D 4F	Warning	No notification registered for event.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00080	00 01 50	Fatal	External non-maskable interrupt detected by MCU.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00081	00 01 51	Fatal	privileged instruction exception detected by MCU.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00082	00 01 52	Fatal	floating-point exception detected by MCU.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00083	00 01 53	Fatal	undefined instruction detected by MCU.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00084	00 01 54	Fatal	timeout of watchdog detected by MCU.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00085	00 01 55	Fatal	timeout of independent watchdog detected by MCU.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00086	00 01 56	Fatal	reset from deep software detected by MCU.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00087	00 01 57	Fatal	low power detection circuit0 reset detected by MCU.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

4. COP

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.001.00088	00 01 58	Fatal	low power detection circuit1 reset detected by MCU.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00089	00 01 59	Fatal	low power detection circuit2 reset detected by MCU.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00090	00 01 5A	Fatal	software reset detected by MCU.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.001.00091	00 01 5B	Fatal	pin reset detected by MCU.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
000.006.00096	00 06 60	Message	Received event during CAN communication.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.000.00160	00 00 A0	Warning	Change to FSE operation mode to { Mode }.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent
000.000.00254	00 00 FE	Warning	Internal warning occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent
000.001.00254	00 01 FE	Warning	Internal warning occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent
000.002.00254	00 02 FE	Warning	Internal warning occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent
000.003.00254	00 03 FE	Warning	Internal warning occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent
000.005.00254	00 05 FE	Warning	Internal warning occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent
000.006.00254	00 06 FE	Warning	Internal warning occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent
000.013.00254	00 0D FE	Warning	Internal warning occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent

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Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.030.00254	00 1E FE	Warning	Internal warning occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent
000.000.00255	00 00 FF	Fatal	Internal error occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent
000.001.00255	00 01 FF	Fatal	Internal error occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent
000.002.00255	00 02 FF	Fatal	Internal error occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent
000.003.00255	00 03 FF	Fatal	Internal error occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent
000.005.00255	00 05 FF	Fatal	Internal error occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent
000.006.00255	00 06 FF	Fatal	Internal error occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent
000.013.00255	00 0D FF	Fatal	Internal error occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
000.030.00255	00 1E FF	Fatal	Internal error occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Subsequent

Table 4–2: Message list, COP

5 Pwr Can Extension

Device ID: 0x10

5.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 5–1: Sub assemblies, Pwr Can Extension

5.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
016.000.00001	10 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
016.000.00002	10 00 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
016.000.00003	10 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
016.000.00004	10 00 04	Critical	An internal error occurred	Electronics • Check fuse • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
016.000.00005	10 00 05	Fatal	A communication error occurred	Electronics • Check fuse • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
016.000.00006	10 00 06	Critical	Voltage is out of range. {SourceSubChannel} , detected voltage {Detected Voltage}	Electronics • Check fuse • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
016.000.00007	10 00 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check fuse • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
016.000.00011	10 00 0B	Fatal	Command terminated abnormally.	Subsequent
016.000.00014	10 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
016.000.00015	10 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
016.000.00016	10 00 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check fuse • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
016.000.00017	10 00 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
016.000.00018	10 00 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
016.000.00019	10 00 13	Critical	The received parameter number needs an authorization which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
016.000.00020	10 00 14	Critical	Received parameter authorization code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
016.000.00021	10 00 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
016.000.00022	10 00 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
016.000.00031	10 00 1F	Warning	Errors on CAN bus detected.	Electronics • Check fuse • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
016.000.00032	10 00 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Table 5–2: Message list, Pwr Can Extension

6 User Access Panel

Device ID: 0x11

6.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 6–1: Sub assemblies, User Access Panel

6.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
017.000.00001	11 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
017.000.00002	11 00 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
017.000.00003	11 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
017.000.00004	11 00 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
017.000.00005	11 00 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

6. User Access Panel

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
017.000.00006	11 00 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
017.000.00007	11 00 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
017.000.00011	11 00 0B	Fatal	Command terminated abnormally.	Subsequent
017.000.00013	11 00 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
017.000.00014	11 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
017.000.00015	11 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
017.000.00016	11 00 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
017.000.00017	11 00 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
017.000.00018	11 00 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
017.000.00019	11 00 13	Critical	The received parameter number needs an authorisation which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
017.000.00020	11 00 14	Critical	Received parameter authorisation code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
017.000.00021	11 00 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
017.000.00022	11 00 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
017.000.00024	11 00 18	Critical	Communication issue: Frame order, or start or end frame number is invalid on long messages.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
017.000.00031	11 00 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
017.000.00032	11 00 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
017.000.00033	11 00 21	Message	The power button was pressed for a short time.	Process Control

6. User Access Panel

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
017.000.00034	11 00 22	Message	The power button was pressed for a long time.	Process Control
017.000.00036	11 00 24	Message	The panel PC is now OFF.	Process Control
017.000.00035	11 00 23	Message	The panel PC is now ON.	Process Control
017.000.00160	11 00 A0	Message	Self diagnostic sends min and max values of the supply voltage.	Process Control
017.000.00161	11 00 A1	Message	Self diagnostic sends how many times the button has been pressed.	Process Control
017.000.00191	11 00 BF	Message	Unexpected error during Self diagnostic.	Process Control • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Electronics
017.000.00176	11 00 B0	Warning	Error during Service Memory access.	Process Control • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Electronics
017.000.00177	11 00 B1	Message	Service Memory read data as long message.	Process Control
017.000.00178	11 00 B2	Critical	Initialize Service Memory values failed.	Process Control • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Electronics
017.000.00179	11 00 B3	Message	Initialize Service Memory values done.	Process Control
017.000.00180	11 00 B4	Warning	Mismatch at Service Memory SD Parameter definition.	Process Control
017.000.00181	11 00 B5	Critical	Wrong Access-Key to modify SD data.	Process Control

Table 6–2: Message list, User Access Panel

7 Incubator

Device ID: 0x12

7.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 7-1: Sub assemblies, Incubator

7.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
018.000.00001	12 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
018.000.00002	12 00 02	Fatal	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
018.000.00003	12 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
018.000.00004	12 00 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
018.000.00005	12 00 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

7. Incubator

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
018.000.00006	12 00 06	Critical	Voltage is out of range. Detected voltage: {DetectedVoltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
018.000.00007	12 00 07	Warning	Voltage is back in range. Detected voltage: {DetectedVoltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
018.000.00008	12 00 08	Fatal	Motor overload	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
018.000.00009	12 00 09	Warning	Target position reached after retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
018.000.00010	12 00 0A	Fatal	Target position not reached by rotor	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
018.000.00011	12 00 0B	Fatal	Command terminated abnormally.	Subsequent

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
018.000.00012	12 00 0C	Fatal	Hardware Error	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
018.000.00013	12 00 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
018.000.00014	12 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
018.000.00015	12 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
018.000.00016	12 00 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
018.000.00017	12 00 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
018.000.00018	12 00 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

7. Incubator

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
018.000.00019	12 00 13	Critical	The received parameter number needs an authorization which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
018.000.00020	12 00 14	Critical	Received parameter authorization code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
018.000.00021	12 00 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
018.000.00022	12 00 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
018.000.00031	12 00 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
018.000.00032	12 00 20	Critical	Incubator temperature is out of range.	Process Control
018.000.00033	12 00 21	Warning	Temperature is back in range.	Process Control
018.000.00034	12 00 22	Fatal	Measured temperature fails plausibility check.	Process Control
018.000.00035	12 00 23	Fatal	Ambient temperature fails plausibility check.	Process Control
018.000.00036	12 00 24	Message	Temperature message.	Process Control
018.000.00038	12 00 26	Fatal	Temperature sensor mismatch.	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
018.000.00040	12 00 28	Critical	Inner Cuvette Sensor failed plausibility test.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
018.000.00041	12 00 29	Critical	Outer Cuvette Sensor failed plausibility test.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
018.000.00042	12 00 2A	Critical	Cuvette sensor failed during init, motor homing ok.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board Exchange module
018.000.00043	12 00 2B	Critical	An Incubator movement problem has been detected	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
018.000.00044	12 00 2C	Warning	Init offset not plausible, check teaching.	Calibration • Calibrate module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
018.000.00162	12 00 A2	Message	Self diagnostic sends min and max values of the supply voltage.	Process Control
018.000.00163	12 00 A3	Message	Self diagnostic cycle counters.	Process Control
018.000.00165	12 00 A5	Message	Self diagnostic min drive current.	Process Control
018.000.00167	12 00 A7	Message	Self diagnostic mechanical calibration.	Process Control

7. Incubator

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
018.000.00166	12 00 A6	Message	Mechanical clearance of the drive evaluated during module self diagnostics.	Process Control
018.000.00191	12 00 BF	Message	An unexpected error occurred during module self diagnostics.	Process Control
018.000.00176	12 00 B0	Warning	An internal error occurred	Calibration • Calibrate module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
018.000.00177	12 00 B1	Message	Service memory data.	Process Control
018.000.00178	12 00 B2	Critical	Initialization of service memory data failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
018.000.00179	12 00 B3	Message	Initialization of service memory data succeeded.	Process Control
018.000.00180	12 00 B4	Warning	An internal error occurred	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
018.000.00181	12 00 B5	Critical	Service memory access denied.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
018.000.00209	12 00 D1	Message	Self-diagnostic Device information: serial, article number and Version.	Process Control

Table 7–2: Message list, Incubator

8 Washer

Device ID: 0x13

8.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General
1	0x01	Probe 1
2	0x02	Probe 2
3	0x03	Probe 3
4	0x04	Ring
5	0x05	Asp Monitoring 1
6	0x06	Asp Monitoring 2
7	0x07	Asp Monitoring 3

Table 8–1: Sub assemblies, Washer

8.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.000.00001	13 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
019.001.00001	13 01 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
019.002.00001	13 02 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
019.003.00001	13 03 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
019.004.00001	13 04 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module

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Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.005.00001	13 05 01	Fatal	Module cannot execute command as module is not initialized.	InstrumentHandling • Check consumable states • Initialize module
019.006.00001	13 06 01	Fatal	Module cannot execute command as module is not initialized.	InstrumentHandling • Check consumable states • Initialize module
019.007.00001	13 07 01	Fatal	Module cannot execute command as module is not initialized.	InstrumentHandling • Check consumable states • Initialize module
019.000.00002	13 00 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.001.00002	13 01 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.002.00002	13 02 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.003.00002	13 03 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.004.00002	13 04 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.005.00002	13 05 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.006.00002	13 06 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.007.00002	13 07 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.000.00003	13 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.000.00004	13 00 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.000.00005	13 00 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

8. Washer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.000.00006	13 00 06	Critical	Voltage is out of range. Detected voltage: {DetectedVoltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.000.00007	13 00 07	Warning	Voltage is back in range. Detected voltage: {DetectedVoltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.001.00009	13 01 09	Warning	Target position reached after retry.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.002.00009	13 02 09	Warning	Target position reached after retry.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.003.00009	13 03 09	Warning	Target position reached after retry.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.004.00009	13 04 09	Warning	Target position reached after retry.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.001.00010	13 01 0A	Fatal	Target position not reached by probe 1	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.002.00010	13 02 0A	Fatal	Target position not reached by probe 2	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.003.00010	13 03 0A	Fatal	Target position not reached by probe 3	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

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Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.004.00010	13 04 0A	Fatal	Target position not reached by rotor	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.000.00011	13 00 0B	Fatal	Command terminated abnormally.	Subsequent
019.001.00011	13 01 0B	Fatal	Command terminated abnormally.	Subsequent
019.002.00011	13 02 0B	Fatal	Command terminated abnormally.	Subsequent
019.003.00011	13 03 0B	Fatal	Command terminated abnormally.	Subsequent
019.004.00011	13 04 0B	Fatal	Command terminated abnormally.	Subsequent
019.005.00011	13 05 0B	Fatal	Command terminated abnormally.	Subsequent
019.006.00011	13 06 0B	Fatal	Command terminated abnormally.	Subsequent
019.007.00011	13 07 0B	Fatal	Command terminated abnormally.	Subsequent
019.000.00013	13 00 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.001.00013	13 01 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.002.00013	13 02 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.003.00013	13 03 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.004.00013	13 04 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.005.00013	13 05 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.006.00013	13 06 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.007.00013	13 07 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.000.00014	13 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.001.00014	13 01 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

8. Washer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.002.00014	13 02 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.003.00014	13 03 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.004.00014	13 04 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.005.00014	13 05 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.006.00014	13 06 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.007.00014	13 07 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.000.00015	13 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.001.00015	13 01 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.002.00015	13 02 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.003.00015	13 03 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.004.00015	13 04 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.005.00015	13 05 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.006.00015	13 06 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.007.00015	13 07 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.000.00016	13 00 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

8. Washer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.001.00016	13 01 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.002.00016	13 02 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.003.00016	13 03 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.004.00016	13 04 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.005.00016	13 05 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.006.00016	13 06 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.007.00016	13 07 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.000.00017	13 00 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.000.00018	13 00 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.000.00019	13 00 13	Critical	The received parameter number needs an authorization which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module

8. Washer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.000.00020	13 00 14	Critical	Received parameter authorization code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
019.000.00021	13 00 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.000.00022	13 00 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.000.00023	13 00 17	Fatal	A movement problem has been detected	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.000.00031	13 00 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.000.00032	13 00 20	Fatal	Embedded software crash occurrence.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.001.00032	13 01 20	Fatal	Embedded software crash occurrence.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.002.00032	13 02 20	Fatal	Embedded software crash occurrence.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.003.00032	13 03 20	Fatal	Embedded software crash occurrence.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.004.00032	13 04 20	Fatal	Embedded software crash occurrence.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.005.00032	13 05 20	Fatal	Embedded software crash occurrence.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.006.00032	13 06 20	Fatal	Embedded software crash occurrence.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.007.00032	13 07 20	Fatal	Embedded software crash occurrence.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.001.00034	13 01 22	Fatal	Initialization of probe 1 failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.002.00034	13 02 22	Fatal	Initialization of probe 2 failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

8. Washer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.003.00034	13 03 22	Fatal	Initialization of probe 3 failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.004.00034	13 04 22	Fatal	Initialization of rotor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.001.00035	13 01 23	Fatal	Initialization of probe 1 failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.002.00035	13 02 23	Fatal	Initialization of probe 2 failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.003.00035	13 03 23	Fatal	Initialization of probe 3 failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.004.00035	13 04 23	Fatal	Initialization of rotor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.000.00048	13 00 30	Fatal	Actuator not connected or defect.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
019.001.00048	13 01 30	Fatal	Actuator not connected or defect.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
019.002.00048	13 02 30	Fatal	Actuator not connected or defect.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
019.003.00048	13 03 30	Fatal	Actuator not connected or defect.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module

8. Washer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.004.00048	13 04 30	Fatal	Actuator not connected or defect.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
019.005.00048	13 05 30	Fatal	Actuator not connected or defect.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
019.006.00048	13 06 30	Fatal	Actuator not connected or defect.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
019.007.00048	13 07 30	Fatal	Actuator not connected or defect.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
019.000.00049	13 00 31	Fatal	Actuator switched off to protect the electronics (subsequent error).	Subsequent
019.001.00049	13 01 31	Fatal	Actuator switched off to protect the electronics (subsequent error).	Subsequent
019.002.00049	13 02 31	Fatal	Actuator switched off to protect the electronics (subsequent error).	Subsequent
019.003.00049	13 03 31	Fatal	Actuator switched off to protect the electronics (subsequent error).	Subsequent
019.004.00049	13 04 31	Fatal	Actuator switched off to protect the electronics (subsequent error).	Subsequent
019.005.00049	13 05 31	Fatal	Actuator switched off to protect the electronics (subsequent error).	Subsequent

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.006.00049	13 06 31	Fatal	Actuator switched off to protect the electronics (subsequent error).	Subsequent
019.007.00049	13 07 31	Fatal	Actuator switched off to protect the electronics (subsequent error).	Subsequent
019.005.00112	13 05 70	Critical	Bubble sensor can not be calibrated or still isn't calibrated.	Calibration • Calibrate module Instrument Handling • Check consumable states • Initialize module
019.006.00112	13 06 70	Critical	Bubble sensor can not be calibrated or still isn't calibrated.	Calibration • Calibrate module Instrument Handling • Check consumable states • Initialize module
019.007.00112	13 07 70	Critical	Bubble sensor can not be calibrated or still isn't calibrated.	Calibration • Calibrate module Instrument Handling • Check consumable states • Initialize module
019.005.00113	13 05 71	Critical	During aspiration the measured volume is out of range.	Process Control
019.006.00113	13 06 71	Critical	During aspiration the measured volume is out of range.	Process Control
019.007.00113	13 07 71	Critical	During aspiration the measured volume is out of range.	Process Control
019.005.00114	13 05 72	Message	Bubble measure is finished and returns measured values.	Process Control
019.006.00114	13 06 72	Message	Bubble measure is finished and returns measured values.	Process Control
019.007.00114	13 07 72	Message	Bubble measure is finished and returns measured values.	Process Control

8. Washer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.005.00115	13 05 73	Warning	During aspiration the max number of bubble measure data is reached.	Process Control
019.006.00115	13 06 73	Warning	During aspiration the max number of bubble measure data is reached.	Process Control
019.007.00115	13 07 73	Warning	During aspiration the max number of bubble measure data is reached.	Process Control
019.001.00067	13 01 43	Fatal	Target position not reached by probe 1	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.002.00067	13 02 43	Fatal	Target position not reached by probe 2	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.003.00067	13 03 43	Fatal	Target position not reached by probe 3	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.004.00080	13 04 50	Fatal	Target position not reached by rotor	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.001.00081	13 01 51	Message	Teach position reached.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.002.00081	13 02 51	Message	Teach position reached.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.003.00081	13 03 51	Message	Teach position reached.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

8. Washer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.004.00081	13 04 51	Message	Teach position reached.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.001.00083	13 01 53	Message	Teach position not reached.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.002.00083	13 02 53	Message	Teach position not reached.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.003.00083	13 03 53	Message	Teach position not reached.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.004.00082	13 04 52	Fatal	Ring init could not be reached.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.000.00162	13 00 A2	Message	Self diagnostic sends min and max values of the supply voltage.	Process Control
019.000.00163	13 00 A3	Message	Cycle counter evaluated during module self diagnostics.	Process Control
019.000.00164	13 00 A4	Message	Cycle counter evaluated during module self diagnostics.	Process Control
019.000.00165	13 00 A5	Message	Minimum motor current evaluated during module self diagnostics.	Process Control
019.000.00166	13 00 A6	Message	Mechanical clearance of the drive evaluated during module self diagnostics.	Process Control
019.000.00167	13 00 A7	Message	Sensor calibration value evaluated during module self diagnostics.	Calibration • Calibrate module
019.000.00191	13 00 BF	Message	An unexpected error occurred during module self diagnostics.	Process Control
019.000.00176	13 00 B0	Warning	An internal error occurred	Calibration • Calibrate module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.000.00177	13 00 B1	Message	Service memory data.	Process Control
019.000.00178	13 00 B2	Critical	Initialization of service memory data failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.000.00179	13 00 B3	Message	Initialization of service memory data succeeded.	Process Control

8. Washer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
019.000.00180	13 00 B4	Warning	An internal error occurred	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
019.000.00181	13 00 B5	Critical	Service memory access denied.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.000.00182	13 00 B6	Fatal	Service memory configuration error.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
019.000.00209	13 00 D1	Message	Self-diagnostic Device information: serial, article number and Version.	Process Control

Table 8–2: Message list, Washer

9 Reader

Device ID: 0x14

9.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General
1	0x01	Measurement
2	0x02	Rotor
3	0x03	Lift
4	0x04	Data Transmission
5	0x05	Bubble Sensor
6	0x06	Pump Starter 1
7	0x07	Pump Starter 2

Table 9–1: Sub assemblies, Reader

9.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.000.00001	14 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
020.001.00001	14 01 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
020.002.00001	14 02 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
020.003.00001	14 03 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
020.004.00001	14 04 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.005.00001	14 05 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
020.000.00002	14 00 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.001.00002	14 01 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.002.00002	14 02 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.003.00002	14 03 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.004.00002	14 04 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.005.00002	14 05 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.000.00003	14 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.000.00004	14 00 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.000.00005	14 00 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.000.00006	14 00 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.000.00007	14 00 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.002.00009	14 02 09	Warning	Target position reached after retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.003.00009	14 03 09	Warning	Target position reached after retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.002.00010	14 02 0A	Fatal	Target position not reached by rotor	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.003.00010	14 03 0A	Fatal	Target position not reached by lift	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.000.00011	14 00 0B	Fatal	Command terminated abnormally.	Subsequent
020.002.00011	14 02 0B	Fatal	Command terminated abnormally.	Subsequent
020.003.00011	14 03 0B	Fatal	Command terminated abnormally.	Subsequent
020.000.00012	14 00 0C	Fatal		Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.000.00013	14 00 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.001.00013	14 01 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.002.00013	14 02 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.003.00013	14 03 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.004.00013	14 04 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.005.00013	14 05 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.000.00014	14 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.001.00014	14 01 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.002.00014	14 02 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.003.00014	14 03 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.004.00014	14 04 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.005.00014	14 05 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.000.00015	14 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.001.00015	14 01 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.002.00015	14 02 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.003.00015	14 03 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.004.00015	14 04 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.005.00015	14 05 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.000.00016	14 00 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.001.00016	14 01 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.002.00016	14 02 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.003.00016	14 03 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.004.00016	14 04 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.005.00016	14 05 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.000.00017	14 00 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.000.00018	14 00 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.000.00019	14 00 13	Critical	The received parameter number needs an authorisation which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.000.00020	14 00 14	Critical	Received parameter authorisation code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer InstrumentHandling • Check consumable states • Initialize module
020.000.00021	14 00 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.000.00022	14 00 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.000.00031	14 00 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.000.00032	14 00 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.001.00032	14 01 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.002.00032	14 02 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.003.00032	14 03 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.004.00032	14 04 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.005.00032	14 05 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.001.00033	14 01 21	Fatal	High Voltage out of range.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.000.00034	14 00 22	Fatal	An Internal error occurred	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.000.00035	14 00 23	Fatal	An Internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.001.00036	14 01 24	Critical	Measured Dark counts are out of range	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.001.00037	14 01 25	Fatal	Measured High voltage reference is out of allowed range	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.001.00038	14 01 26	Message	High Voltage restarted and in range	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.006.00039	14 06 27	Fatal	No response from submodule within time limit	Subsequent
020.007.00039	14 07 27	Fatal	No response from submodule within time limit	Subsequent
020.003.00048	14 03 30	Fatal	Initialization of lift failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.002.00048	14 02 30	Fatal	Initialization of rotor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.003.00049	14 03 31	Fatal	Initialization of lift failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.002.00049	14 02 31	Fatal	Initialization of rotor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.002.00053	14 02 35	Critical	Rotor movement not possible due lift is not initialized	Instrument Handling • Check consumable states • Initialize module
020.002.00054	14 02 36	Critical	Lift blocks rotor. Move lift up first.	Process Control • Check if process sequence was correct • Process information
020.003.00055	14 03 37	Critical	Lift movement not possible due rotor is not initialized	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Instrument Handling • Check consumable states • Initialize module
020.003.00056	14 03 38	Critical	Rotor not in position to move lift down. Move rotor to aspirate / eject position first	Process Control • Check if process sequence was correct • Process information
020.002.00057	14 02 39	Fatal	Target position not reached by rotor	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.002.00058	14 02 3A	Critical	From this position movement backward to target position is not allowed	Process Control • Check if process sequence was correct • Process information
020.002.00059	14 02 3B	Fatal	Target position not reached by rotor	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.002.00060	14 02 3C	Message	Additional Info to Rotor LB change count error	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.000.00065	14 00 41	Warning	Detected Fan frequency out of allowed range.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.000.00066	14 00 42	Warning	Detected Fan frequency is back in range.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.000.00067	14 00 43	Warning	Detected Fan frequency is invalid.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.000.00081	14 00 51	Message	Teach position reached.	Process Control • Check if process sequence was correct • Process information
020.003.00082	14 03 52	Message	Lift has lost some steps during auto teach.	Process Control • Check if process sequence was correct • Process information
020.003.00083	14 03 53	Fatal	Target position not reached by lift	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Calibration • Calibrate module
020.001.00101	14 01 65	Warning	Values are Simulated.	
020.001.00133	14 01 85	Warning	Spikes detected during measurement	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.001.00134	14 01 86	Critical	Number of spikes exceeds defined limit	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.005.00144	14 05 90	Critical	Sensor not calibrated.	Calibration • Calibrate module Instrument Handling • Check consumable states • Initialize module
020.005.00145	14 05 91	Critical	Aspirated volume out of range.	Process Control • Check if process sequence was correct • Process information
020.005.00146	14 05 92	Message	Bubble measure is finished and returns measured values.	Process Control • Check if process sequence was correct • Process information
020.005.00147	14 05 93	Critical	During aspiration the max number of bubble measure data is reached.	Process Control • Check if process sequence was correct • Process information
020.000.00160	14 00 A0	Message	PMW value evaluated during module self diagnostics.	Process Control • Check if process sequence was correct • Process information
020.000.00161	14 00 A1	Message	Self diagnostic measures dark counts at dark count and RLU position.	Process Control • Check if process sequence was correct • Process information
020.000.00162	14 00 A2	Message	Self diagnostic sends min and max values of the supply voltage.	Process Control • Check if process sequence was correct • Process information
020.000.00163	14 00 A3	Message	Cycle counter evaluated during module self diagnostics.	Process Control • Check if process sequence was correct • Process information
020.000.00164	14 00 A4	Message	Cycle counter evaluated during module self diagnostics.	Process Control • Check if process sequence was correct • Process information

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.000.00165	14 00 A5	Message	Minimum motor current evaluated during module self diagnostics.	Process Control • Check if process sequence was correct • Process information
020.000.00166	14 00 A6	Message	Mechanical clearance of the drive evaluated during module self diagnostics.	Process Control • Check if process sequence was correct • Process information
020.000.00167	14 00 A7	Message	Sensor calibration value evaluated during module self diagnostics.	Process Control • Check if process sequence was correct • Process information
020.000.00176	14 00 B0	Warning	Error during Service Memory access.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.000.00177	14 00 B1	Message	Service Memory read data as long message.	Process Control • Check if process sequence was correct • Process information
020.000.00178	14 00 B2	Critical	Initialize Service Memory values failed.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.000.00179	14 00 B3	Message	Initialize Service Memory values done.	Process Control • Check if process sequence was correct • Process information
020.000.00180	14 00 B4	Warning	Mismatch at Service Memory SD Parameter definition.	Process Control • Check if process sequence was correct • Process information
020.000.00181	14 00 B5	Critical	Wrong Access-Key to modify SD data.	Process Control • Check if process sequence was correct • Process information Instrument Handling • Check consumable states • Initialize module

9. Reader

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.000.00191	14 00 BF	Message	Unexpected error during Self diagnostic.	Process Control • Check if process sequence was correct • Process information
020.000.00209	14 00 D1	Message	Self-diagnostic Device information: serial, article number and Version.	Process Control • Check if process sequence was correct • Process information

Table 9–2: Message list, Reader

10 Starter Dispense Pump 1

Device ID: 0x14

10.1 Sub assemblies

ID (dec)	ID (hex)	Name
6	0x06	General

Table 10–1: Sub assemblies, Starter Dispense Pump 1

10.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.006.00001	14 06 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
020.006.00002	14 06 02	Fatal	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.006.00003	14 06 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.006.00004	14 06 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.006.00005	14 06 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

10. Starter Dispense Pump 1

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.006.00006	14 06 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.006.00007	14 06 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.006.00008	14 06 08	Fatal	Motor overload	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.006.00009	14 06 09	Warning	Target position reached after retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.006.00010	14 06 0A	Fatal	Target position not reached by pump	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.006.00011	14 06 0B	Warning	Command terminated abnormally.	Subsequent

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.006.00012	14 06 0C	Fatal	Hardware Error	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.006.00013	14 06 0D	Fatal	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.006.00014	14 06 0E	Fatal	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.006.00015	14 06 0F	Fatal	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.006.00016	14 06 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.006.00017	14 06 11	Fatal	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.006.00018	14 06 12	Fatal	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

10. Starter Dispense Pump 1

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.006.00019	14 06 13	Fatal	The received parameter number needs an authorization which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
020.006.00020	14 06 14	Fatal	Received parameter authorization code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
020.006.00021	14 06 15	Fatal	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.006.00022	14 06 16	Fatal	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.006.00031	14 06 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.006.00032	14 06 20	Critical	Initialization of pump failed	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.006.00033	14 06 21	Critical	Initialization of pump failed	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.006.00162	14 06 A2	Message	Self diagnostic sends min and max values of the supply voltage.	Process Control
020.006.00163	14 06 A3	Message	Self diagnostic cycle counters.	Process Control
020.006.00165	14 06 A5	Message	Self diagnostic min drive current.	Process Control
020.006.00166	14 06 A6	Message	Self diagnostic mechanical calibration.	Process Control
020.006.00191	14 06 BF	Message	An unexpected error occurred during module self diagnostics.	Process Control
020.006.00176	14 06 B0	Warning	An internal error occurred	Calibration • Calibrate module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.006.00177	14 06 B1	Message	Service memory data.	Process Control
020.006.00178	14 06 B2	Critical	Initialization of service memory data failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.006.00179	14 06 B3	Message	Initialization of service memory data succeeded.	Process Control

10. Starter Dispense Pump 1

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.006.00180	14 06 B4	Warning	An internal error occurred	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.006.00181	14 06 B5	Critical	Service memory access denied.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Table 10–2: Message list, Starter Dispense Pump 1

11 Starter Dispense Pump 2

Device ID: 0x14

11.1 Sub assemblies

ID (dec)	ID (hex)	Name
7	0x07	General

Table 11–1: Sub assemblies, Starter Dispense Pump 2

11.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.007.00001	14 07 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
020.007.00002	14 07 02	Fatal	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.007.00003	14 07 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.007.00004	14 07 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.007.00005	14 07 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

11. Starter Dispense Pump 2

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.007.00006	14 07 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.007.00007	14 07 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.007.00008	14 07 08	Fatal	Motor overload	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.007.00009	14 07 09	Warning	Target position reached after retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.007.00010	14 07 0A	Fatal	Target position not reached by pump	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.007.00011	14 07 0B	Warning	Command terminated abnormally.	Subsequent

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.007.00012	14 07 0C	Fatal	Hardware Error	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.007.00013	14 07 0D	Fatal	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.007.00014	14 07 0E	Fatal	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.007.00015	14 07 0F	Fatal	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.007.00016	14 07 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.007.00017	14 07 11	Fatal	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.007.00018	14 07 12	Fatal	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

11. Starter Dispense Pump 2

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.007.00019	14 07 13	Fatal	The received parameter number needs an authorization which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
020.007.00020	14 07 14	Fatal	Received parameter authorization code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
020.007.00021	14 07 15	Fatal	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.007.00022	14 07 16	Fatal	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.007.00031	14 07 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.007.00032	14 07 20	Critical	Initialization of pump failed	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.007.00033	14 07 21	Critical	Initialization of pump failed	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
020.007.00162	14 07 A2	Message	Self diagnostic sends min and max values of the supply voltage.	Process Control
020.007.00163	14 07 A3	Message	Self diagnostic cycle counters.	Process Control
020.007.00165	14 07 A5	Message	Self diagnostic min drive current.	Process Control
020.007.00166	14 07 A6	Message	Self diagnostic mechanical calibration.	Process Control
020.007.00191	14 07 BF	Message	An unexpected error occurred during module self diagnostics.	Process Control
020.007.00176	14 07 B0	Warning	An internal error occurred	Calibration • Calibrate module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.007.00177	14 07 B1	Message	Service memory data.	Process Control
020.007.00178	14 07 B2	Critical	Initialization of service memory data failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
020.007.00179	14 07 B3	Message	Initialization of service memory data succeeded.	Process Control

11. Starter Dispense Pump 2

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
020.007.00180	14 07 B4	Warning	An internal error occurred	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
020.007.00181	14 07 B5	Critical	Service memory access denied.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Table 11–2: Message list, Starter Dispense Pump 2

12 Loading Bay

Device ID: 0x16

12.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General
1	0x01	Loading
2	0x02	RFID
3	0x03	Stirrer
4	0x04	Cooling
5	0x05	Scanner

Table 12–1: Sub assemblies, Loading Bay

12.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.000.00001	16 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
022.001.00001	16 01 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
022.002.00001	16 02 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
022.003.00001	16 03 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
022.004.00001	16 04 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module

12. Loading Bay

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.005.00001	16 05 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
022.000.00002	16 00 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.001.00002	16 01 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.002.00002	16 02 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.003.00002	16 03 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.004.00002	16 04 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.005.00002	16 05 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.000.00003	16 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.001.00003	16 01 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.002.00003	16 02 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.003.00003	16 03 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.004.00003	16 04 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.005.00003	16 05 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.000.00004	16 00 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

12. Loading Bay

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.000.00005	16 00 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
022.000.00006	16 00 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
022.000.00007	16 00 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
022.003.00009	16 03 09	Warning	Target position reached after retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
022.003.00010	16 03 0A	Fatal	Target position not reached by stirrer	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
022.000.00011	16 00 0B	Fatal	Command terminated abnormally.	Subsequent
022.001.00011	16 01 0B	Fatal	Command terminated abnormally.	Subsequent
022.002.00011	16 02 0B	Fatal	Command terminated abnormally.	Subsequent

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.003.00011	16 03 0B	Fatal	Command terminated abnormally.	Subsequent
022.004.00011	16 04 0B	Fatal	Command terminated abnormally.	Subsequent
022.005.00011	16 05 0B	Fatal	Command terminated abnormally.	Subsequent
022.000.00013	16 00 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.001.00013	16 01 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.002.00013	16 02 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.003.00013	16 03 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.004.00013	16 04 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.005.00013	16 05 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.000.00014	16 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

12. Loading Bay

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.001.00014	16 01 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.002.00014	16 02 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.003.00014	16 03 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.004.00014	16 04 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.005.00014	16 05 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.000.00015	16 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.001.00015	16 01 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.002.00015	16 02 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.003.00015	16 03 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.004.00015	16 04 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.005.00015	16 05 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.000.00016	16 00 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.001.00016	16 01 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

12. Loading Bay

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.002.00016	16 02 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.003.00016	16 03 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.004.00016	16 04 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.005.00016	16 05 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.000.00017	16 00 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.000.00018	16 00 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.000.00019	16 00 13	Critical	The received parameter number needs an authorisation which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
022.000.00020	16 00 14	Critical	Received parameter authorisation code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
022.000.00021	16 00 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.000.00022	16 00 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.003.00023	16 03 17	Fatal	A movement problem has been detected	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
022.000.00031	16 00 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
022.000.00032	16 00 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

12. Loading Bay

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.001.00032	16 01 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.002.00032	16 02 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.003.00032	16 03 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.004.00032	16 04 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.005.00032	16 05 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.000.00036	16 00 24	Message	Temperature report.	Process Control
022.000.00033	16 00 21	Critical	Actual temperature is not in allowed range defined by parameters.	Environmental • Check module environment (temperature, humidity) • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.004.00033	16 04 21	Critical	Actual temperature is not in allowed range defined by parameters.	Environmental • Check module environment (temperature, humidity) • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.000.00034	16 00 22	Warning	Actual temperature is back in allowed range defined by parameters.	Environmental • Check module environment (temperature, humidity) • Exchange module Process Control • Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.004.00034	16 04 22	Warning	Actual temperature is back in allowed range defined by parameters.	Environmental • Check module environment (temperature, humidity) • Exchange module Process Control • Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.000.00035	16 00 23	Fatal	Temperature sensor fails plausibility check	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.004.00035	16 04 23	Fatal	Temperature sensor fails plausibility check	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.004.00037	16 04 25	Warning	Seebeck voltage too low	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
022.004.00038	16 04 26	Critical	Seebeck voltage too high	Electronics • Check fan • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Check fan • Clean module • Check calibration • Check mechanic • Exchange module
022.004.00039	16 04 27	Warning	Difference between monitored seebeck voltages exceeds limits. Peltier elements defect or not connected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
022.004.00040	16 04 28	Warning	Warm side fan not connected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
022.004.00041	16 04 29	Critical	All monitored seebeck voltages exceeds limits. All peltiers are switched off.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.004.00042	16 04 2A	Warning	Icing protection temperature reached.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.004.00043	16 04 2B	Critical	Temperature sensor fails plausibility check	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.004.00044	16 04 2C	Critical	Temperature control is currently disabled.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.003.00048	16 03 30	Fatal	Initialization of stirrer failed	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
022.003.00049	16 03 31	Fatal	Initialization of stirrer failed	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
022.003.00050	16 03 32	Message	Stirring started	Process Control
022.003.00051	16 03 33	Message	Stirring stopped	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.003.00052	16 03 34	Critical	Stirrer motor rotation error frequency too low	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
022.000.00064	16 00 40	Message	Rack occupation change detected without scanning any barcode.	Process Control
022.000.00066	16 00 42	Message	Rack occupation change detected without scanning any barcode.	Process Control
022.000.00067	16 00 43	Message	Rack occupation changed due to insertion of rack and scanning of valid barcode information.	Process Control
022.000.00068	16 00 44	Message	Reagent flap recently opened or closed	Process Control
022.000.00069	16 00 45	Message	Sample flap recently opened or closed	Process Control
022.001.00080	16 01 50	Message	Scanner activated	Process Control
022.001.00081	16 01 51	Message	Scanner deactivated	Process Control
022.001.00082	16 01 52	Warning	Scanner laser on time exceeds limits. Laser is switched off.	Process Control
022.001.00083	16 01 53	Warning	Not plausible barcode read. It may be that rack insertion took too long.	Instrument Handling • Check consumable states • Initialize module
022.001.00084	16 01 54	Warning	All lanes are already occupied.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.001.00085	16 01 55	Warning	Implausible lane occupation. Besides the current rack insertion additional occupation changes detected within a short period of time.	Instrument Handling • Check consumable states • Initialize module
022.001.00095	16 01 5F	Fatal	Embedded software crash occurs (loading).	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.005.00096	16 05 60	Critical	An Internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
022.005.00097	16 05 61	Critical	An Internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.005.00098	16 05 62	Warning	Response from scanner too long.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.005.00099	16 05 63	Critical	Sending to scanner takes too long.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
022.005.00100	16 05 64	Critical	Answer from scanner takes too long.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
022.005.00101	16 05 65	Critical	Answer from scanner has wrong format.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

12. Loading Bay

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.005.00102	16 05 66	Message	Scanner polygon wheel on time exceeds limits. Polygon wheel is switched off.	Process Control
022.005.00103	16 05 67	Critical	Error occurred during scanner communication.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
022.005.00111	16 05 6F	Fatal	Embedded software crash occurs (scanner).	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.001.00112	16 01 70	Warning	First position barcode missing (/A)	Consumable • Exchange consumable • Exchange module Instrument Handling • Check consumable states • Initialize module
022.001.00113	16 01 71	Warning	Rack ID barcode missing	Consumable • Exchange consumable • Exchange module Instrument Handling • Check consumable states • Initialize module
022.001.00114	16 01 72	Warning	Any position barcode missing	Consumable • Exchange consumable • Exchange module Instrument Handling • Check consumable states • Initialize module
022.001.00115	16 01 73	Warning	Conversion error of tube status and/or barcode type information	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.001.00116	16 01 74	Warning	Barcode length exceeds limits	Consumable • Exchange consumable • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.001.00117	16 01 75	Warning	Doubling of 0x26 character within barcode failed as barcode length after doubling exceeds limits	Consumable • Exchange consumable • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.001.00118	16 01 76	Warning	Rack position barcodes not continuously present or more barcodes read than rack position barcodes	Consumable • Exchange consumable • Exchange module
022.001.00119	16 01 77	Message	Scanner raw data string as input for firmware decoder function.	Process Control
022.002.00128	16 02 80	Warning	Access to VICC disturbed, retry needed.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Consumable • Exchange consumable • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module

12. Loading Bay

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.002.00129	16 02 81	Critical	Access to VICC failed.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Consumable • Exchange consumable • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
022.002.00130	16 02 82	Critical	An error occurred while accessing RFTag	Consumable • Exchange consumable • Exchange module
022.002.00131	16 02 83	Critical	No tag detected.	Consumable • Exchange consumable • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Instrument Handling • Check consumable states • Initialize module
022.002.00132	16 02 84	Critical	Tag type neither defined as module parameter nor as command parameter.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.002.00133	16 02 85	Critical	Mismatch of expected and detected tag type.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Consumable • Exchange consumable • Exchange module
022.002.00134	16 02 86	Critical	Written data does not match to reread data.	Consumable • Exchange consumable • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.002.00135	16 02 87	Critical	First block out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.002.00136	16 02 88	Critical	Number of blocks out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.002.00137	16 02 89	Critical	Requested data from tag does not fit into command response buffer.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.002.00138	16 02 8A	Critical	Mismatch of block parameters and amount of data, both provided by command.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.002.00139	16 02 8B	Warning	Timeout appeared during access to tag. Tag response to read/write command takes too long time. Communication issue between reader IC and tag.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.002.00143	16 02 8F	Warning	Reader library error message.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Consumable • Exchange consumable • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
022.002.00144	16 02 90	Critical	Antenna not connected to specified reader IC	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

12. Loading Bay

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.002.00145	16 02 91	Critical	I2C communication failed	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
022.002.00159	16 02 9F	Fatal	Internal error of RFID components. Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
022.004.00160	16 04 A0	Message	Self diagnostic cyclic report of essential cooling data.	Process Control
022.000.00161	16 00 A1	Message	Self diagnostic sends min and max values of the supply voltage.	Process Control
022.000.00162	16 00 A2	Message	Self diagnostic cycle counter.	Process Control
022.000.00163	16 00 A3	Message	Self diagnostic cycle counter.	Process Control
022.000.00164	16 00 A4	Message	Self diagnostic operation time.	Process Control
022.000.00165	16 00 A5	Message	Self diagnostic operation time.	Process Control
022.000.00166	16 00 A6	Message	Self diagnostic min drive current.	Process Control
022.000.00167	16 00 A7	Message	Self-diagnostic Device information: serial, article number and Version.	Process Control
022.000.00175	16 00 AF	Message	Error exception during SD is wrapped into an SD message.	Process Control
022.000.00176	16 00 B0	Warning	Error during ServiceMemory access.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
022.000.00191	16 00 BF	Message	Successful read or write access to Service Memory.	Process Control
022.000.00177	16 00 B1	Message	Service Memory read data as long message.	Process Control
022.000.00178	16 00 B2	Critical	Initialize Service Memory values failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
022.000.00179	16 00 B3	Message	Initialize Service Memory values done.	Process Control
022.000.00180	16 00 B4	Warning	Mismatch at Service Memory SD Parameter definition.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
022.000.00181	16 00 B5	Critical	Wrong Access-Key to modify SD data.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Table 12-2: Message list, Loading Bay

13 Liquid Support

Device ID: 0x18

13.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General
1	0x01	Bubble Sensor 1
2	0x02	Bubble Sensor 2
3	0x03	Heat Tube 1
4	0x04	Heat Tube 2
5	0x05	RFID Starter

Table 13–1: Sub assemblies, Liquid Support

13.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.000.00001	18 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
024.001.00001	18 01 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
024.002.00001	18 02 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
024.003.00001	18 03 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
024.004.00001	18 04 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module

13. Liquid Support

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.005.00001	18 05 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
024.000.00002	18 00 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.001.00002	18 01 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.002.00002	18 02 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.003.00002	18 03 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.004.00002	18 04 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.005.00002	18 05 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.000.00003	18 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.003.00003	18 03 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.004.00003	18 04 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.000.00004	18 00 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
024.000.00005	18 00 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
024.000.00006	18 00 06	Critical	Voltage is out of range. Detected voltage: {DetectedVoltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
024.000.00007	18 00 07	Warning	Voltage is back in range. Detected voltage: {DetectedVoltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

13. Liquid Support

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.000.00011	18 00 0B	Fatal	Command terminated abnormally.	Subsequent
024.000.00012	18 00 0C	Fatal	An Internal error occurred during self test	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
024.000.00013	18 00 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.001.00013	18 01 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.002.00013	18 02 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.003.00013	18 03 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.004.00013	18 04 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.005.00013	18 05 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.000.00014	18 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.001.00014	18 01 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.002.00014	18 02 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.003.00014	18 03 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.004.00014	18 04 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.005.00014	18 05 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.000.00015	18 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.001.00015	18 01 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

13. Liquid Support

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.002.00015	18 02 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.003.00015	18 03 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.004.00015	18 04 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.005.00015	18 05 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.000.00016	18 00 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.001.00016	18 01 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.002.00016	18 02 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.003.00016	18 03 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.004.00016	18 04 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.005.00016	18 05 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.000.00017	18 00 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.000.00018	18 00 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.000.00019	18 00 13	Critical	The received parameter number needs an authorisation which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
024.000.00020	18 00 14	Critical	Received parameter authorisation code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
024.000.00021	18 00 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.000.00022	18 00 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.000.00032	18 00 20	Message	Bottle occupation state changed	Process Control
024.000.00048	18 00 30	Message	Flap Status state changed	Process Control
024.000.00064	18 00 40	Message	Liquid Level state changed	Process Control
024.001.00080	18 01 50	Critical	Bubble sensor 1 calibration error	Calibration • Calibrate module Instrument Handling • Check consumable states • Initialize module
024.002.00080	18 02 50	Critical	Bubble sensor 2 calibration error	Calibration • Calibrate module Instrument Handling • Check consumable states • Initialize module
024.001.00081	18 01 51	Critical	During aspiration the measured volume is out of range.	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.002.00081	18 02 51	Critical	During aspiration the measured volume is out of range.	Process Control
024.001.00082	18 01 52	Message	Bubble measure is finished and returns measured values.	Process Control
024.002.00082	18 02 52	Message	Bubble measure is finished and returns measured values.	Process Control
024.001.00084	18 01 54	Warning	During aspiration the max number of bubble measure data is reached.	Process Control
024.002.00084	18 02 54	Warning	During aspiration the max number of bubble measure data is reached.	Process Control
024.001.00085	18 01 55	Critical	Sensor voltage is out of range.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
024.002.00085	18 02 55	Critical	Sensor voltage is out of range.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
024.005.00128	18 05 80	Warning	Access to VICC disturbed, retry needed.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Consumable • Exchange consumable • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.005.00129	18 05 81	Critical	Access to VICC failed.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Consumable • Exchange consumable • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
024.005.00130	18 05 82	Critical	An error occurred while accessing RFTag	Consumable • Exchange consumable • Exchange module
024.005.00131	18 05 83	Critical	No tag detected.	Consumable • Exchange consumable • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Instrument Handling • Check consumable states • Initialize module
024.005.00132	18 05 84	Critical	Tag type neither defined as module parameter nor as command parameter.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.005.00133	18 05 85	Critical	Mismatch of expected and detected tag type.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.005.00134	18 05 86	Critical	Written data does not match to reread data.	Consumable • Exchange consumable • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.005.00135	18 05 87	Critical	First block out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.005.00136	18 05 88	Critical	Number of blocks out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.005.00137	18 05 89	Critical	Requested data from tag does not fit into command response buffer.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.005.00138	18 05 8A	Critical	Mismatch of block parameters and amount of data, both provided by command.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.005.00143	18 05 8F	Warning	Reader library error message.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Consumable • Exchange consumable • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
024.005.00112	18 05 70	Critical	Antenna not connected to specified reader IC	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.005.00113	18 05 71	Message	Reader library error message.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Consumable • Exchange consumable • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
024.005.00144	18 05 90	Message	Starter bottle state has changed.	
024.000.00031	18 00 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
024.000.00023	18 00 17	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.001.00023	18 01 17	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.002.00023	18 02 17	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.003.00023	18 03 17	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.004.00023	18 04 17	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.005.00023	18 05 17	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
024.003.00160	18 03 A0	Warning	Define valid temperature control parameters.	Process Control
024.004.00160	18 04 A0	Warning	Define valid temperature control parameters.	Process Control
024.003.00161	18 03 A1	Warning	Temperature is out of range	Process Control
024.004.00161	18 04 A1	Warning	Temperature is out of range	Process Control
024.003.00162	18 03 A2	Warning	Temperature is in range	Process Control
024.004.00162	18 04 A2	Warning	Temperature is in range	Process Control
024.003.00163	18 03 A3	Critical	Measured temperature value failed plausibility check	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
024.004.00163	18 04 A3	Critical	Measured temperature value failed plausibility check	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
024.003.00164	18 03 A4	Critical	The heater drive IC reported a failure	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
024.004.00164	18 04 A4	Critical	The heater drive IC reported a failure	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
024.003.00165	18 03 A5	Critical	Temperature watchdog switched off	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

13. Liquid Support

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.004.00165	18 04 A5	Critical	Temperature watchdog switched off.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
024.003.00166	18 03 A6	Message	Cyclical temperature report for selected heater.	
024.004.00166	18 04 A6	Message	Cyclical temperature report for selected heater.	
024.003.00167	18 03 A7	Critical	Target temperature not reached with in the defined time.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module System • Check module environment (temperature, humidity) • Exchange module
024.004.00167	18 04 A7	Critical	Target temperature not reached with in the defined time.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module System • Check module environment (temperature, humidity) • Exchange module
024.003.00168	18 03 A8	Message	temperature at the start of self test and at the end of self test.	
024.004.00168	18 04 A8	Message	temperature at the start of self test and at the end of self test.	
024.003.00169	18 03 A9	Warning	Heater not connected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.004.00169	18 04 A9	Warning	Heater not connected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
024.000.00176	18 00 B0	Message	Self diagnostic sends min and max values of the supply voltage.	
024.000.00177	18 00 B1	Message	Self diagnostic cycle counter.	
024.000.00178	18 00 B2	Message	Self diagnostic cycle counter.	
024.000.00179	18 00 B3	Message	Self diagnostic cycle counter.	
024.000.00180	18 00 B4	Message	Self diagnostic cycle counter.	
024.000.00181	18 00 B5	Message	Self diagnostic cycle counter.	
024.000.00182	18 00 B6	Message	Self diagnostic cycle counter.	
024.000.00183	18 00 B7	Message	Self diagnostic cycle counter.	
024.003.00184	18 03 B8	Message	Self diagnostic cycle counter.	
024.004.00184	18 04 B8	Message	Self diagnostic cycle counter.	
024.000.00191	18 00 BF	Message	Unexpected error during Self diagnostic.	
024.000.00192	18 00 C0	Warning	Error during Service Memory access.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
024.000.00193	18 00 C1	Message	Service Memory read data as long message.	
024.000.00194	18 00 C2	Critical	Initialize Service Memory values failed.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

13. Liquid Support

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
024.000.00195	18 00 C3	Message	Initialize Service Memory values done.	Process Control • Check consumable states • Initialize module Instrument Handling
024.000.00196	18 00 C4	Warning	Mismatch at Service Memory SD Parameter definition.	
024.000.00197	18 00 C5	Critical	Wrong Access-Key to modify SD data.	
024.000.00209	18 00 D1	Message	Self-diagnostic Device information: serial, article number and Version.	

Table 13–2: Message list, Liquid Support

14 Wash Buffer Dispense Pump

Device ID: **0x32**

14.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 14–1: Sub assemblies, Wash Buffer Dispense Pump

14.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
050.000.00001	32 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
050.000.00002	32 00 02	Fatal	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
050.000.00003	32 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
050.000.00004	32 00 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
050.000.00005	32 00 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

14. Wash Buffer Dispense Pump

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
050.000.00006	32 00 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
050.000.00007	32 00 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
050.000.00008	32 00 08	Fatal	Motor overload	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
050.000.00009	32 00 09	Warning	Target position reached after retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
050.000.00010	32 00 0A	Fatal	Target position not reached by washer pump	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
050.000.00011	32 00 0B	Warning	Command terminated abnormally.	Subsequent

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
050.000.00012	32 00 0C	Fatal	Hardware Error	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
050.000.00013	32 00 0D	Fatal	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
050.000.00014	32 00 0E	Fatal	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
050.000.00015	32 00 0F	Fatal	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
050.000.00016	32 00 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
050.000.00017	32 00 11	Fatal	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
050.000.00018	32 00 12	Fatal	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

14. Wash Buffer Dispense Pump

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
050.000.00019	32 00 13	Fatal	The received parameter number needs an authorization which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
050.000.00020	32 00 14	Fatal	Received parameter authorization code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
050.000.00021	32 00 15	Fatal	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
050.000.00022	32 00 16	Fatal	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
050.000.00031	32 00 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
050.000.00032	32 00 20	Critical	Initialization of pump failed	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
050.000.00033	32 00 21	Critical	Initialization of pump failed	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
050.000.00162	32 00 A2	Message	Self diagnostic sends min and max values of the supply voltage.	Process Control
050.000.00163	32 00 A3	Message	Self diagnostic cycle counters.	Process Control
050.000.00165	32 00 A5	Message	Self diagnostic min drive current.	Process Control
050.000.00166	32 00 A6	Message	Self diagnostic mechanical calibration.	Process Control
050.000.00191	32 00 BF	Message	An unexpected error occurred during module self diagnostics.	Process Control
050.000.00176	32 00 B0	Warning	An internal error occurred	Calibration • Calibrate module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
050.000.00177	32 00 B1	Message	Service memory data.	Process Control
050.000.00178	32 00 B2	Critical	Initialization of service memory data failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
050.000.00179	32 00 B3	Message	Initialization of service memory data succeeded.	Process Control

14. Wash Buffer Dispense Pump

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
050.000.00180	32 00 B4	Warning	An internal error occurred	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
050.000.00181	32 00 B5	Critical	Service memory access denied.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
050.000.00209	32 00 D1	Message	Self-diagnostic Device information: serial, article number and Version.	Process Control

Table 14–2: Message list, Wash Buffer Dispense Pump

15 WAP

Device ID: 0x35

15.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 15–1: Sub assemblies, WAP

15.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
053.000.00001	35 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
053.000.00002	35 00 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
053.000.00003	35 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
053.000.00004	35 00 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
053.000.00005	35 00 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
053.000.00006	35 00 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
053.000.00007	35 00 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
053.000.00008	35 00 08	Fatal	Motor overload	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
053.000.00009	35 00 09	Warning	Target position reached after retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
053.000.00010	35 00 0A	Fatal	Target position not reached by washer pump	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
053.000.00011	35 00 0B	Fatal	Command terminated abnormally.	Subsequent

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
053.000.00012	35 00 0C	Fatal	Hardware Error	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
053.000.00013	35 00 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
053.000.00014	35 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
053.000.00015	35 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
053.000.00016	35 00 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
053.000.00017	35 00 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
053.000.00018	35 00 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
053.000.00019	35 00 13	Critical	The received parameter number needs an authorization which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
053.000.00020	35 00 14	Critical	Received parameter authorization code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
053.000.00021	35 00 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
053.000.00022	35 00 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
053.000.00031	35 00 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
053.000.00033	35 00 21	Warning	Move interrupted.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
053.000.00034	35 00 22	Message	Move completely finished.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
053.000.00162	35 00 A2	Message	Self diagnostic sends min and max values of the supply voltage.	Process Control
053.000.00163	35 00 A3	Message	Self diagnostic cycle counters.	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
053.000.00165	35 00 A5	Message	Self diagnostic min drive current.	Process Control
053.000.00191	35 00 BF	Message	An unexpected error occurred during module selfdiagnostics.	Process Control
053.000.00176	35 00 B0	Warning	An internal error occurred	Calibration • Calibrate module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
053.000.00177	35 00 B1	Message	Service memory data.	Process Control
053.000.00178	35 00 B2	Critical	Initialization of service memory data failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
053.000.00179	35 00 B3	Message	Initialization of service memory data succeeded.	Process Control
053.000.00180	35 00 B4	Warning	An internal error occurred	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
053.000.00181	35 00 B5	Critical	Service memory access denied.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
053.000.00209	35 00 D1	Message	Self-diagnostic Device information: serial, article number and Version.	Process Control

Table 15–2: Message list, WAP

16 Front Cover

Device ID: 0x40

16.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 16–1: Sub assemblies, Front Cover

16.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
064.000.00001	40 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
064.000.00002	40 00 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
064.000.00003	40 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
064.000.00004	40 00 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
064.000.00005	40 00 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
064.000.00006	40 00 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
064.000.00007	40 00 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
064.000.00011	40 00 0B	Fatal	Command terminated abnormally.	Subsequent
064.000.00013	40 00 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
064.000.00014	40 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
064.000.00015	40 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
064.000.00016	40 00 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
064.000.00017	40 00 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
064.000.00018	40 00 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
064.000.00019	40 00 13	Critical	The received parameter number needs an authorisation which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
064.000.00020	40 00 14	Critical	Received parameter authorisation code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
064.000.00021	40 00 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
064.000.00022	40 00 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
064.000.00024	40 00 18	Critical	Communication issue: Frame order, or start or end frame number is invalid on long messages.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
064.000.00031	40 00 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
064.000.00032	40 00 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
064.000.00048	40 00 30	Message	Flap recently opened or closed.	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
064.000.00176	40 00 B0	Message	Self diagnostic sends min and max values of the supply voltage.	Process Control
064.000.00177	40 00 B1	Message	Self diagnostic cycle counter.	Process Control
064.000.00178	40 00 B2	Message	Self diagnostic cycle counter.	Process Control
064.000.00179	40 00 B3	Message	Self diagnostic cycle counter.	Process Control
064.000.00180	40 00 B4	Message	Self diagnostic cycle counter.	Process Control
064.000.00181	40 00 B5	Message	Self diagnostic cycle counter.	Process Control
064.000.00191	40 00 BF	Message	Unexpected error during Self diagnostic.	Process Control
064.000.00192	40 00 C0	Warning	Error during Service Memory access.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
064.000.00193	40 00 C1	Message	Service Memory read data as long message.	Process Control
064.000.00194	40 00 C2	Critical	Initialize Service Memory values failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
064.000.00195	40 00 C3	Message	Initialize Service Memory values done.	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
064.000.00196	40 00 C4	Warning	Mismatch at Service Memory SD Parameter definition.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
064.000.00197	40 00 C5	Critical	Wrong Access-Key to modify SD data.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
064.000.00209	40 00 D1	Message	Self-diagnostic Device information: serial, article number and Version.	Process Control

Table 16–2: Message list, Front Cover

17 Can Bus Support

Device ID: 0xD0

17.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 17–1: Sub assemblies, Can Bus Support

17.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
208.000.00003	D0 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
208.000.00014	D0 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
208.000.00015	D0 00 0F	Fatal	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
208.000.00032	D0 00 20	Message	Can Bus Device State Report.	Process Control

Table 17–2: Message list, Can Bus Support

18 Application Manager

Device ID: 0xD1

18.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 18–1: Sub assemblies, Application Manager

18.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
209.000.00003	D1 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
209.000.00011	D1 00 0B	Fatal	Command terminated abnormally.	Subsequent
209.000.00014	D1 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
209.000.00015	D1 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
209.000.00016	D1 00 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connection • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
209.000.00032	D1 00 20	Message	Global state has changed.	Process Control
209.000.00035	D1 00 23	Fatal	Current off line state of instrument does not allow command execution. Power-up instrument.	Instrument Handling • Check consumable states • Initialize module
209.000.00036	D1 00 24	Fatal	Instrument stopped by command.	Process Control
209.000.00037	D1 00 25	Critical	The operation (e.g. priming) was aborted by exception and AM returned to idle state. Detailed description for the reason is given by reported exception(s).	Process Control Subsequent
209.000.00038	D1 00 26	Critical	Current state of instrument does not allow mandatory switch to another state. Perhaps another operation is still in process.	Process Control
209.000.00080	D1 00 50	Warning	Internal error memory nearly full. Please collect data with SSW and erase error memory.	Process Control
209.000.00081	D1 00 51	Message	Instrument's error memory entry.	Process Control
209.000.00128	D1 00 80	Fatal	Synchronous Messaging Interface in use	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
209.000.00129	D1 00 81	Fatal	Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
209.000.00130	D1 00 82	Fatal	Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
209.000.00131	D1 00 83	Fatal	IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
209.000.00132	D1 00 84	Fatal	IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
209.000.00134	D1 00 86	Critical	Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
209.000.00135	D1 00 87	Fatal	Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
209.000.00190	D1 00 BE	Message	Self-diagnostic Device information: serial, article number and Version.	
209.000.00207	D1 00 CF	Message	Process for self diagnostic on instrument has finished.	

Table 18–2: Message list, Application Manager

19 Job Manager

Device ID: 0xD2

19.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 19–1: Sub assemblies, Job Manager

19.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
210.000.00001	D2 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
210.000.00002	D2 00 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00011	D2 00 0B	Fatal	Command terminated abnormally.	Subsequent
210.000.00013	D2 00 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00014	D2 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
210.000.00015	D2 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00016	D2 00 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00032	D2 00 20	Fatal	Embedded Software Crash: Memory inconsistency in the job list	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00033	D2 00 21	Critical	Embedded Software Crash: Job memory full. Wait until one job is executed. JobId {Job Identifier}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00034	D2 00 22	Critical	Start cycle smaller than current cycle (Job sent too late). JobId {Job Identifier}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00035	D2 00 23	Critical	Same start cycle as a job already programmed. JobId {Job Identifier}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00036	D2 00 24	Critical	An internal error occurred while processing Job Id {Job Identifier}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00037	D2 00 25	Critical	Size of assay structure does not fit to the length of the command	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00038	D2 00 26	Critical	An internal error occurred while processing Job Id {Job Identifier}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00039	D2 00 27	Critical	Required dilution pipetting sequence not stored. Job Id {Job Identifier}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
210.000.00040	D2 00 28	Critical	The number of resources does not match to the number required by the assay. JobId {Job Identifier}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00041	D2 00 29	Critical	Reagent identifier is unknown to the assay. JobId {Job Identifier}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00042	D2 00 2A	Critical	Maximum number of reagents exceeded. JobId {Job Identifier}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00043	D2 00 2B	Critical	Invalid reagent position. JobId {Job Identifier}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00044	D2 00 2C	Critical	Maximum number of pre-treatments exceeded. Job Id {Job Identifier}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00045	D2 00 2D	Critical	Dilution volume out of range. JobId {Job Identifier}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00048	D2 00 30	Fatal	Special event list is full.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00049	D2 00 31	Fatal	Schedule conflict: Phase is still in use. JobId {JobIdentifier1} and Job Id {JobIdentifier2}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00050	D2 00 32	Critical	Job rejected: Assay sequence not plausible. Job Id {Job Identifier}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00052	D2 00 34	Critical	An internal error occurred while processing	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
210.000.00051	D2 00 33	Critical	Job {Job Identifier} aborted: Needed resource {Resource Id} was suspended.	Instrument Handling • Check consumable states • Initialize module
210.000.00080	D2 00 50	Critical	Resource {Resource Id} is empty	Instrument Handling • Check consumable states • Initialize module
210.000.00097	D2 00 61	Critical	Job canceled due user request. JobId {Job Identifier}	Instrument Handling • Check consumable states • Initialize module
210.000.00128	D2 00 80	Fatal	Embedded Software Crash: Synchronous Messaging Interface in use	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00129	D2 00 81	Fatal	Embedded Software Crash: Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00130	D2 00 82	Fatal	Embedded Software Crash: Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00131	D2 00 83	Fatal	Embedded Software Crash: IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00132	D2 00 84	Fatal	Embedded Software Crash: IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00134	D2 00 86	Critical	Embedded Software Crash: Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00135	D2 00 87	Fatal	Embedded Software Crash	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00144	D2 00 90	Message	First Job stored in Job List	Process Control
210.000.00145	D2 00 91	Message	Last Job in Job List recently started	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
210.000.00146	D2 00 92	Message	Last Job in Job List recently finished	Process Control
210.000.00147	D2 00 93	Critical	Job Id {Job Identifier} unknown.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
210.000.00148	D2 00 94	Message	Job Id {Job Identifier} changed the state to {Job State}.	Process Control
210.000.00149	D2 00 95	Message	Test {Test Number} of Job Id {Job Identifier} changed the state to {Test State}.	Process Control
210.000.00151	D2 00 97	Fatal	Unknown event in the special event list	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Table 19–2: Message list, Job Manager

20 Assay Manager

Device ID: 0xD3

20.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 20–1: Sub assemblies, Assay Manager

20.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
211.000.00011	D3 00 0B	Fatal	Command terminated abnormally.	Subsequent
211.000.00013	D3 00 0D	Critical	Check syntax, sequence and values.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00014	D3 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00015	D3 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00024	D3 00 18	Critical	Embedded Software Crash: Invalid frame number given for long message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00032	D3 00 20	Warning	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
211.000.00033	D3 00 21	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00034	D3 00 22	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00035	D3 00 23	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00036	D3 00 24	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00037	D3 00 25	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00038	D3 00 26	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00039	D3 00 27	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00040	D3 00 28	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00041	D3 00 29	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00042	D3 00 2A	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
211.000.00043	D3 00 2B	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00044	D3 00 2C	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00045	D3 00 2D	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00046	D3 00 2E	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00048	D3 00 30	Warning	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00049	D3 00 31	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00050	D3 00 32	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00051	D3 00 33	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00052	D3 00 34	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00053	D3 00 35	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
211.000.00054	D3 00 36	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00055	D3 00 37	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00056	D3 00 38	Critical	It was not possible to correctly handle the Assay protocol	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00057	D3 00 39	Critical	Sample pipetting step is used more than once	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00128	D3 00 80	Fatal	Embedded software crash: Synchronous Messaging Interface in use	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00129	D3 00 81	Fatal	Embedded software crash: Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00130	D3 00 82	Fatal	Embedded software crash: Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00131	D3 00 83	Fatal	Embedded software crash: IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00132	D3 00 84	Fatal	Embedded software crash: IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
211.000.00134	D3 00 86	Critical	Embedded software crash: Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
211.000.00135	D3 00 87	Fatal	Embedded software crashed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Table 20–2: Message list, Assay Manager

21 Dispatcher

Device ID: 0xD4

21.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General
1	0x01	LeftArm
2	0x02	RightArm

Table 21–1: Sub assemblies, Dispatcher

21.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
212.000.00003	D4 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
212.000.00011	D4 00 0B	Fatal	Command terminated abnormally.	Subsequent
212.000.00014	D4 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
212.000.00015	D4 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
212.001.00015	D4 01 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

21. Dispatcher

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
212.002.00015	D4 02 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
212.000.00034	D4 00 22	Fatal	System cycle belated by {Delay}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
212.000.00035	D4 00 23	Critical	system delay by {Delay} in work list queue {Queue Id} occurred	Process Control
212.000.00048	D4 00 30	Message	Job Id {Job Id} has new cuvette position {Carrier Pos}	Process Control
212.001.00051	D4 01 33	Message	Pipetting Step: JobId {Job Id} Index {PipStepReport.PipStepParamIndex} Type {PipStepReport.StepType} Phase {PipStepReport.PipPhase} Target {PipStepReport.Target} LLDResult {PipStepReport.LldResult} Z-Height {PipStepReport.Z_Height} SensorResult {PipStepReport.SensorResult}	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
212.002.00051	D4 02 33	Message	Pipetting Step: JobId {Job Id} Index {PipStepReport.PipStepParamIndex} Type {PipStepReport.StepType} Phase {PipStepReport.PipPhase} Target {PipStepReport.Target} LLDResult {PipStepReport.LldResult} Z-Height {PipStepReport.Z_Height} SensorResult {PipStepReport.SensorResult}	Process Control
212.000.00052	D4 00 34	Message	result for completed measurement available	Process Control
212.000.00053	D4 00 35	Message	result for dark count measurement available	Process Control
212.000.00064	D4 00 40	Fatal	Schedule mismatch error detected affecting job.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
212.000.00065	D4 00 41	Critical	No cuvette available for job	Process Control
212.000.00067	D4 00 43	Critical	Starter reagent temperature is out of range for job while injection	Process Control
212.000.00068	D4 00 44	Critical	Aspiration at washer probe failed for job	Process Control
212.000.00069	D4 00 45	Critical	Aspiration at waste probe failed for job	Process Control
212.000.00080	D4 00 50	Critical	Irregular result for completed measurement received	Process Control
212.001.00096	D4 01 60	Critical	Pickup of disposable tip failed	Process Control
212.001.00097	D4 01 61	Critical	Not enough liquid found by probe (LLD) for Job Id {Job Id}	Process Control

21. Dispatcher

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
212.002.00097	D4 02 61	Critical	Not enough liquid found by probe (LLD) for Job Id {Job Id}	Process Control
212.001.00098	D4 01 62	Critical	No liquid found by probe (LLD) for Job Id {Job Id}	Process Control
212.002.00098	D4 02 62	Critical	No liquid found by probe (LLD) for Job Id {Job Id}	Process Control
212.001.00102	D4 01 66	Critical	Z Move was denied for Job Id {Job Id}	Process Control
212.002.00102	D4 02 66	Critical	Z Move was denied for Job Id {Job Id}	Process Control
212.001.00103	D4 01 67	Critical	Pip result not okay for Job Id {Job Id}	Process Control
212.002.00103	D4 02 67	Critical	Pip result not okay for Job Id {Job Id}	Process Control
212.001.00104	D4 01 68	Critical	Pressure threshold exceeded by probe (ADPM) for Job Id {Job Id}	Process Control
212.001.00105	D4 01 69	Critical	Clot detected by probe (ADPM) for Job Id {Job Id}	Process Control
212.001.00106	D4 01 6A	Critical	Overpressure detected found by probe (ADPM) for Job Id {Job Id}	Process Control
212.002.00106	D4 02 6A	Critical	Overpressure detected found by probe (ADPM) for Job Id {Job Id}	Process Control
212.002.00112	D4 02 70	Critical	Insufficient volume detected by probe (OLV) for Job Id {Job Id}	Process Control
212.000.00128	D4 00 80	Fatal	Synchronous Messaging Interface in use	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
212.000.00129	D4 00 81	Fatal	Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
212.000.00130	D4 00 82	Fatal	Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
212.000.00131	D4 00 83	Fatal	IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
212.000.00132	D4 00 84	Fatal	IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
212.000.00134	D4 00 86	Critical	Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
212.000.00135	D4 00 87	Fatal	Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
212.000.00144	D4 00 90	Fatal	Memory inconsistency in data structure	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
212.000.00145	D4 00 91	Fatal	No memory available for additional data	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
212.000.00192	D4 00 C0	Message	Job Id {Job Identifier} changed the state to {JobState}.	Process Control
212.000.00193	D4 00 C1	Message	Test {Test Number} of Job Id {Job Identifier} for system test changed the state to {Test State}.	Process Control
212.000.00196	D4 00 C4	Critical	Resource {Resource Id} is empty	Process Control
212.000.00255	D4 00 FF	Fatal	Inconsistency in managed data detected.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Table 21–2: Message list, Dispatcher

22 Cuvette Handler

Device ID: 0xD5

22.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 22–1: Sub assemblies, Cuvette Handler

22.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
213.000.00003	D5 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
213.000.00011	D5 00 0B	Fatal	Command terminated abnormally.	Subsequent
213.000.00014	D5 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
213.000.00015	D5 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
213.000.00032	D5 00 20	Message	occupation has changed for positions of incubation and/or washer ring	Process Control
213.000.00033	D5 00 21	Critical	Occupation status not yet determined	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

22. Cuvette Handler

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
213.000.00034	D5 00 22	Message	initial occupation status determined for cuvette positions whereas present cuvettes set to status 'Unknown'	Process Control
213.000.00048	D5 00 30	Fatal	Operation failed due to missing cuvette	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
213.000.00129	D5 00 81	Fatal	Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
213.000.00130	D5 00 82	Fatal	Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
213.000.00131	D5 00 83	Fatal	IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
213.000.00132	D5 00 84	Fatal	IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
213.000.00134	D5 00 86	Critical	Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
213.000.00135	D5 00 87	Fatal	Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
213.000.00144	D5 00 90	Fatal	Memory inconsistency in data structure	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
213.000.00145	D5 00 91	Fatal	No memory available for additional data	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
213.000.00225	D5 00 E1	Warning	warning received from managed assembly	Subsequent
213.000.00226	D5 00 E2	Critical	critical error received from managed assembly	Subsequent
213.000.00227	D5 00 E3	Fatal	fatal error received from managed assembly	Subsequent
213.000.00255	D5 00 FF	Fatal	Inconsistency in managed data detected.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Table 22–2: Message list, Cuvette Handler

23 Loading Manager

Device ID: 0xD6

23.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 23–1: Sub assemblies, Loading Manager

23.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
214.000.00003	D6 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
214.000.00011	D6 00 0B	Fatal	Command terminated abnormally.	Subsequent
214.000.00014	D6 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
214.000.00015	D6 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
214.000.00032	D6 00 20	Critical	Safety check for open instrument cover disabled while an operation is ongoing	Instrument Handling • Instrument is running, pay attention while operating with the front cover open
214.000.00040	D6 00 28	Message	{Changed At} {Changed To}.	Instrument Handling • Check consumable states • Initialize module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
214.000.00041	D6 00 29	Message	{Changed At} {Changed To}.	Instrument Handling • Check consumable states • Initialize module
214.000.00042	D6 00 2A	Fatal	Instrument cover open while an operation is in progress	Instrument Handling • Check the front cover sensor • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
214.000.00043	D6 00 2B	Message	led display mode changed for lane	Process Control
214.000.00048	D6 00 30	Message	Occupation changed due to insertion or removal of a rack	Instrument Handling • Check consumable states • Initialize module Process Control
214.000.00049	D6 00 31	Message	state for present resource set to off line	Process Control
214.000.00050	D6 00 32	Message	Barcode reading deactivated due to an external event	Process Control
214.000.00051	D6 00 33	Message	barcode scanner ready for reading	Instrument Handling • Check consumable states • Initialize module
214.000.00064	D6 00 40	Message	Updated status for containers alternating ranges	Process Control
214.000.00065	D6 00 41	Message	Updated status for containers alternating filling levels	Process Control
214.000.00066	D6 00 42	Fatal	Notification about overflow has occurred in solid waste	Instrument Handling • Check consumable states • Initialize module
214.000.00067	D6 00 43	Fatal	Solid waste has to be present while operation	Instrument Handling • Check consumable states • Initialize module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
214.000.00068	D6 00 44	Fatal	Liquid waste has to be present while operation	Instrument Handling • Check consumable states • Initialize module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
214.000.00128	D6 00 80	Fatal	Synchronous Messaging Interface in use	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
214.000.00129	D6 00 81	Fatal	Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
214.000.00130	D6 00 82	Fatal	Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
214.000.00131	D6 00 83	Fatal	IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
214.000.00132	D6 00 84	Fatal	IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
214.000.00134	D6 00 86	Critical	Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
214.000.00135	D6 00 87	Fatal	Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
214.000.00144	D6 00 90	Fatal	Memory inconsistency in data structure	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
214.000.00145	D6 00 91	Fatal	No memory available for additional data	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
214.000.00225	D6 00 E1	Warning	warning received from managed assembly	Subsequent
214.000.00226	D6 00 E2	Critical	critical error received from managed assembly	Subsequent
214.000.00227	D6 00 E3	Fatal	fatal error received from managed assembly	Subsequent
214.000.00255	D6 00 FF	Fatal	Inconsistency found in managed data structures.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Table 23–2: Message list, Loading Manager

24 DiTi Manager

Device ID: 0xD7

24.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 24–1: Sub assemblies, DiTi Manager

24.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
215.000.00003	D7 00 03	Fatal	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
215.000.00011	D7 00 0B	Fatal	Command terminated abnormally.	Subsequent
215.000.00014	D7 00 0E	Fatal	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
215.000.00015	D7 00 0F	Fatal	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
215.000.00128	D7 00 80	Fatal	Embedded Software Crash: Synchronous Messaging Interface in use	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
215.000.00129	D7 00 81	Fatal	Embedded software crash: Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
215.000.00131	D7 00 83	Fatal	Embedded Software Crash: IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
215.000.00132	D7 00 84	Fatal	Embedded Software Crash: IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
215.000.00041	D7 00 29	Message	Consumable presence changed	Process Control
215.000.00048	D7 00 30	Message	Consumable tray(s) unlocked for user access	Process Control
215.000.00133	D7 00 85	Fatal	Embedded Software Crash: Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
215.000.00134	D7 00 86	Critical	Embedded Software Crash: Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
215.000.00135	D7 00 87	Fatal	Embedded Software crashed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Table 24–2: Message list, DiTi Manager

25 System Monitor Driver

Device ID: 0xDF

25.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 25–1: Sub assemblies, System Monitor Driver

25.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
223.000.00001	DF 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
223.000.00002	DF 00 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
223.000.00003	DF 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
223.000.00011	DF 00 0B	Fatal	Command terminated abnormally.	Subsequent
223.000.00014	DF 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
223.000.00015	DF 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
223.000.00032	DF 00 20	Message	Diagnostics for {CanBus} succeeded - CAN_L={LineCAN_L}, CAN_H={LineCAN_H}	Process Control
223.000.00033	DF 00 21	Critical	Shutdown of instrument for CAN bus diagnostic has failed	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
223.000.00034	DF 00 22	Critical	Possible short to ground detected for {CanBus}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
223.000.00035	DF 00 23	Critical	Possible short between lines detected for {CanBus}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
223.000.00036	DF 00 24	Critical	Possibly misconfigured termination detected for {CanBus}	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
223.000.00048	DF 00 30	Critical	Circuit {CircuitId} voltage out of range ({Value})	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
223.000.00049	DF 00 31	Warning	Circuit {CircuitId} voltage back in range ({Value})	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
223.000.00128	DF 00 80	Fatal	Synchronous Messaging Interface in use	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
223.000.00129	DF 00 81	Fatal	Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
223.000.00130	DF 00 82	Fatal	Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
223.000.00131	DF 00 83	Fatal	IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
223.000.00132	DF 00 84	Fatal	IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
223.000.00134	DF 00 86	Critical	Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

25. System Monitor Driver

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
223.000.00135	DF 00 87	Fatal	Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Table 25–2: Message list, System Monitor Driver

26 Driver Framework Support Driver

Device ID: **0xE0**

26.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 26–1: Sub assemblies, Driver Framework Support Driver

26.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
224.000.00003	E0 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
224.000.00014	E0 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
224.000.00015	E0 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
224.000.00032	E0 00 20	Message	Unknown Device Address	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
224.000.00203	E0 00 CB	Message	COP/IPC/CAN Frame	Process Control
224.000.00219	E0 00 DB	Message	COP Debug Monitor Line	Process Control

Table 26–2: Message list, Driver Framework Support Driver

27 User Access Driver

Device ID: 0xE1

27.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 27–1: Sub assemblies, User Access Driver

27.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
225.000.00001	E1 00 01	Fatal	Send Init command.	
225.000.00002	E1 00 02	Fatal	Check command ID and command length.	
225.000.00011	E1 00 0B	Fatal	Command terminated abnormally.	
225.000.00013	E1 00 0D	Critical	Check syntax, sequence and values.	
225.000.00014	E1 00 0E	Critical	Command not allowed in this state. Perhaps another command is in process.	
225.000.00015	E1 00 0F	Critical	Command ID not implemented or wrong command length.	
225.000.00113	E1 00 71	Fatal	Work list load error.	
225.000.00128	E1 00 80	Fatal	Synchronous Messaging Interface in use.	
225.000.00129	E1 00 81	Fatal	Device is under exclusive driver access.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
225.000.00130	E1 00 82	Fatal	Lost event due to ring buffer overflow.	
225.000.00131	E1 00 83	Fatal	IPC result error.	
225.000.00132	E1 00 84	Fatal	IPC response length error.	
225.000.00134	E1 00 86	Critical	Received unknown message.	
225.000.00135	E1 00 87	Fatal	Process in unknown state.	
225.000.00225	E1 00 E1	Warning	Warning received from managed assembly.	
225.000.00226	E1 00 E2	Critical	Critical error received from managed assembly.	
225.000.00227	E1 00 E3	Fatal	Fatal error received from managed assembly.	
225.000.00096	E1 00 60	Message	Soft power button was pressed.	
225.000.00160	E1 00 A0	Message	Received self diagnostic event.	

Table 27–2: Message list, User Access Driver

28 Service Driver

Device ID: 0xE2

28.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 28–1: Sub assemblies, Service Driver

28.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
226.000.00001	E2 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
226.000.00002	E2 00 02	Fatal	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
226.000.00011	E2 00 0B	Fatal	Command terminated abnormally.	Subsequent
226.000.00013	E2 00 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
226.000.00014	E2 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
226.000.00015	E2 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
226.000.00128	E2 00 80	Fatal	Embedded Software Crash: Synchronous Messaging Interface in use	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
226.000.00129	E2 00 81	Fatal	Embedded software crash: Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
226.000.00130	E2 00 82	Fatal	Embedded Software Crash: Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
226.000.00131	E2 00 83	Fatal	Embedded Software Crash: IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
226.000.00132	E2 00 84	Fatal	Embedded Software Crash: IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
226.000.00134	E2 00 86	Critical	Embedded Software Crash: Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
226.000.00135	E2 00 87	Fatal	Embedded Software crashed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
226.000.00225	E2 00 E1	Warning	Embedded Software Crash: Exception received from managed assembly	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
226.000.00032	E2 00 20	Critical	Insertion of cuvette to position {Position} failed. Cuvette stat is {CuvState}	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
226.000.00033	E2 00 21	Critical	Prime of {Module} failed.	Instrument Handling • Check consumable states • Initialize module Process Control
226.000.00160	E2 00 A0	Message	Received self diagnostic event.	Process Control

Table 28–2: Message list, Service Driver

29 Incubator Driver

Device ID: 0xE3

29.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 29–1: Sub assemblies, Incubator Driver

29.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
227.000.00001	E3 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
227.000.00002	E3 00 02	Fatal	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
227.000.00003	E3 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
227.000.00011	E3 00 0B	Fatal	Command terminated abnormally.	Subsequent
227.000.00013	E3 00 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
227.000.00014	E3 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
227.000.00015	E3 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
227.000.00032	E3 00 20	Critical	Incubator (reaction) temperature is out of range.	Environmental • Check module environment (temperature, humidity) • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
227.000.00033	E3 00 21	Warning	Incubator (reaction) temperature is back in range	Environmental • Check module environment (temperature, humidity) • Exchange module
227.000.00034	E3 00 22	Fatal	Measured temperature fails plausibility check.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
227.000.00035	E3 00 23	Fatal	Ambient temperature fails plausibility check.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
227.000.00037	E3 00 25	Fatal	Incubator (reaction) temperature is out of range for too long.	Environmental • Check module environment (temperature, humidity) • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
227.000.00036	E3 00 24	Message	Temperature message.	Process Control
227.000.00038	E3 00 26	Fatal	Temperature sensor mismatch.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
227.000.00042	E3 00 2A	Fatal	Cuvette sensor failed during init, motor homing ok.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
227.000.00128	E3 00 80	Fatal	Embedded Software Crash: Synchronous Messaging Interface in use	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
227.000.00129	E3 00 81	Fatal	Embedded software crash: Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
227.000.00130	E3 00 82	Fatal	Embedded Software Crash: Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
227.000.00131	E3 00 83	Fatal	Embedded Software Crash: IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
227.000.00132	E3 00 84	Fatal	Embedded Software Crash: IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
227.000.00134	E3 00 86	Critical	Embedded Software Crash: Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
227.000.00135	E3 00 87	Fatal	Embedded Software crashed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

29. Incubator Driver

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
227.000.00048	E3 00 30	Critical	Embedded Software Crash: Worklist load error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
227.000.00176	E3 00 B0	Message	Received self diagnostic event.	Process Control

Table 29–2: Message list, Incubator Driver

30 Washer Driver

Device ID: 0xE4

30.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General
1	0x01	Probe1
2	0x02	Probe2
3	0x03	Probe3

Table 30–1: Sub assemblies, Washer Driver

30.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
228.000.00003	E4 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.001.00003	E4 01 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.002.00003	E4 02 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.003.00003	E4 03 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.000.00011	E4 00 0B	Fatal	Command terminated abnormally.	Subsequent

30. Washer Driver

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
228.001.00011	E4 01 0B	Fatal	Command terminated abnormally.	Subsequent
228.002.00011	E4 02 0B	Fatal	Command terminated abnormally.	Subsequent
228.003.00011	E4 03 0B	Fatal	Command terminated abnormally.	Subsequent
228.000.00014	E4 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.001.00014	E4 01 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.002.00014	E4 02 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.003.00014	E4 03 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.000.00015	E4 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.001.00015	E4 01 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
228.002.00015	E4 02 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.003.00015	E4 03 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.000.00129	E4 00 81	Fatal	Embedded software crash: Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.000.00130	E4 00 82	Fatal	Embedded Software Crash: Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.000.00131	E4 00 83	Fatal	Embedded Software Crash: IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.000.00132	E4 00 84	Fatal	Embedded Software Crash: IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.000.00134	E4 00 86	Critical	Embedded Software Crash: Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.000.00135	E4 00 87	Fatal	Embedded Software crashed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
228.001.00144	E4 01 90	Critical	Aspiration failed single event.	
228.002.00144	E4 02 90	Critical	Aspiration failed single event.	
228.003.00144	E4 03 90	Critical	Aspiration failed single event.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
228.001.00145	E4 01 91	Fatal	Aspiration failed for 3 consecutive measurements at washer port 1	
228.002.00145	E4 02 91	Fatal	Aspiration failed for 3 consecutive measurements at washer port 2	
228.003.00145	E4 03 91	Fatal	Aspiration failed for 3 consecutive measurements at washer port 3	
228.001.00146	E4 01 92	Message	Aspiration result post as a series of consecutive auto messages for Job Id {Job Id}	
228.002.00146	E4 02 92	Message	Aspiration result post as a series of consecutive auto messages for Job Id {Job Id}	
228.003.00146	E4 03 92	Message	Aspiration result post as a series of consecutive auto messages for Job Id {Job Id}	
228.000.00160	E4 00 A0	Message	Received self diagnostic event.	Process Control

Table 30–2: Message list, Washer Driver

31 Reader Driver

Device ID: 0xE5

31.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General
1	0x01	Bubble Sensor

Table 31–1: Sub assemblies, Reader Driver

31.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
229.000.00003	E5 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
229.001.00003	E5 01 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
229.000.00011	E5 00 0B	Fatal	Command terminated abnormally.	
229.001.00011	E5 01 0B	Fatal	Command terminated abnormally.	
229.000.00012	E5 00 0C	Fatal		
229.000.00013	E5 00 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
229.001.00013	E5 01 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
229.000.00014	E5 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
229.001.00014	E5 01 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
229.000.00015	E5 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
229.001.00015	E5 01 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
229.000.00048	E5 00 30	Critical	Work list load error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
229.000.00080	E5 00 50	Message	Measure results	Process Control
229.000.00081	E5 00 51	Fatal	Process not possible at this cuvette state	Process Control • Check consumable states • Initialize module Instrument Handling
229.000.00083	E5 00 53	Message	Measure results	Process Control
229.001.00084	E5 01 54	Message	Aspiration result post as a series of consecutive auto messages for Job Id {Identifier}	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
229.000.00085	E5 00 55	Fatal	Dark count events occurred	Process Control • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Electronics • Clean module • Check calibration • Check mechanic • Exchange module Mechanics
229.000.00128	E5 00 80	Fatal	Embedded Software Crash: Synchronous Messaging Interface in use	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
229.000.00131	E5 00 83	Fatal	Embedded Software Crash: IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
229.000.00132	E5 00 84	Fatal	Embedded Software Crash: IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
229.000.00133	E5 00 85	Fatal	Embedded Software Crash: Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
229.000.00134	E5 00 86	Critical	Embedded Software Crash: Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
229.000.00135	E5 00 87	Fatal	Embedded Software crashed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
229.001.00144	E5 01 90	Critical	Aspiration failed single event.	Process Control
229.001.00145	E5 01 91	Fatal	Aspiration failed for 3 consecutive measurements at reader port	Process Control
229.000.00160	E5 00 A0	Message	Received self diagnostic event.	Process Control

31. Reader Driver

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
229.000.00240	E5 00 F0	Fatal	Embedded software crash: Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Table 31–2: Message list, Reader Driver

32 Reagent Sample Bay Driver

Device ID: 0xE6

32.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General
1	0x01	Reagent
2	0x02	Sample

Table 32–1: Sub assemblies, Reagent Sample Bay Driver

32.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
230.000.00003	E6 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.001.00003	E6 01 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.002.00003	E6 02 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.000.00011	E6 00 0B	Fatal	Command terminated abnormally.	Subsequent
230.001.00011	E6 01 0B	Fatal	Command terminated abnormally.	Subsequent
230.002.00011	E6 02 0B	Fatal	Command terminated abnormally.	Subsequent

32. Reagent Sample Bay Driver

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
230.000.00014	E6 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.001.00014	E6 01 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.002.00014	E6 02 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.000.00015	E6 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.001.00015	E6 01 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.002.00015	E6 02 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.000.00032	E6 00 20	Critical	Work list load error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.000.00036	E6 00 24	Message	Temperature report.	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
230.000.00037	E6 00 25	Critical	Reagent bay temperature is out of range for too long time.	Environmental • Check module environment (temperature, humidity) • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.000.00048	E6 00 30	Message	Update notification of volume and fill height of specified position(s)	Process Control
230.001.00052	E6 01 34	Critical	Stirrer rotation frequency too low	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
230.000.00068	E6 00 44	Message	Reagent flap recently opened or closed	Process Control
230.000.00069	E6 00 45	Message	Sample flap recently opened or closed	Process Control
230.002.00080	E6 02 50	Message	Sample scanner activated	Process Control
230.002.00081	E6 02 51	Message	Sample scanner deactivated	Process Control
230.002.00082	E6 02 52	Warning	Scanner active for too long time and hence is switched off	Process Control
230.002.00102	E6 02 66	Message	Scanner polygon wheel is currently switched off.	Process Control
230.000.00128	E6 00 80	Fatal	Embedded Software Crash: Synchronous Messaging Interface in use	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.000.00129	E6 00 81	Fatal	Embedded software crash: Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

32. Reagent Sample Bay Driver

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
230.000.00130	E6 00 82	Fatal	Embedded Software Crash: Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.000.00131	E6 00 83	Fatal	Embedded Software Crash: IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.000.00132	E6 00 84	Fatal	Embedded Software Crash: IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.000.00134	E6 00 86	Critical	Embedded Software Crash: Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.000.00135	E6 00 87	Fatal	Embedded Software crashed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
230.000.00160	E6 00 A0	Message	Received self diagnostic event.	Process Control

Table 32–2: Message list, Reagent Sample Bay Driver

33 Liquid Support Driver

Device ID: 0xE8

33.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General
1	0x01	BubbleSensor1
2	0x02	BubbleSensor2
3	0x03	HeatTube1
4	0x04	HeatTube2
5	0x05	RFIDStarter

Table 33–1: Sub assemblies, Liquid Support Driver

33.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
232.000.00001	E8 00 01	Fatal	Send Init command.	
232.001.00001	E8 01 01	Fatal	Send Init command.	
232.002.00001	E8 02 01	Fatal	Send Init command.	
232.003.00001	E8 03 01	Fatal	Send Init command.	
232.004.00001	E8 04 01	Fatal	Send Init command.	
232.005.00001	E8 05 01	Fatal	Send Init command.	
232.000.00002	E8 00 02	Fatal	Check command ID and command length.	
232.001.00002	E8 01 02	Fatal	Check command ID and command length.	
232.002.00002	E8 02 02	Fatal	Check command ID and command length.	
232.003.00002	E8 03 02	Fatal	Check command ID and command length.	

33. Liquid Support Driver

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
232.004.00002	E8 04 02	Fatal	Check command ID and command length.	
232.005.00002	E8 05 02	Fatal	Check command ID and command length.	
232.000.00011	E8 00 0B	Fatal	Command terminated abnormally.	
232.001.00011	E8 01 0B	Fatal	Command terminated abnormally.	
232.002.00011	E8 02 0B	Fatal	Command terminated abnormally.	
232.003.00011	E8 03 0B	Fatal	Command terminated abnormally.	
232.004.00011	E8 04 0B	Fatal	Command terminated abnormally.	
232.005.00011	E8 05 0B	Fatal	Command terminated abnormally.	
232.000.00013	E8 00 0D	Critical	Check syntax, sequence and values.	
232.001.00013	E8 01 0D	Critical	Check syntax, sequence and values.	
232.002.00013	E8 02 0D	Critical	Check syntax, sequence and values.	
232.003.00013	E8 03 0D	Critical	Check syntax, sequence and values.	
232.004.00013	E8 04 0D	Critical	Check syntax, sequence and values.	
232.005.00013	E8 05 0D	Critical	Check syntax, sequence and values.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
232.000.00014	E8 00 0E	Critical	Command not allowed in this state. Perhaps another command is in process.	
232.001.00014	E8 01 0E	Critical	Command not allowed in this state. Perhaps another command is in process.	
232.002.00014	E8 02 0E	Critical	Command not allowed in this state. Perhaps another command is in process.	
232.003.00014	E8 03 0E	Critical	Command not allowed in this state. Perhaps another command is in process.	
232.004.00014	E8 04 0E	Critical	Command not allowed in this state. Perhaps another command is in process.	
232.005.00014	E8 05 0E	Critical	Command not allowed in this state. Perhaps another command is in process.	
232.000.00015	E8 00 0F	Critical	Command ID not implemented or wrong command length.	
232.001.00015	E8 01 0F	Critical	Command ID not implemented or wrong command length.	
232.002.00015	E8 02 0F	Critical	Command ID not implemented or wrong command length.	
232.003.00015	E8 03 0F	Critical	Command ID not implemented or wrong command length.	

33. Liquid Support Driver

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
232.004.00015	E8 04 0F	Critical	Command ID not implemented or wrong command length.	
232.005.00015	E8 05 0F	Critical	Command ID not implemented or wrong command length.	
232.000.00128	E8 00 80	Fatal	Synchronous Messaging Interface in use	
232.000.00129	E8 00 81	Fatal	Device is under exclusive driver access	
232.000.00130	E8 00 82	Fatal	Lost event due to ring buffer overflow	
232.000.00131	E8 00 83	Fatal	IPC Result Error	
232.000.00132	E8 00 84	Fatal	IPC Response Length Error	
232.000.00134	E8 00 86	Critical	Received unknown message.	
232.000.00135	E8 00 87	Fatal	Received unknown message.	
232.000.00033	E8 00 21	Critical	Worklist load error	
232.000.00032	E8 00 20	Message	Received unknown message.	
232.000.00048	E8 00 30	Message	Received unknown message.	
232.000.00064	E8 00 40	Message	Received unknown message.	
232.001.00084	E8 01 54	Message	Measure results	
232.002.00084	E8 02 54	Message	Measure results	
232.000.00082	E8 00 52	Message	Bubble measure finished returns measured values.	
232.001.00082	E8 01 52	Message	Bubble measure finished returns measured values.	
232.002.00082	E8 02 52	Message	Bubble measure finished returns measured values.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
232.003.00082	E8 03 52	Message	Bubble measure finished returns measured values.	
232.004.00082	E8 04 52	Message	Bubble measure finished returns measured values.	
232.005.00082	E8 05 52	Message	Bubble measure finished returns measured values.	
232.003.00160	E8 03 A0	Warning	Define valid temperature control parameters.	
232.004.00160	E8 04 A0	Warning	Define valid temperature control parameters.	
232.003.00161	E8 03 A1	Warning	Starter 1 (injection) temperature is out of range	
232.004.00161	E8 04 A1	Warning	Starter 2 (injection) temperature is out of range	
232.003.00162	E8 03 A2	Warning	Starter 1 (injection) temperature is back in range	
232.004.00162	E8 04 A2	Warning	Starter 2 (injection) temperature is back in range	
232.003.00163	E8 03 A3	Critical	Measured temperature value failed plausibility check	
232.004.00163	E8 04 A3	Critical	Measured temperature value failed plausibility check	
232.003.00164	E8 03 A4	Critical	The heater drive IC reported a failure	
232.004.00164	E8 04 A4	Critical	The heater drive IC reported a failure	
232.003.00165	E8 03 A5	Critical	Temperature watchdog switched off	

33. Liquid Support Driver

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
232.004.00165	E8 04 A5	Critical	Temperature watchdog switched off	
232.003.00166	E8 03 A6	Message	Cyclical temperature report for selected heater	
232.004.00166	E8 04 A6	Message	Cyclical temperature report for selected heater	
232.003.00167	E8 03 A7	Critical	Target temperature not reached with in the defined time	
232.004.00167	E8 04 A7	Critical	Target temperature not reached with in the defined time	
232.003.00168	E8 03 A8	Message	temperature at the start of self test and at the end of self test	
232.004.00168	E8 04 A8	Message	temperature at the start of self test and at the end of self test	
232.003.00169	E8 03 A9	Message	Send actual temperature from heater to pc software.	
232.004.00169	E8 04 A9	Message	Send actual temperature from heater to pc software.	
232.005.00128	E8 05 80	Warning	Read access to VICC disturbed, retry needed.	
232.005.00129	E8 05 81	Critical	Read access to VICC failed.	
232.005.00130	E8 05 82	Warning	Write access to VICC disturbed, retry needed.	
232.005.00131	E8 05 83	Critical	Write access to VICC failed.	
232.005.00132	E8 05 84	Warning	Lock access to VICC disturbed, retry needed.	
232.005.00133	E8 05 85	Critical	Lock access to VICC failed.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
232.005.00134	E8 05 86	Warning	Access to VICC disturbed, retry needed.	
232.005.00135	E8 05 87	Critical	Access to VICC failed.	
232.005.00143	E8 05 8F	Warning	Reader lib error message.	
232.005.00113	E8 05 71	Critical	Starter bottle has no more shots.	
232.000.00114	E8 00 72	Message	Update notification of volume and fill height of specified position(s)	
232.005.00112	E8 05 70	Critical	Antennalin to reader failed.	
232.000.00176	E8 00 B0	Message	Received self diagnostic event.	

Table 33–2: Message list, Liquid Support Driver

34 Front Cover Driver

Device ID: 0xE9

34.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General

Table 34–1: Sub assemblies, Front Cover Driver

34.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
233.000.00001	E9 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
233.000.00002	E9 00 02	Fatal	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
233.000.00011	E9 00 0B	Fatal	Command terminated abnormally.	Subsequent
233.000.00013	E9 00 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
233.000.00014	E9 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

34. Front Cover Driver

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
233.000.00015	E9 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
233.000.00128	E9 00 80	Fatal	Embedded Software Crash: Synchronous Messaging Interface in use	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
233.000.00129	E9 00 81	Fatal	Embedded software crash: Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
233.000.00130	E9 00 82	Fatal	Embedded Software Crash: Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
233.000.00131	E9 00 83	Fatal	Embedded Software Crash: IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
233.000.00132	E9 00 84	Fatal	Embedded Software Crash: IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
233.000.00134	E9 00 86	Critical	Embedded Software Crash: Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
233.000.00135	E9 00 87	Fatal	Embedded Software crashed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
233.000.00048	E9 00 30	Message	Front Cover is {Flap Drawer State}	Process Control
233.000.00049	E9 00 31	Message	Consumable drawer is {Flap Drawer State}.	Process Control
233.000.00050	E9 00 32	Message	Starter reagent flap is {Flap Drawer State}.	Process Control
233.000.00051	E9 00 33	Message	Reagent bay flap is {Flap Drawer State}.	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
233.000.00052	E9 00 34	Message	Sample bay flap is {Flap Drawer State}.	Process Control
233.000.00053	E9 00 35	Message	Solid waste drawer is {Flap Drawer State}.	Process Control
233.000.00054	E9 00 36	Critical	Reagent bay flap is open for a too long time.	Process Control
233.000.00160	E9 00 A0	Message	Received self diagnostic event.	Process Control

Table 34–2: Message list, Front Cover Driver

35 Com Gen Driver

Device ID: 0xEF

35.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	General
1	0x01	Left Arm
2	0x02	Right Arm

Table 35–1: Sub assemblies, Com Gen Driver

35.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
239.000.00001	EF 00 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
239.001.00001	EF 01 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
239.002.00001	EF 02 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
239.000.00002	EF 00 02	Fatal	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
239.001.00002	EF 01 02	Fatal	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.002.00002	EF 02 02	Fatal	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.000.00011	EF 00 0B	Fatal	Command terminated abnormally.	Subsequent
239.001.00011	EF 01 0B	Fatal	Command terminated abnormally.	Subsequent
239.002.00011	EF 02 0B	Fatal	Command terminated abnormally.	Subsequent
239.000.00013	EF 00 0D	Fatal	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.001.00013	EF 01 0D	Fatal	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.002.00013	EF 02 0D	Fatal	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.000.00014	EF 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
239.001.00014	EF 01 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.002.00014	EF 02 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.001.00028	EF 01 1C	Message	Pressure trace data report	
239.002.00029	EF 02 1D	Message	OLV trace data report	
239.001.00030	EF 01 1E	Message	BLLD trace data report	
239.002.00031	EF 02 1F	Message	CLLD trace data report	
239.002.00032	EF 02 20	Message	Pressure trace data report during wash	
239.002.00033	EF 02 21	Message	Start for new OLV trace	
239.002.00034	EF 02 22	Message	Pressure trace analyse report during wash.	
239.002.00035	EF 02 23	Message	Report of single OLV dead volume verification step.	
239.000.00026	EF 00 1A	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
239.001.00026	EF 01 1A	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.002.00026	EF 02 1A	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.000.00015	EF 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.001.00015	EF 01 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.002.00015	EF 02 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.000.00022	EF 00 16	Fatal	Module cannot execute command as number of received command parameter is invalid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.001.00022	EF 01 16	Fatal	Module cannot execute command as number of received command parameter is invalid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.002.00022	EF 02 16	Fatal	Module cannot execute command as number of received command parameter is invalid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.002.00027	EF 02 1B	Message	OLV verification result	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
239.001.00147	EF 01 93	Fatal	Tip eject failed	Instrument Handling • Check consumable states • Initialize module
239.001.00144	EF 01 90	Critical	No DiTi available	
239.001.00152	EF 01 98	Critical	DiTi pickup failed	
239.001.00154	EF 01 9A	Fatal	Cuvette sensor unexpected ON	Process Control
239.001.00155	EF 01 9B	Critical	Gripper detected no cuvette	Instrument Handling • Check consumable states • Initialize module
239.001.00156	EF 01 9C	Critical	No new cuvette available	Instrument Handling • Check consumable states • Initialize module
239.001.00157	EF 01 9D	Critical	New cuvette pickup failed	Instrument Handling • Check consumable states • Initialize module
239.001.00145	EF 01 91	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.002.00145	EF 02 91	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.000.00239	EF 00 EF	Fatal	Nak received from submodule	
239.001.00239	EF 01 EF	Fatal	Nak received from submodule	
239.002.00239	EF 02 EF	Fatal	Nak received from submodule	
239.002.00149	EF 02 95	Fatal	OLV sensor calibration failed	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
239.002.00150	EF 02 96	Fatal	OLV dead volume verification failed	Initialize the system and perform the resume; If the problem persists call Service.
239.002.00151	EF 02 97	Warning	OLV dead volume verification succeeded with retries.	
239.001.00158	EF 01 9E	Critical	Pump move canceled as excessive pressure was detected	
239.002.00158	EF 02 9E	Critical	Pump move canceled as excessive pressure was detected	Process Control
239.001.00159	EF 01 9F	Critical	Clot detected	
239.001.00160	EF 01 A0	Critical	Command skipped, due {Skip Reason}	
239.002.00160	EF 02 A0	Critical	Command skipped, due {Skip Reason}	Process Control
				• Check consumable states • Initialize module
				Instrument Handling
239.001.00161	EF 01 A1	Critical	Not enough liquid detected	Instrument Handling
239.001.00166	EF 01 A6	Critical	No liquid detected	• Check consumable states • Initialize module
				Instrument Handling
239.001.00162	EF 01 A2	Critical	LLD signal failure detected	Process Control
239.000.00163	EF 00 A3	Critical	ZMove is denied because resource is not available	Instrument Handling
239.001.00163	EF 01 A3	Critical	ZMove is denied because resource is not available	• Check consumable states • Initialize module
				Instrument Handling
239.002.00163	EF 02 A3	Critical	ZMove is denied because resource is not available	Instrument Handling
				• Check consumable states • Initialize module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
239.001.00164	EF 01 A4	Critical	Detected pressure values are out of defined range	Process Control
239.002.00165	EF 02 A5	Critical	OLV sensor detected insufficient volume	Process Control
239.001.00174	EF 01 AE	Critical	DiTi pickup failed multiple times	Process Control
239.001.00175	EF 01 AF	Critical	Pickup of new cuvette failed multiple times	Process Control
239.000.00176	EF 00 B0	Message	Received self diagnostic event.	Process Control
239.000.00112	EF 00 70	Fatal	Memory inconsistency in data structure	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.001.00112	EF 01 70	Fatal	Memory inconsistency in data structure	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.002.00112	EF 02 70	Fatal	Memory inconsistency in data structure	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.000.00113	EF 00 71	Fatal	No memory available for additional data	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.001.00113	EF 01 71	Fatal	No memory available for additional data	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.002.00113	EF 02 71	Fatal	No memory available for additional data	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.000.00128	EF 00 80	Fatal	Embedded Software Crash: Synchronous Messaging Interface in use	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
239.000.00129	EF 00 81	Fatal	Embedded software crash: Device is under exclusive driver access	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.000.00130	EF 00 82	Fatal	Embedded Software Crash: Lost event due to ring buffer overflow	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.000.00131	EF 00 83	Fatal	Embedded Software Crash: IPC Result Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.000.00132	EF 00 84	Fatal	Embedded Software Crash: IPC Response Length Error	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.000.00134	EF 00 86	Critical	Embedded Software Crash: Received unknown message.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
239.000.00135	EF 00 87	Fatal	Embedded Software crashed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Table 35–2: Message list, Com Gen Driver

36 Air Pipettor Left Arm

Device ID: 0xFD

36.1 Sub assemblies

ID (dec)	ID (hex)	Name
14	0x0E	General
15	0x0F	SctICU
16	0x10	APHeadCU

Table 36–1: Sub assemblies, Air Pipettor Left Arm

36.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.014.00001	FD 0E 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.015.00001	FD 0F 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.016.00001	FD 10 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.014.00002	FD 0E 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

36. Air Pipettor Left Arm

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.015.00002	FD 0F 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.016.00002	FD 10 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.014.00003	FD 0E 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.014.00004	FD 0E 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.014.00005	FD 0E 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.014.00006	FD 0E 06	Critical	Supply voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.014.00007	FD 0E 07	Warning	Supply voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.014.00010	FD 0E 0A	Fatal	Syringe motor overload	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.015.00010	FD 0F 0A	Fatal	Syringe motor overload	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.014.00011	FD 0E 0B	Fatal	Command terminated abnormally.	Subsequent
253.015.00011	FD 0F 0B	Fatal	Command terminated abnormally.	Subsequent
253.014.00013	FD 0E 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.015.00013	FD 0F 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.016.00013	FD 10 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.014.00014	FD 0E 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.015.00014	FD 0F 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.016.00014	FD 10 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.014.00016	FD 0E 10	Fatal	Execution time of command exceeds limits or connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.015.00016	FD 0F 10	Fatal	Execution time of command exceeds limits or connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.016.00016	FD 10 10	Fatal	Execution time of command exceeds limits or connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.014.00017	FD 0E 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.014.00018	FD 0E 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.014.00019	FD 0E 13	Critical	The received parameter number needs an authorization which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
253.014.00020	FD 0E 14	Critical	Received parameter authorization code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
253.014.00021	FD 0E 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.014.00022	FD 0E 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.014.00031	FD 0E 1F	Warning	Communication errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.014.00032	FD 0E 20	Message	Trigger was set by liquid level detection (LLD).	Process Control
253.014.00033	FD 0E 21	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.014.00034	FD 0E 22	Fatal	SCTL timeout: interrupt not received within expected time or internal communication error occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.014.00035	FD 0E 23	Fatal	Initialization of syringe failed	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.014.00036	FD 0E 24	Critical	Valve 1/2 switching Test during init command failed.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.014.00037	FD 0E 25	Warning	Syringe motor overload	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.014.00038	FD 0E 26	Warning	Pressure sensor noise bad at self-test during init command.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.014.00039	FD 0E 27	Warning	Init self-test: Pressure sensor ambient level out of range.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.014.00040	FD 0E 28	Warning	No airflow on init move detected, system might be leaky or clogged.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.014.00041	FD 0E 29	Critical	Self-Test during init: Pressure sensor signal not valid.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

36. Air Pipettor Left Arm

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.014.00048	FD 0E 30	Fatal	Target position not reached by syringe	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.014.00049	FD 0E 31	Fatal	SCTL Watchdog reset, internal communication error.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.014.00050	FD 0E 32	Warning	Init-Test: Frequency of C-LLD level oscillator out of range.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.014.00051	FD 0E 33	Critical	No or insufficient air loaded for a b-LLD.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.014.00052	FD 0E 34	Fatal	Initialization of syringe failed	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.014.00053	FD 0E 35	Fatal	Syringe overpressure	Calibration • Calibrate module Consumable • Exchange consumable • Exchange module
253.014.00054	FD 0E 36	Fatal	Parameter iVP9101PosConv not set. Module needs suitable parameter first.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.014.00055	FD 0E 37	Message	State of tip sensor has changed.	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.014.00056	FD 0E 38	Warning	Verification of Init position needed retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.014.00064	FD 0E 40	Critical	Communication error ({APHErrorEvent}) with APH μ C via internal RS232 interface.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.014.00065	FD 0E 41	Warning	Communication Problem ({APHWarningEvent}) with APH μ C via internal RS232 interface.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.014.00066	FD 0E 42	Warning	No pressure built-up detected at start of bLLD.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.014.00067	FD 0E 43	Warning	Self-test error occurred ({DevProtErrorID}d), retry needed for complete init: {NrOfRetry} retries	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

36. Air Pipettor Left Arm

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.014.00068	FD 0E 44	Critical	Calibration result = 0, calibration criterias not met	Calibration • Calibrate module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.014.00070	FD 0E 46	Warning	Verification result after BLLD-trigger.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Consumable • Exchange consumable • Exchange module Process Control
253.014.00096	FD 0E 60	Fatal	Reset of μ C has occurred, internal Software crash or external reset invoked: {ResetExceptions} at {ResetAddress}.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.014.00160	FD 0E A0	Message	Self-diagnostic: min and max values of the 24V supply voltage.	
253.014.00161	FD 0E A1	Message	Self-diagnostic: min and max values of the 7V supply voltage.	
253.014.00162	FD 0E A2	Message	Self-diagnostic: min and max values of the 3.6V supply voltage.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.014.00163	FD 0E A3	Message	Self-diagnostic: PWM Average at full range move.	
253.014.00164	FD 0E A4	Message	Self-diagnostic: percentiles of position deviation at full range moves.	
253.014.00176	FD 0E B0	Message	Self-diagnostic: deviation at full range dispense move.	
253.014.00177	FD 0E B1	Message	Self-diagnostic Device information: serial, article number and Version.	
253.014.00165	FD 0E A5	Message	Self-diagnostic: move and valve switching cycle counter.	
253.014.00168	FD 0E A8	Message	Self-diagnostic: result of pressure sensor test with open valve.	
253.014.00169	FD 0E A9	Message	Self-diagnostic: pressure trace during dispense move.	
253.014.00170	FD 0E AA	Message	Self-diagnostic: pressure trace during aspirate move.	
253.014.00171	FD 0E AB	Message	Self-diagnostic: pressure trace during barometric LLD move.	
253.014.00172	FD 0E AC	Message	Self-diagnostic: Valve test results.	
253.014.00173	FD 0E AD	Message	Self-diagnostic: deviation at retries on move to block.	
253.014.00174	FD 0E AE	Message	Self-diagnostic: Final position errors.	
253.014.00175	FD 0E AF	Message	Self-diagnostic: Tip state during self-diagnostic command.	

36. Air Pipettor Left Arm

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.014.00192	FD 0E C0	Warning	Error during ServiceMemory access.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.014.00193	FD 0E C1	Message	ServiceMemory read data.	
253.014.00255	FD 0E FF	Fatal	Embedded software crash occurs.	
253.015.00255	FD 0F FF	Fatal	Embedded software crash occurs.	
253.016.00255	FD 10 FF	Fatal	Embedded software crash occurs.	

Table 36–2: Message list, Air Pipettor Left Arm

37 Dual Sledge Y

Device ID: 0xFD

37.1 Sub assemblies

ID (dec)	ID (hex)	Name
5	0x05	General
6	0x06	Y_Drive

Table 37–1: Sub assemblies, Dual Sledge Y

37.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.005.00001	FD 05 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.006.00001	FD 06 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.005.00002	FD 05 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.006.00002	FD 06 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

37. Dual Sledge Y

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.005.00003	FD 05 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.005.00004	FD 05 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.005.00005	FD 05 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.005.00006	FD 05 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.005.00007	FD 05 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.006.00009	FD 06 09	Warning	Target position reached after retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.006.00010	FD 06 0A	Fatal	Target position not reached by X motor	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.006.00011	FD 06 0B	Fatal	Command terminated abnormally.	Subsequent
253.005.00013	FD 05 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.006.00013	FD 06 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.005.00014	FD 05 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.006.00014	FD 06 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.005.00015	FD 05 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.006.00015	FD 06 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.005.00016	FD 05 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.006.00016	FD 06 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.005.00017	FD 05 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.005.00018	FD 05 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.005.00019	FD 05 13	Critical	The received parameter number needs an authorisation which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.005.00020	FD 05 14	Critical	Received parameter authorisation code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
253.005.00021	FD 05 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.005.00022	FD 05 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.005.00023	FD 05 17	Fatal	A movement problem has been detected	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.006.00023	FD 06 17	Fatal	A movement problem has been detected	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.005.00031	FD 05 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.005.00032	FD 05 20	Fatal	Embedded software crash occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.006.00032	FD 06 20	Fatal	Embedded software crash occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.006.00050	FD 06 32	Fatal	Initialization of Y motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.006.00051	FD 06 33	Fatal	Initialization of Y motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.006.00052	FD 06 34	Fatal	Unexpected change of internal position.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.006.00053	FD 06 35	Fatal	Unexpected change of target position.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.006.00080	FD 06 50	Message	Entered function MODE3_SetElectricalPosition.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.006.00081	FD 06 51	Message	Left function MODE3_SetElectricalPosition.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.005.00160	FD 05 A0	Message	Minimum/maximum supply voltage detected.	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.005.00161	FD 05 A1	Message	Number of moves executed.	Process Control
253.005.00162	FD 05 A2	Message	Minimum motor current evaluated during module self diagnostics.	Process Control
253.005.00163	FD 05 A3	Message	Mechanical clearance of the drive evaluated during module self diagnostics.	Process Control
253.005.00191	FD 05 BF	Message	An unexpected error occurred during module self diagnostics.	Process Control
253.005.00209	FD 05 D1	Message	Self-diagnostic Device information: serial, article number and Version.	Process Control

Table 37–2: Message list, Dual Sledge Y

38 Dual Sledge Z

Device ID: 0xFD

38.1 Sub assemblies

ID (dec)	ID (hex)	Name
9	0x09	General
10	0x0A	ZDrive1 (Sample pipettor)
11	0x0B	ZDrive2 (Gripper)

Table 38–1: Sub assemblies, Dual Sledge Z

38.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.009.00001	FD 09 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.010.00001	FD 0A 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.011.00001	FD 0B 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.009.00002	FD 09 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.010.00002	FD 0A 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.011.00002	FD 0B 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.009.00003	FD 09 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.009.00004	FD 09 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.009.00005	FD 09 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.009.00011	FD 09 0B	Fatal	Command terminated abnormally.	Subsequent
253.010.00011	FD 0A 0B	Fatal	Command terminated abnormally.	Subsequent
253.011.00011	FD 0B 0B	Fatal	Command terminated abnormally.	Subsequent
253.009.00013	FD 09 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.010.00013	FD 0A 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.011.00013	FD 0B 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.009.00014	FD 09 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.010.00014	FD 0A 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.011.00014	FD 0B 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.009.00015	FD 09 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.010.00015	FD 0A 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.011.00015	FD 0B 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.009.00016	FD 09 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.010.00016	FD 0A 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.011.00016	FD 0B 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.009.00017	FD 09 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.009.00018	FD 09 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.009.00019	FD 09 13	Critical	The received parameter number needs an authorisation which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.009.00020	FD 09 14	Critical	Received parameter authorisation code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
253.009.00021	FD 09 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.009.00022	FD 09 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.009.00031	FD 09 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.010.00033	FD 0A 21	Warning	The required disposable tip pick-up movement is started form below z-start.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.011.00033	FD 0B 21	Warning	The required disposable tip pick-up movement is started form below z-start.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.010.00034	FD 0A 22	Fatal	Servo controller communication error	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.011.00034	FD 0B 22	Fatal	Servo controller communication error	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.010.00037	FD 0A 25	Fatal	Z motor overload	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.011.00037	FD 0B 25	Fatal	Z motor overload	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.010.00038	FD 0A 26	Critical	The drive was stopped after exceeding a given force limitation.	ProcessControl
253.011.00038	FD 0B 26	Critical	The drive was stopped after exceeding a given force limitation.	ProcessControl
253.010.00048	FD 0A 30	Fatal	Target position not reached by Z motor	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.011.00048	FD 0B 30	Fatal	Target position not reached by Z motor	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.010.00050	FD 0A 32	Fatal	Initialization of Z motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.011.00050	FD 0B 32	Fatal	Initialization of Z motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.010.00051	FD 0A 33	Fatal	Initialization of Z motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

38. Dual Sledge Z

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.011.00051	FD 0B 33	Fatal	Initialization of Z motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.010.00052	FD 0A 34	Warning	Friction calibration is not possible as the current position is below z-start.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.011.00052	FD 0B 34	Warning	Friction calibration is not possible as the current position is below z-start.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.010.00053	FD 0A 35	Message	Control deviation reported by the servo controller	ProcessControl
253.011.00053	FD 0B 35	Message	Control deviation reported by the servo controller	ProcessControl
253.009.00160	FD 09 A0	Message	Report of executed moves.	ProcessControl
253.009.00161	FD 09 A1	Message	Minimum servo controller saturation threshold evaluated during module selfdiagnostics.	ProcessControl
253.009.00162	FD 09 A2	Message	Mechanical clearance of the drive evaluated during module selfdiagnostics.	ProcessControl
253.009.00191	FD 09 BF	Message	An unexpected error occurred during module selfdiagnostics.	ProcessControl
253.009.00083	FD 09 53	Fatal	Embedded software crash occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.010.00083	FD 0A 53	Fatal	Embedded software crash occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.011.00083	FD 0B 53	Fatal	Embedded software crash occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.009.00209	FD 09 D1	Message	Self-diagnostic Device information: serial, article number and Version.	

Table 38–2: Message list, Dual Sledge Z

39 Gripper

Device ID: 0xFD

39.1 Sub assemblies

ID (dec)	ID (hex)	Name
20	0x14	General

Table 39–1: Sub assemblies, Gripper

39.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.020.00001	FD 14 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.020.00002	FD 14 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.020.00003	FD 14 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.020.00004	FD 14 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.020.00005	FD 14 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.020.00006	FD 14 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.020.00007	FD 14 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.020.00008	FD 14 08	Fatal	Motor overload	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.020.00009	FD 14 09	Warning	Target position reached after retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.020.00010	FD 14 0A	Fatal	Target position not reached by gripper	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.020.00011	FD 14 0B	Fatal	Command terminated abnormally.	Subsequent

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.020.00012	FD 14 0C	Fatal	Hardware Error	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.020.00013	FD 14 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.020.00014	FD 14 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.020.00015	FD 14 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.020.00016	FD 14 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.020.00017	FD 14 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.020.00018	FD 14 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.020.00019	FD 14 13	Critical	The received parameter number needs an authorization which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
253.020.00020	FD 14 14	Critical	Received parameter authorization code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
253.020.00021	FD 14 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.020.00022	FD 14 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.020.00030	FD 14 1E	Fatal	Init failed.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.020.00031	FD 14 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.020.00032	FD 14 20	Critical	Load was not found after Hold command executed	Process Control
253.020.00033	FD 14 21	Critical	Load was still detected after Release command executed	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.020.00034	FD 14 22	Critical	Load was lost	Process Control
253.020.00162	FD 14 A2	Message	Self diagnostic sends min and max values of the supply voltage.	Process Control
253.020.00163	FD 14 A3	Message	Self diagnostic cycle counters.	Process Control
253.020.00165	FD 14 A5	Message	Self diagnostic min drive current.	Process Control
253.020.00167	FD 14 A7	Message	Self diagnostic mechanical calibration.	Process Control
253.020.00191	FD 14 BF	Message	An unexpected error occurred during module self diagnostics.	Process Control
253.020.00176	FD 14 B0	Warning	An internal error occurred	Calibration • Calibrate module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.020.00177	FD 14 B1	Message	Service memory data.	Process Control
253.020.00178	FD 14 B2	Critical	Initialization of service memory data failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.020.00179	FD 14 B3	Message	Initialization of service memory data succeeded.	Process Control
253.020.00180	FD 14 B4	Warning	An internal error occurred	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

39. Gripper

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.020.00181	FD 14 B5	Critical	Service memory access denied.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.020.00209	FD 14 D1	Message	Self-diagnostic Device information: serial, article number and Version.	Process Control

Table 39–2: Message list, Gripper

40 LWP

Device ID: 0xFD

40.1 Sub assemblies

ID (dec)	ID (hex)	Name
21	0x15	General

Table 40–1: Sub assemblies, LWP

40.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.021.00001	FD 15 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.021.00002	FD 15 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.021.00003	FD 15 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.021.00004	FD 15 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.021.00005	FD 15 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.021.00006	FD 15 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.021.00007	FD 15 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.021.00008	FD 15 08	Fatal	Motor overload	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.021.00009	FD 15 09	Warning	Target position reached after retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.021.00010	FD 15 0A	Fatal	Target position not reached by pump	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.021.00011	FD 15 0B	Fatal	Command terminated abnormally.	Subsequent

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.021.00012	FD 15 0C	Fatal	Hardware Error	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.021.00013	FD 15 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.021.00014	FD 15 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.021.00015	FD 15 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.021.00016	FD 15 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.021.00017	FD 15 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.021.00018	FD 15 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.021.00019	FD 15 13	Critical	The received parameter number needs an authorization which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
253.021.00020	FD 15 14	Critical	Received parameter authorization code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
253.021.00021	FD 15 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.021.00022	FD 15 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.021.00031	FD 15 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.021.00033	FD 15 21	Warning	Move interrupted.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.021.00034	FD 15 22	Message	Move completely finished.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.021.00162	FD 15 A2	Message	Self diagnostic sends min and max values of the supply voltage.	Process Control
253.021.00163	FD 15 A3	Message	Self diagnostic cycle counters.	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.021.00165	FD 15 A5	Message	Self diagnostic min drive current.	Process Control
253.021.00191	FD 15 BF	Message	An unexpected error occurred during module self diagnostics.	Process Control
253.021.00176	FD 15 B0	Warning	An internal error occurred	Calibration • Calibrate module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.021.00177	FD 15 B1	Message	Service memory data.	Process Control
253.021.00178	FD 15 B2	Critical	Initialization of service memory data failed.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.021.00179	FD 15 B3	Message	Initialization of service memory data succeeded.	Process Control
253.021.00180	FD 15 B4	Warning	An internal error occurred	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.021.00181	FD 15 B5	Critical	Service memory access denied.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.021.00209	FD 15 D1	Message	Self-diagnostic Device information: serial, article number and Version.	Process Control

Table 40–2: Message list, LWP

41 Pipettor

Device ID: 0xFD

41.1 Sub assemblies

The column link leads to the corresponding message list.

ID (dec)	ID (hex)	Name	Link to message list
1	0x01	Left Arm	
2	0x02	Right Arm	
3	0x03	Left Y Motor	
4	0x04	Right Y Motor	
0	0x00	General	
5	0x05	Left X Main	37. Dual Sledge Y
6	0x06	Left X Motor	37. Dual Sledge Y
7	0x07	Right X Main	42. Single Sledge Y
8	0x08	Right X Motor	42. Single Sledge Y
9	0x09	Left Z Main	38. Dual Sledge Z
10	0x0A	Left Z Motor 1	38. Dual Sledge Z
11	0x0B	Left Z Motor 2	38. Dual Sledge Z
12	0x0C	Right Z Main	43. Single Sledge Z
13	0x0D	Right Z Motor1	43. Single Sledge Z
14	0x0E	Left Zspip Main	36. Air Pipettor Left Arm
15	0x0F	Left Zspip Sctl	36. Air Pipettor Left Arm
16	0x10	Left Zspip Head	36. Air Pipettor Left Arm
17	0x11	Right Spolv Main	44. SPOLV Right Arm
18	0x12	Right Spolv Sctl	44. SPOLV Right Arm
19	0x13	Right Spolv Head	44. SPOLV Right Arm
20	0x14	Left Gripper	39. Gripper
21	0x15	Drain Pump	40. LWP

Table 41–1: Sub assemblies, Pipettor

41.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.001.00001	FD 01 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.002.00001	FD 02 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.003.00001	FD 03 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.004.00001	FD 04 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.000.00002	FD 00 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.001.00002	FD 01 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.002.00002	FD 02 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.003.00002	FD 03 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.004.00002	FD 04 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.000.00003	FD 00 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.000.00004	FD 00 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.000.00005	FD 00 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.000.00006	FD 00 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.003.00006	FD 03 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.004.00006	FD 04 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.000.00007	FD 00 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.003.00007	FD 03 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.004.00007	FD 04 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.003.00009	FD 03 09	Warning	Target position reached after retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.004.00009	FD 04 09	Warning	Target position reached after retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.003.00010	FD 03 0A	Fatal	Target position not reached by Y motor	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.004.00010	FD 04 0A	Fatal	Target position not reached by Y motor	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.000.00011	FD 00 0B	Fatal	Command terminated abnormally.	Subsequent
253.001.00011	FD 01 0B	Fatal	Command terminated abnormally.	Subsequent
253.002.00011	FD 02 0B	Fatal	Command terminated abnormally.	Subsequent
253.003.00011	FD 03 0B	Fatal	Command terminated abnormally.	Subsequent
253.004.00011	FD 04 0B	Fatal	Command terminated abnormally.	Subsequent
253.000.00013	FD 00 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.001.00013	FD 01 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

41. Pipettor

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.002.00013	FD 02 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.000.00014	FD 00 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.001.00014	FD 01 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.002.00014	FD 02 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.003.00014	FD 03 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.004.00014	FD 04 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.000.00015	FD 00 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.001.00015	FD 01 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.002.00015	FD 02 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.003.00015	FD 03 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.004.00015	FD 04 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.000.00016	FD 00 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.001.00016	FD 01 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.002.00016	FD 02 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.003.00016	FD 03 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.004.00016	FD 04 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.000.00017	FD 00 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.000.00018	FD 00 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.000.00019	FD 00 13	Critical	The received parameter number needs an authorization which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.000.00020	FD 00 14	Critical	Received parameter authorization code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
253.000.00021	FD 00 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.000.00022	FD 00 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.003.00023	FD 03 17	Fatal	A movement problem has been detected	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.004.00023	FD 04 17	Fatal	A movement problem has been detected	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.002.00028	FD 02 1C	Message	OLV signal	Process Control
253.000.00031	FD 00 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.000.00032	FD 00 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.001.00032	FD 01 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.002.00032	FD 02 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.003.00032	FD 03 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.004.00032	FD 04 20	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.003.00034	FD 03 22	Fatal	Initialization of Y motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.004.00034	FD 04 22	Fatal	Initialization of Y motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.003.00035	FD 03 23	Fatal	Initialization of Y motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.004.00035	FD 04 23	Fatal	Initialization of Y motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.003.00036	FD 03 24	Fatal	Unexpected change of internal position.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.004.00036	FD 04 24	Fatal	Unexpected change of internal position.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.003.00037	FD 03 25	Fatal	Unexpected change of target position.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.004.00037	FD 04 25	Fatal	Unexpected change of target position.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.001.00048	FD 01 30	Fatal	The pipettor is missing target coordinates.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

41. Pipettor

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.002.00048	FD 02 30	Fatal	The pipettor is missing target coordinates.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.001.00049	FD 01 31	Warning	The required z target position exceeds the range of motion. The target position had to be changed in order to prevent a crash.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.002.00049	FD 02 31	Warning	The required z target position exceeds the range of motion. The target position had to be changed in order to prevent a crash.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.001.00050	FD 01 32	Warning	The amount of air the SPOLV module has previously aspirated is not enough to execute a pressure based Liquid Level Detection for the required z distance. The z speed is set to maximum in order to cover as much as possible of the required z distance.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.001.00051	FD 01 33	Warning	The required z speed is too low to execute a pressure based Liquid Level Detection given the amount of air the SPOLV module has previously aspirated. The z speed is increased to the minimum value necessary to cover the required z distance.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.001.00052	FD 01 34	Fatal	Test of the Liquid Level Detection sensor system has failed.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.002.00052	FD 02 34	Fatal	Test of the Liquid Level Detection sensor system has failed.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.002.00053	FD 02 35	Warning	Optical Liquid Verification data trace might be corrupt.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Process Control
253.001.00054	FD 01 36	Warning	Pipettor is aspirating/dispensing from/in an undefined bottle. Tracking of the water surface (in z direction) might not be accurate.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.002.00054	FD 02 36	Warning	Pipettor is aspirating/dispensing from/in an undefined bottle. Tracking of the water surface (in z direction) might not be accurate.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.001.00055	FD 01 37	Fatal	Pipettor has lost a disposable tip.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.001.00056	FD 01 38	Warning	A pipettor submodule has sent a warning message previously.	Subsequent
253.002.00056	FD 02 38	Warning	A pipettor submodule has sent a warning message previously.	Subsequent
253.001.00057	FD 01 39	Critical	A pipettor submodule has sent a critical error message previously.	Subsequent

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Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.002.00057	FD 02 39	Critical	A pipettor submodule has sent a critical error message previously.	Subsequent
253.001.00058	FD 01 3A	Fatal	A pipettor submodule has sent a fatal error message previously.	Subsequent
253.002.00058	FD 02 3A	Fatal	A pipettor submodule has sent a fatal error message previously.	Subsequent
253.001.00059	FD 01 3B	Fatal	Gripper has lost a cuvette.	
253.005.00060	FD 05 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.006.00060	FD 06 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.007.00060	FD 07 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.008.00060	FD 08 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.009.00060	FD 09 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.010.00060	FD 0A 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.011.00060	FD 0B 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

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Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.012.00060	FD 0C 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.013.00060	FD 0D 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.014.00060	FD 0E 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.015.00060	FD 0F 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.016.00060	FD 10 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.017.00060	FD 11 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.018.00060	FD 12 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.019.00060	FD 13 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

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Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.020.00060	FD 14 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.021.00060	FD 15 3C	Fatal	A command response sent by a pipettor submodule was received delayed or not received at all.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.002.00061	FD 02 3D	Fatal	The pipettor wash pump is not connected or broken.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.001.00062	FD 01 3E	Fatal	Disposable tip eject failed.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.001.00063	FD 01 3F	Warning	Eject of disposable tip failed. A retry will be executed.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.000.00069	FD 00 45	Warning	The main cycle of the pipettor embedded software exceeded a critical time.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.003.00069	FD 03 45	Warning	The main cycle of the pipettor embedded software exceeded a critical time.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.004.00069	FD 04 45	Warning	The main cycle of the pipettor embedded software exceeded a critical time.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.003.00080	FD 03 50	Message	Entered function MODE3_SetElectricalPosition.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.004.00080	FD 04 50	Message	Entered function MODE3_SetElectricalPosition.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.003.00081	FD 03 51	Message	Left function MODE3_SetElectricalPosition.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.004.00081	FD 04 51	Message	Left function MODE3_SetElectricalPosition.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.002.00096	FD 02 60	Fatal	Undefined OLV dead volume reference value.	Configuration • Check instrument configuration • Run Instrument Setup Call manufacturer
253.000.00160	FD 00 A0	Message	Optical Liquid Verification sensor calibration results gathered during module selfdiagnostics.	
253.000.00161	FD 00 A1	Message	Optical Liquid Verification dead volume calibration results gathered during module selfdiagnostics.	
253.000.00162	FD 00 A2	Message	Minimum/maximum supply voltage detected.	
253.000.00163	FD 00 A3	Message	Number of wash cycles executed.	
253.000.00164	FD 00 A4	Message	Number of moves executed.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.000.00165	FD 00 A5	Message	Minimum motor current evaluated during module self diagnostics.	Calibration • Calibrate module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.000.00166	FD 00 A6	Message	Mechanical clearance of the drive evaluated during module self diagnostics.	
253.000.00191	FD 00 BF	Message	An unexpected error occurred during module self diagnostics.	
253.000.00176	FD 00 B0	Warning	An internal error occurred	
253.000.00177	FD 00 B1	Message	Service memory data.	
253.000.00178	FD 00 B2	Critical	Initialization of service memory data failed.	
253.000.00179	FD 00 B3	Message	Initialization of service memory data succeeded.	
253.000.00180	FD 00 B4	Warning	An internal error occurred.	
253.000.00181	FD 00 B5	Critical	Service memory access denied.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.000.00209	FD 00 D1	Message	Self-diagnostic Device information: serial, article number and Version.	

Table 41–2: Message list, Pipettor

42 Single Sledge Y

Device ID: 0xFD

42.1 Sub assemblies

ID (dec)	ID (hex)	Name
7	0x07	General
8	0x08	Y_Drive

Table 42–1: Sub assemblies, Single Sledge Y

42.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.007.00001	FD 07 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.008.00001	FD 08 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.007.00002	FD 07 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.008.00002	FD 08 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.007.00003	FD 07 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.007.00004	FD 07 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.007.00005	FD 07 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.007.00006	FD 07 06	Critical	Voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.007.00007	FD 07 07	Warning	Voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.008.00009	FD 08 09	Warning	Target position reached after retry.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.008.00010	FD 08 0A	Fatal	Target position not reached by Y motor	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.008.00011	FD 08 0B	Fatal	Command terminated abnormally.	Subsequent
253.007.00013	FD 07 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.008.00013	FD 08 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.007.00014	FD 07 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.008.00014	FD 08 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.007.00015	FD 07 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.008.00015	FD 08 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.007.00016	FD 07 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.008.00016	FD 08 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.007.00017	FD 07 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.007.00018	FD 07 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.007.00019	FD 07 13	Critical	The received parameter number needs an authorisation which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.007.00020	FD 07 14	Critical	Received parameter authorisation code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
253.007.00021	FD 07 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.007.00022	FD 07 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.007.00023	FD 07 17	Fatal	A movement problem has been detected	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.008.00023	FD 08 17	Fatal	A movement problem has been detected	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.007.00031	FD 07 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.007.00032	FD 07 20	Fatal	Embedded software crash occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

42. Single Sledge Y

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.008.00032	FD 08 20	Fatal	Embedded software crash occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.008.00050	FD 08 32	Fatal	Initialization of Y motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.008.00051	FD 08 33	Fatal	Initialization of Y motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.008.00052	FD 08 34	Fatal	Unexpected change of internal position.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.008.00053	FD 08 35	Fatal	Unexpected change of target position.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.008.00080	FD 08 50	Message	Entered function MODE3_SetElectricalPosition.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.008.00081	FD 08 51	Message	Left function MODE3_SetElectricalPosition.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.007.00160	FD 07 A0	Message	Minimum/maximum supply voltage detected.	Process Control

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.007.00161	FD 07 A1	Message	Number of moves executed.	Process Control
253.007.00162	FD 07 A2	Message	Minimum motor current evaluated during module self diagnostics.	Process Control
253.007.00163	FD 07 A3	Message	Mechanical clearance of the drive evaluated during module self diagnostics.	Process Control
253.007.00191	FD 07 BF	Message	An unexpected error occurred during module self diagnostics.	Process Control
253.007.00209	FD 07 D1	Message	Self-diagnostic Device information: serial, article number and Version.	Process Control

Table 42–2: Message list, Single Sledge Y

43 Single Sledge Z

Device ID: 0xFD

43.1 Sub assemblies

ID (dec)	ID (hex)	Name
12	0x0C	General
13	0x0D	ZDrive1
255	0xFF	ZDrive2 (not connected)

Table 43–1: Sub assemblies, Single Sledge Z

43.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.012.00001	FD 0C 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.013.00001	FD 0D 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.255.00001	FD FF 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.012.00002	FD 0C 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.013.00002	FD 0D 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.255.00002	FD FF 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.012.00003	FD 0C 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.012.00004	FD 0C 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.012.00005	FD 0C 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.012.00011	FD 0C 0B	Fatal	Command terminated abnormally.	Subsequent
253.013.00011	FD 0D 0B	Fatal	Command terminated abnormally.	Subsequent
253.255.00011	FD FF 0B	Fatal	Command terminated abnormally.	Subsequent
253.012.00013	FD 0C 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.013.00013	FD 0D 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.255.00013	FD FF 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.012.00014	FD 0C 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.013.00014	FD 0D 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.255.00014	FD FF 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.012.00015	FD 0C 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.013.00015	FD 0D 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.255.00015	FD FF 0F	Critical	Module cannot execute command as received command is not supported yet.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

43. Single Sledge Z

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.012.00016	FD 0C 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.013.00016	FD 0D 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.255.00016	FD FF 10	Fatal	Execution time of command exceeds limits as connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.012.00017	FD 0C 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.012.00018	FD 0C 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.012.00019	FD 0C 13	Critical	The received parameter number needs an authorisation which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer InstrumentHandling • Check consumable states • Initialize module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.012.00020	FD 0C 14	Critical	Received parameter authorisation code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
253.012.00021	FD 0C 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.012.00022	FD 0C 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.012.00031	FD 0C 1F	Warning	Errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.013.00033	FD 0D 21	Warning	The required disposable tip pick-up movement is started form below z-start.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.255.00033	FD FF 21	Warning	The required disposable tip pick-up movement is started form below z-start.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.013.00034	FD 0D 22	Fatal	Servo controller communication error	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.255.00034	FD FF 22	Fatal	Servo controller communication error	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

43. Single Sledge Z

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.013.00037	FD 0D 25	Fatal	Z motor overload	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.255.00037	FD FF 25	Fatal	Z motor overload	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.013.00038	FD 0D 26	Critical	The drive was stopped after exceeding a given force limitation.	Process Control
253.255.00038	FD FF 26	Critical	The drive was stopped after exceeding a given force limitation.	Process Control
253.013.00048	FD 0D 30	Fatal	Target position not reached by Z motor	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.255.00048	FD FF 30	Fatal	Target position not reached by Z motor	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.013.00050	FD 0D 32	Fatal	Initialization of Z motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.255.00050	FD FF 32	Fatal	Initialization of Z motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.013.00051	FD 0D 33	Fatal	Initialization of Z motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

43. Single Sledge Z

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.255.00051	FD FF 33	Fatal	Initialization of Z motor failed	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.013.00052	FD 0D 34	Warning	Friction calibration is not possible as the current position is below z-start.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.255.00052	FD FF 34	Warning	Friction calibration is not possible as the current position is below z-start.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.013.00053	FD 0D 35	Message	Control deviation reported by the servo controller	Process Control
253.255.00053	FD FF 35	Message	Control deviation reported by the servo controller	Process Control
253.012.00160	FD 0C A0	Message	Report of executed moves.	Process Control
253.012.00161	FD 0C A1	Message	Minimum servo controller saturation threshold evaluated during module self diagnostics.	Process Control
253.012.00162	FD 0C A2	Message	Mechanical clearance of the drive evaluated during module selfdiagnostics.	Process Control
253.012.00191	FD 0C BF	Message	An unexpected error occurred during module self diagnostics.	Process Control
253.012.00083	FD 0C 53	Fatal	Embedded software crash occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.013.00083	FD 0D 53	Fatal	Embedded software crash occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.255.00083	FD FF 53	Fatal	Embedded software crash occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.012.00209	FD 0C D1	Message	Self-diagnostic Device information: serial, article number and Version.	

Table 43–2: Message list, Single Sledge Z

44 SPOLV Right Arm

Device ID: 0xFD

44.1 Sub assemblies

ID (dec)	ID (hex)	Name
17	0x11	General
18	0x12	SctICU
19	0x13	APHeadCU

Table 44–1: Sub assemblies, SPOLV Right Arm

44.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.017.00001	FD 11 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.018.00001	FD 12 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.019.00001	FD 13 01	Fatal	Module cannot execute command as module is not initialized.	Instrument Handling • Check consumable states • Initialize module
253.017.00002	FD 11 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.018.00002	FD 12 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.019.00002	FD 13 02	Critical	Module cannot execute command as received command or subassembly identifier is unknown or command length is incorrect.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.017.00003	FD 11 03	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.017.00004	FD 11 04	Critical	An internal error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.017.00005	FD 11 05	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.017.00006	FD 11 06	Critical	Supply voltage is out of range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.017.00007	FD 11 07	Warning	Supply voltage is back in range. Detected voltage: {Detected Voltage}	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.017.00010	FD 11 0A	Fatal	Syringe motor overload	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.018.00010	FD 12 0A	Fatal	Syringe motor overload	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.017.00011	FD 11 0B	Fatal	Command terminated abnormally.	Subsequent
253.018.00011	FD 12 0B	Fatal	Command terminated abnormally.	Subsequent
253.017.00013	FD 11 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.018.00013	FD 12 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.019.00013	FD 13 0D	Critical	Module cannot execute command as received command parameter is out of range.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.017.00014	FD 11 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.018.00014	FD 12 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.019.00014	FD 13 0E	Critical	Current state of module or subassembly does not allow command execution. Perhaps another command is still in process.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.017.00016	FD 11 10	Fatal	Execution time of command exceeds limits or connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.018.00016	FD 12 10	Fatal	Execution time of command exceeds limits or connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.019.00016	FD 13 10	Fatal	Execution time of command exceeds limits or connection to submodule is lost.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.017.00017	FD 11 11	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.017.00018	FD 11 12	Critical	Parameter number of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.017.00019	FD 11 13	Critical	The received parameter number needs an authorization which is currently not set.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
253.017.00020	FD 11 14	Critical	Received parameter authorization code is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Instrument Handling • Check consumable states • Initialize module
253.017.00021	FD 11 15	Critical	Parameter type of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.017.00022	FD 11 16	Critical	Parameter length of received parameter command is not valid.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.017.00031	FD 11 1F	Warning	Communication errors on CAN bus detected.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.017.00032	FD 11 20	Message	Trigger was set by liquid level detection (LLD).	Process Control
253.017.00033	FD 11 21	Fatal	A communication error occurred	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.017.00034	FD 11 22	Fatal	SCTL timeout: interrupt not received within expected time or internal communication error occurred.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.017.00035	FD 11 23	Fatal	Initialization of syringe failed	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.017.00037	FD 11 25	Warning	Syringe motor overload	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.017.00039	FD 11 27	Warning	Pressure sensor ambient level (zero offset) out of range during init sensor test.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.017.00040	FD 11 28	Warning	No pressure change on init move detected, module not primed or sensor failure.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.017.00048	FD 11 30	Fatal	Target position not reached by syringe	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module
253.017.00049	FD 11 31	Fatal	SCTL Watchdog reset, internal communication error.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.017.00050	FD 11 32	Warning	Frequency of cLLD oscillator out of range during init sensor test.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.017.00052	FD 11 34	Fatal	Initialization of syringe failed	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.017.00053	FD 11 35	Fatal	Syringe overpressure	Calibration • Calibrate module Consumable • Exchange consumable • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.017.00054	FD 11 36	Fatal	Parameter iVP9101Conv not set. Module needs suitable parameter first.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.017.00056	FD 11 38	Warning	Init position verification needed retry.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.017.00064	FD 11 40	Critical	Communication error ({APHErrorEvent}) with LPH board via interface cable.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.017.00065	FD 11 41	Warning	Communication warning ({LPHWarningEvent}) with LPH board via interface cable.	Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.017.00068	FD 11 44	Critical	Calibration result = 0, calibration criterias not met.	Calibration • Calibrate module Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.017.00096	FD 11 60	Fatal	Reset of μ C has occurred, internal Software crash or external reset invoked: {Reset Exceptions} at {Reset Address}.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.017.00128	FD 11 80	Critical	Evaluation result of pressure curve not plausible. Wash process could be insufficient	
253.017.00130	FD 11 82	Warning	Analyse wash pressure: Expected wash time length longer than recorded buffer length. Analysis not possible.	
253.017.00131	FD 11 83	Warning	Analyse wash pressure: Wash setting matches no defined pattern. Analysis not possible.	
253.017.00160	FD 11 A0	Message	Self diagnostic sends min and max values of the supply voltage.	
253.017.00161	FD 11 A1	Message	Move status message.	Mechanics • Clean module • Check calibration • Check mechanic • Exchange module Electronics • Check cable and connections • Check instrument configuration • Exchange printed circuit board • Exchange module
253.017.00163	FD 11 A3	Message	Self-diagnostic: PWM Average at full range move.	
253.017.00164	FD 11 A4	Message	Self-diagnostic: percentiles of deviation at full range moves.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
253.017.00177	FD 11 B1	Message	Self-diagnostic Device information: serial, article number and Version.	
253.017.00165	FD 11 A5	Message	Self-diagnostic: move and valve switching cycle counter.	
253.017.00168	FD 11 A8	Message	Self-diagnostic: pressure trace w/o moving, pressure sensor noise.	
253.017.00169	FD 11 A9	Message	Self-diagnostic: pressure, deviation or count trace of a dedicated test-step.	
253.017.00170	FD 11 AA	Message	Self-diagnostic: analyse data of a dedicated move.	
253.017.00173	FD 11 AD	Message	Self-diagnostic: deviation at retries on move to block.	
253.017.00174	FD 11 AE	Message	Self-diagnostic: Final position errors.	
253.017.00192	FD 11 C0	Warning	Error during ServiceMemory access.	
253.017.00193	FD 11 C1	Message	ServiceMemory read data.	
253.017.00255	FD 11 FF	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.018.00255	FD 12 FF	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer
253.019.00255	FD 13 FF	Fatal	Embedded software crash occurs.	Configuration • Check instrument configuration • Run Instrument Setup • Call manufacturer

Table 44–2: Message list, SPOLV Right Arm

45 Application

Device ID: 0x1F4

45.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	Started
999	0x3E7	ResearchAndDevelopmentException

Table 45–1: Sub assemblies, Application

45.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
500.000.00000	1F4 00 00	Message	Application started. Version (product/file): {0}/{1}	
500.000.00001	1F4 00 01	Message	Application ended.	
500.000.00002	1F4 00 02	Message	User {0} logged in at level {1}.	
500.000.00003	1F4 00 03	Message	User {0} logged off.	
500.000.00004	1F4 00 04	Problem	File {0} has wrong format	
500.000.00005	1F4 00 05	Problem	File {0} has wrong checksum.	
500.000.00006	1F4 00 06	Problem	Aborted import of file {0}.	
500.000.00007	1F4 00 07	Problem	File export error.	
500.000.00008	1F4 00 08	Problem	File {0} could not be exported.	
500.000.00009	1F4 00 09	Catastrophic	Unexpected software error. Application must be closed.	
500.000.00010	1F4 00 0A	Critical	An internal error occurred	
500.000.00011	1F4 00 0B	Critical	An internal error occurred	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
500.000.00012	1F4 00 0C	Warning	An internal error occurred	
500.000.00013	1F4 00 0D	Catastrophic	A timeout has been reached while waiting for a response from the instrument	
500.000.00015	1F4 00 0F	Warning	The system failed to start '{0}'	
500.000.00016	1F4 00 10	Problem	Assay with Abbreviation {0}, Article Number {1}, Name {2} LIS alias {3} could not be imported because of existing Assay with Abbreviation {4}, Article Number {5}, LIS Alias {7}	
500.000.00017	1F4 00 11	Problem	Assay '{0}' with article number '{1}', name '{2}' and LIS alias '{3}' could not be deleted because it is still referenced by assay group '{4}'.	
500.000.00018	1F4 00 12	Problem	Assay '{0}' with article number '{1}', name '{2}' and LIS alias '{3}' could not be deleted because it is still referenced by assay profile '{4}'.	
500.000.00019	1F4 00 13	Problem	Assay '{0}' with article number '{1}', name '{2}' and LIS alias '{3}' could not be deleted because it is still referenced by control definition '{4}', lot number '{5}'.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
500.000.00020	1F4 00 14	Problem	Assay '{0}' with article number '{1}', name '{2}' and LIS alias '{3}' could not be deleted because it is still referenced by rerun rule '{4}'.	
500.000.00021	1F4 00 15	Problem	Assay '{0}' with article number '{1}', name '{2}' and LIS alias '{3}' could not be deleted because it is still referenced by one or more browser results.	
500.000.00022	1F4 00 16	Message	Generic sequences import finished successfully.	
500.000.00023	1F4 00 17	Problem	Generic sequences import failed.	
500.000.00024	1F4 00 18	Problem	Generic sequences not loaded.	
500.000.00025	1F4 00 19	Message	Dilution sequences import finished successfully.	
500.000.00026	1F4 00 1A	Problem	Dilution sequences import failed.	
500.000.00027	1F4 00 1B	Critical	A problem occurred during the archiving of the old system log files	
500.000.00028	1F4 00 1C	Message	System Settings import finished successfully.	
500.000.00029	1F4 00 1D	Catastrophic	System Settings import failed.	
500.000.00030	1F4 00 1E	Message	System Settings Export finished successfully.	
500.000.00031	1F4 00 1F	Problem	System Settings Export failed.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
500.000.00032	1F4 00 20	Problem	Could not delete calibration with Calibration Id {0}. The calibration is still referenced by at least one result.	
500.000.00033	1F4 00 21	Problem	Versions Export failed.	
500.000.00034	1F4 00 22	Message	Versions Export finished successfully.	
500.000.00035	1F4 00 23	Problem	Firmware Paramter Export failed.	
500.000.00036	1F4 00 24	Message	Firmware Paramter Export finished successfully.	
500.000.00037	1F4 00 25	Problem	Release Configuration Export failed.	
500.000.00038	1F4 00 26	Message	Release Configuration Export finished successfully.	
500.000.00040	1F4 00 28	Message	Assay deletion failed. The following assays are still referenced by other assays: {0}	
500.000.00041	1F4 00 29	Warning	Online Help initialization failed. Could not link the software views with the Online Help chapters.	
500.000.00042	1F4 00 2A	Message	Deleted reference to unknown Assay '{0}' in imported group '{1}'.	
500.000.00043	1F4 00 2B	Message	Group '{0}' has been overwritten by import.	
500.000.00044	1F4 00 2C	Warning	Patient database could not be opened and has been recreated.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
500.000.00045	1F4 00 2D	Message	Removed Rerun Rule '{0}' with unknown Assay(s) '{1}' from import.	
500.000.00046	1F4 00 2E	Message	Rerun Rule '{0}' has been overwritten by import.	
500.000.00050	1F4 00 32	Message	At least one combi family was not archived due to a result not in Done or Failed status.	
500.000.00051	1F4 00 33	Message	The {0} manuals have been successfully downloaded.	
500.000.00052	1F4 00 34	Message	The {0} instrument package for the consumable with article no. {1} and lot no. {2} has been successfully downloaded.	
500.000.00053	1F4 00 35	Message	The {0} instrument package for the assay with article no. {1} and lot no. {2} has been successfully downloaded.	
500.000.00054	1F4 00 36	Message	The {0} instrument package for the control with control name {1} and control lot {2} have been successfully downloaded.	
500.000.00055	1F4 00 37	Message	The {0} manuals could not be downloaded.	
500.000.00056	1F4 00 38	Message	The {0} instrument package for the consumable with article no. {1} and lot no. {2} could not be downloaded.	

45. Application

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
500.000.00057	1F4 00 39	Message	The {0} instrument package for the assay with article no. {1} and lot no. {2} could not be downloaded.	
500.000.00058	1F4 00 3A	Message	The {0} instrument package for the control with control name {1} and control lot {2} could not be downloaded.	
500.000.00059	1F4 00 3B	Message	All instrument package were successfully acquired.	
500.000.00060	1F4 00 3C	Message	One or more instrument package could not be acquired.	Try to download and acquire the instrument package again; if the problem persists, call Service.
500.000.00061	1F4 00 3D	Message	Expected Country Code {0} but got {1}.	In case the file was downloaded from DiaSorin online repository, ensure that the country selected for the package matches with the country set in the Main SW; If the problem persists, call Service.
500.000.00062	1F4 00 3E	Message	The instrument package to be imported has a wrong structure.	Try to download and acquire the instrument package again; if the problem persists, call Service.
500.000.00063	1F4 00 3F	Message	The connection to the cloud is currently interrupted. The download will start once the connection is established.	Try to download package again; if the problem persists, check the network cable; if the problem persists, contact the Laboratory IT to check that the network policy have been correctly set to allow the connection for the IoT services; if the problem persists, call Service
500.000.00100	1F4 00 64	Warning	{0} replicates found in the Results All page. Please, archive or delete Results to maintain proper software performance.	
500.999.99999	1F4 3E7 1869F	Catastrophic	Software has to close. {0}	

Table 45–2: Message list, Application

46 COM Protocol

Device ID: 0x1F4

46.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	ComProtocolException

Table 46–1: Sub assemblies, COM Protocol

46.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
500.000.00039	1F4 00 27	Fatal	A communication error occurred	

Table 46–2: Message list, COM Protocol

47 Database

Device ID: 0x1F5

47.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	RepositoryException

Table 47–1: Sub assemblies, Database

47.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
501.000.00000	1F5 00 00	Fatal	General database error.	
501.000.00001	1F5 00 01	Catastrophic	Failed to connect to the database.	
501.000.00003	1F5 00 03	Catastrophic	No database available or connection not possible.	
501.000.00004	1F5 00 04	Catastrophic	The database is not compatible with the installed SW version.	
501.000.00005	1F5 00 05	Problem	The checksum of assay '{0}' is wrong.	
501.000.00010	1F5 00 0A	Problem	A cycle mismatch occurred. Backup Cycle: {0} Database Cycle {1} COP Cycle {2} new Cycle {3}.	
501.000.00011	1F5 00 0B	Message	Write Cycle {0} to COP.	
501.000.00012	1F5 00 0C	Catastrophic	Database check error occurred: {0}. Please call Customer Service.	

Table 47–2: Message list, Database

48 Instrument Control

Device ID: 0x1F6

48.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	CoordinateFileNotAvailable

Table 48–1: Sub assemblies, Instrument Control

48.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
502.000.00000	1F6 00 00	Catastrophic	Coordinate file could not be loaded.	
502.000.00001	1F6 00 01	Fatal	An internal error occurred	
502.000.00002	1F6 00 02	Message	Instrument state has changed from {0} to {1}.	
502.000.00003	1F6 00 03	Problem	One or more instrument modules did not initialize properly.	
502.000.00004	1F6 00 04	Fatal	Instrument halted!	
502.000.00005	1F6 00 05	Message	Instrument Serial Number: {0}.	
502.000.00006	1F6 00 06	Message	Module: {0}, Version: {1}	
502.000.00007	1F6 00 07	Problem	Transparent mode required by service software is active.	
502.000.00008	1F6 00 08	Problem	Incompatible firmware version: {0}. Expected one of the following: {1}.	
502.000.00009	1F6 00 09	Problem	A communication error occurred	
502.000.00010	1F6 00 0A	Problem	A communication error occurred	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
502.000.00011	1F6 00 0B	Problem	A communication error occurred	
502.000.00012	1F6 00 0C	Fatal	Dynamic coordinates could not be downloaded to the system.	
502.000.00013	1F6 00 0D	Critical	Too late to transmit Job with JobID {0}.	
502.000.00014	1F6 00 0E	Critical	A scheduling error occurred	
502.000.00015	1F6 00 0F	Warning	The reported incubator ring status does not match the internal status. Incubator ring will be cleared and reloaded.	
502.000.00016	1F6 00 10	Critical	An internal error occurred	
502.000.00017	1F6 00 11	Message	Resource state has changed: {0} changed to {1}.	
502.000.00018	1F6 00 12	Warning	The ring is completely cleared as status Ready was not entered after Running or Maintenance.	
502.000.00019	1F6 00 13	Warning	After the Instrument entered halted, the instrument communication and database access was checked, incubator status is correctly stored. No complete incubator cleaning needed.	
502.000.00020	1F6 00 14	Warning	The import was aborted because the instrument is in Running.	
502.000.00050	1F6 00 32	Fatal	A fatal error occurred on Instrument. (EventID: {0})	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
502.000.00051	1F6 00 33	Fatal	An internal error occurred.	
502.000.00052	1F6 00 34	Message	Instrument System Cycle changed to {0}.	
502.000.00053	1F6 00 35	Message	FSE Mode disabled.	
502.000.00054	1F6 00 36	Message	FSE Mode enabled.	
502.000.00100	1F6 00 64	Message	SelfDiagnostics finished with success.	
502.000.00101	1F6 00 65	Message	SelfDiagnostics finished with failure.	

Table 48-2: Message list, Instrument Control

49 Process Control

Device ID: 0x1F7

49.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	JobScheduled
1	0x01	AbortOngoingJobOnSWStartOrInitialization

Table 49–1: Sub assemblies, Process Control

49.2 Message list for internal-tanks configuration

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.000.00000	1F7 00 00	Message	Scheduled Job {0} {1} for Cycle {2} with JobId {3} (Replicate: {4}, Dilution: {5}, Finish {6}, Integral Lane {7}).	
503.000.00001	1F7 00 01	Message	Unscheduled Job {0} {1} {2} {3} for Cycle {4} JobId {5}.	
503.000.00002	1F7 00 02	Message	Started Job {0} {1} {2} {3} for Cycle {4} JobId {5}.	
503.000.00003	1F7 00 03	Message	Started Mitigation {0} for cycle {1} phase {2} JobId {3}.	
503.000.00004	1F7 00 04	Message	The Start button was pressed	
503.000.00005	1F7 00 05	Message	ScheduleJobs.	
503.000.00006	1F7 00 06	Message	Unschedule Jobs.	
503.000.00007	1F7 00 07	Message	Reschedule jobs.	
503.000.00010	1F7 00 0A	Message	Start Jobs that were created by a rerun rule.	
503.000.00011	1F7 00 0B	Warning	A scheduling error occurred	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.000.00012	1F7 00 0C	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no wash buffer.	
503.000.00013	1F7 00 0D	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no cuvette.	
503.000.00014	1F7 00 0E	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no disposable tip.	
503.000.00015	1F7 00 0F	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no solid waste.	
503.000.00016	1F7 00 10	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no starter.	
503.000.00017	1F7 00 11	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no integral.	
503.000.00018	1F7 00 12	Message	Scheduled Starter Prime Job for cycle {0} JobId {1}.	
503.000.00019	1F7 00 13	Warning	No jobs could be scheduled for start.	
503.000.00020	1F7 00 14	Problem	Calibration Data incorrect.	
503.000.00021	1F7 00 15	Warning	A needed Dilution Sequence was not found. Schedule aborted.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.000.00022	1F7 00 16	Message	Aspirated {0} for Job {1} {2} in Cycle {3} Phase {4} (Replicate {5}, Dilution {6}, Coordinate {7}, JobId {8}).	
503.000.00023	1F7 00 17	Critical	Aspirated from onboard expired integral.	
503.000.00024	1F7 00 18	Critical	Aspirated from onboard expired ancillary.	
503.000.00025	1F7 00 19	Critical	Aspirated from expired control vial.	
503.000.00026	1F7 00 1A	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no ancillary.	
503.000.00027	1F7 00 1B	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), reagent {4} depleted.	
503.000.00030	1F7 00 1E	Message	Needed Control {0} not available.	
503.000.00031	1F7 00 1F	Warning	Prime failed.	
503.000.00040	1F7 00 28	Message	Received Rlu {0} for JobId {1} (With Spikes {2}, Without Spikes {3}, DarkCount {4}, DarkCountEvents {5}, DarkCountAverage {6}, MeasureTime {7})	
503.000.00041	1F7 00 29	Message	Scheduled Job {0} {1} for Cycle {2} with JobId {3} (Replicate: {4}, Dilution: {5}, Finish {6}, Integral Lane {7}, agitation time insufficient).	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.000.00050	1F7 00 32	Critical	Job {0} {1} {2} {3} for Cycle {4} Jobld {5} above Assay Range.	
503.000.00051	1F7 00 33	Critical	Job {0} {1} {2} {3} for Cycle {4} Jobld {5} below Assay Range.	
503.000.00052	1F7 00 34	Critical	Job {0} {1} {2} {3} for Cycle {4} Jobld {5} above Normal Range.	
503.000.00053	1F7 00 35	Critical	Job {0} {1} {2} {3} for Cycle {4} Jobld {5} below Normal Range.	
503.000.00054	1F7 00 36	Critical	Job {0} {1} {2} {3} for Cycle {4} Jobld {5} manipulated.	
503.000.00055	1F7 00 37	Critical	Job {0} {1} {2} {3} for Cycle {4} Jobld {5} recalculated.	
503.000.00056	1F7 00 38	Critical	Job {0} {1} {2} {3} for Cycle {4} Jobld {5} flagged with Control out of Manufacturer Range.	
503.000.00057	1F7 00 39	Critical	Job {0} {1} {2} {3} for Cycle {4} Jobld {5} flagged with No Calibration.	
503.000.00058	1F7 00 3A	Warning	Manipulation aborted because no applicable integral found in database.	
503.000.00060	1F7 00 3C	Critical	Dark Count Average exceeds envelop.	
503.000.00061	1F7 00 3D	Critical	Dark Count was out of Range.	
503.000.00062	1F7 00 3E	Fatal	Three times in a row DarkCount Exceeds Envelop.	
503.000.00063	1F7 00 3F	Fatal	Three times in a row DarkCount out of Range.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.000.00064	1F7 00 40	Warning	Spikes Detected for Job {0} {1} {2} {3} for Cycle {4} Jobld {5}.	
503.000.00067	1F7 00 43	Message	Rerun Rule {0} did not create test {1} for {2} because dilution factor {3} is not defined in the assay.	
503.000.00068	1F7 00 44	Message	Rerun Rule {0} did not create test {1} because sample {2} has been unloaded.	
503.000.00069	1F7 00 45	Message	Flagged Result {0} {1} with 'caused rerun'.	
503.000.00070	1F7 00 46	Critical	Result {0} {1} flagged with QC Above Manufacturer Range.	
503.000.00071	1F7 00 47	Critical	Result {0} {1} flagged with QC Below Manufacturer Range.	
503.000.00072	1F7 00 48	Critical	Result {0} {1} flagged with QC Above User Range.	
503.000.00073	1F7 00 49	Critical	Result {0} {1} flagged with QC Below User Range.	
503.000.00074	1F7 00 4A	Critical	Result {0} {1} flagged with Above Assay Range.	
503.000.00075	1F7 00 4B	Critical	Result {0} {1} flagged with Below Assay Range.	
503.000.00076	1F7 00 4C	Critical	Result {0} {1} flagged with Above Normal Range.	
503.000.00077	1F7 00 4D	Critical	Result {0} {1} flagged with Below Normal Range.	
503.000.00078	1F7 00 4E	Critical	Result {0} {1} flagged with Control out of Manufacturer Range.	
503.000.00079	1F7 00 4F	Critical	Result {0} {1} flagged with Divided By Zero.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.000.00081	1F7 00 51	Message	Recalculated Result {0} {1}.	
503.000.00082	1F7 00 52	Message	Could not recalculated Result {0} {1}, undo changes.	
503.000.00090	1F7 00 5A	Critical	Job {0} {1} {2} {3} for Cycle {4} Jobld {5} flagged with Control Expired.	
503.001.00001	1F7 01 01	Critical	Abort Job {0} {1} {2} {3} for Cycle {4} Jobld {5} on SW Start or during Initialization.	Start again the aborted job; If the problems persists call Service.
503.001.00002	1F7 01 02	Warning	An internal error occurred	Call Service.
503.001.00003	1F7 01 03	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to darkcount subtraction.	Start again the aborted job; If the problems persists call Service.
503.001.00004	1F7 01 04	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to unexpected Result Cycle.	Start again the aborted job; If the problems persists call Service.
503.001.00005	1F7 01 05	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to Overdilution.	Start again the aborted job; If the problems persists call Service.
503.001.00006	1F7 01 06	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to not plausible Aspiration.	Start again the aborted job; If the problems persists call Service.
503.001.00007	1F7 01 07	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to starter prime not performed.	Start again the aborted job; If the problems persists call Service.
503.001.00008	1F7 01 08	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to mitigation not performed.	Start again the aborted job; If the problems persists call Service.

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.001.00009	1F7 01 09	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to divided by zero.	Start again the aborted job; If the problems persists call Service.
503.001.00010	1F7 01 0A	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to math error.	Start again the aborted job; If the problem persists call Service.
503.001.00011	1F7 01 0B	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to sent too late.	Start again the aborted job; If the problem persists call Service.
503.001.00020	1F7 01 14	Critical	Aborted Job {0} {1} {2} {3} due to starter lot switch.	Start again the aborted job; If the problem persists call Service.
503.001.00022	1F7 01 16	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to Sample Integrity Error.	Ensure the tube contains the needed amount of Sample; Check Sample integrity; Load the Sample tube in the appropriate rack; Start again the aborted job; If the problem persists call Service.
503.001.00023	1F7 01 17	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to clot detected in the sample probe	Ensure the tube contains the needed amount of Sample; Check Sample integrity; Load the Sample tube in the appropriate rack; Start again the aborted job; If the problem persists call Service.
503.001.00025	1F7 01 19	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing Cuvette.	Ensure the needed Cuvettes are available and assigned on the system; Start again the aborted job; If the problem persists call Service.
503.001.00026	1F7 01 1A	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing Disposable Tip.	Ensure the needed Tips are available and assigned on the system; Start again the aborted job; If the problem persists call Service.
503.001.00027	1F7 01 1B	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to Tip Pickup Failure.	Check the assignment of the Tips on the SW; Restart the job; If the problem persists call Service.
503.001.00028	1F7 01 1C	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing Starter.	Ensure the Starter bottles are loaded and primed; Check liquid integrity; In case the reagent was set off line reset the LLD; Start again the aborted job; If the problem persists call Service.

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.001.00029	1F7 01 1D	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing Wash Buffer.	Ensure that the Wash Buffer tank is loaded, recognized and primed; Restart the job; If the problem persists call Service.
503.001.00030	1F7 01 1E	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to Washer Aspiration Error at Port {6}.	Start again the aborted job; If the problem persists call Service.
503.001.00031	1F7 01 1F	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to not enough or no liquid found in sample tube.	Start again the aborted job; If the problem persists call Service.
503.001.00032	1F7 01 20	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing Sample.	Start again the aborted job; If the problem persists call Service.
503.001.00033	1F7 01 21	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to not enough or no reagent found in the Integral vial {6}.	Start again the aborted job; If the problem persists call Service.
503.001.00034	1F7 01 22	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing Reagent {6}.	Start again the aborted job; If the problem persists call Service.
503.001.00035	1F7 01 23	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to not enough or no reagent found in the Ancillary vial {6}.	Start again the aborted job; If the problem persists call Service.
503.001.00036	1F7 01 24	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing Ancillary {6}.	Start again the aborted job; If the problem persists call Service.
503.001.00037	1F7 01 25	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to starter reagent temperature out of range.	Start again the aborted job; If the problem persists call Service.

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.001.00039	1F7 01 27	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to OLV Integrity Error during dispense.	Start again the aborted job; If the problem persists call Service.
503.001.00040	1F7 01 28	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to Starter Dispense Verification Check.	Start again the aborted job; If the problem persists call Service.
503.001.00041	1F7 01 29	Critical	Aborted Job {0} {1} {2} because calibrator can not be scheduled and its calibration was already started.	Start again the aborted job; If the problem persists call Service.
503.001.00042	1F7 01 2A	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to user request.	Start again the aborted job; If the problem persists call Service.
503.001.00043	1F7 01 2B	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to OLV Integrity Error at Integral Lane {6} Position {7}.	Start again the aborted job; If the problem persists call Service.
503.001.00044	1F7 01 2C	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to OLV Integrity Error at Ancillary Position {6}.	Start again the aborted job; If the problem persists call Service.
503.001.00045	1F7 01 2D	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to DarkCountOutOfRange.	Start again the aborted job; If the problem persists call Service.
503.001.00046	1F7 01 2E	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to DarkCount Average Invalid.	Start again the aborted job; If the problem persists call Service.
503.001.00047	1F7 01 2F	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to DarkCount Average Exceeds Envelop.	Start again the aborted job; If the problem persists call Service.

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.001.00048	1F7 01 30	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} JobId {5} due to too many Spikes detected.	Start again the aborted job; If the problem persists call Service.
503.001.00050	1F7 01 32	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} JobId {5} due to overpressure during Sample Pipetting.	Ensure the tube contains the needed amount of Sample; Check Sample integrity; Load the Sample tube in the appropriate rack; Start again the aborted job; If the problem persists call Service.
503.001.00100	1F7 01 64	Message	Aborted Job {0} {1} {2} {3} for Cycle {4} JobId {5}, due to event {6} {7}	Start again the aborted job; If the problem persists call Service.
503.001.00101	1F7 01 65	Message	Job with JobId {0} should be aborted due to critical event {1} {2} with Flag 'Nothing'.	

Table 49–2: Message list, Process Control for internal-tanks configuration

49.3 Message list for external tanks-configuration

The following message list applies from SW version 1.3.7.

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.000.00000	1F7 00 00	Message	Scheduled Job {0} {1} for Cycle {2} with JobId {3} (Replicate: {4}, Dilution: {5}, Finish {6}, Integral Lane {7}).	
503.000.00001	1F7 00 01	Message	Unscheduled Job {0} {1} {2} {3} for Cycle {4} JobId {5}.	
503.000.00002	1F7 00 02	Message	Started Job {0} {1} {2} {3} for Cycle {4} JobId {5}.	
503.000.00003	1F7 00 03	Message	Started Mitigation {0} for cycle {1} phase {2} JobId {3}.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.000.00004	1F7 00 04	Message	The Start button was pressed	
503.000.00005	1F7 00 05	Message	ScheduleJobs.	
503.000.00006	1F7 00 06	Message	Unschedule Jobs.	
503.000.00007	1F7 00 07	Message	Reschedule jobs.	
503.000.00009	1F7 00 09	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no system liquid.	
503.000.00010	1F7 00 0A	Message	Start Jobs that were created by a rerun rule.	
503.000.00011	1F7 00 0B	Warning	A scheduling error occurred	
503.000.00012	1F7 00 0C	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no wash buffer.	
503.000.00013	1F7 00 0D	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no cuvette.	
503.000.00014	1F7 00 0E	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no disposable tip.	
503.000.00015	1F7 00 0F	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no solid waste.	
503.000.00016	1F7 00 10	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no starter.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.000.00017	1F7 00 11	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no integral.	
503.000.00018	1F7 00 12	Message	Scheduled Starter Prime Job for cycle {0} JobId {1}.	
503.000.00019	1F7 00 13	Warning	No jobs could be scheduled for start.	
503.000.00020	1F7 00 14	Problem	Calibration Data incorrect.	
503.000.00021	1F7 00 15	Warning	A needed Dilution Sequence was not found. Schedule aborted.	
503.000.00022	1F7 00 16	Message	Aspirated {0} for Job {1} {2} in Cycle {3} Phase {4} (Replicate {5}, Dilution {6}, Coordinate {7}, JobId {8}).	
503.000.00023	1F7 00 17	Critical	Aspirated from onboard expired integral.	
503.000.00024	1F7 00 18	Critical	Aspirated from onboard expired ancillary.	
503.000.00025	1F7 00 19	Critical	Aspirated from expired control vial.	
503.000.00026	1F7 00 1A	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), no ancillary.	
503.000.00027	1F7 00 1B	Message	Could not schedule Job {0} {1} (Replicate: {2} Dilution: {3}), reagent {4} depleted.	
503.000.00030	1F7 00 1E	Message	Needed Control {0} not available.	
503.000.00031	1F7 00 1F	Warning	Prime failed.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.000.00040	1F7 00 28	Message	Received Rlu {0} for JobId {1} (With Spikes {2}, Without Spikes {3}, DarkCount {4}, DarkCountEvents {5}, DarkCountAverage {6}, MeasureTime {7})	
503.000.00041	1F7 00 29	Message	Scheduled Job {0} {1} for Cycle {2} with JobId {3} (Replicate: {4}, Dilution: {5}, Finish {6}, Integral Lane {7}, agitation time insufficient).	
503.000.00050	1F7 00 32	Critical	Job {0} {1} {2} {3} for Cycle {4} JobId {5} above Assay Range.	
503.000.00051	1F7 00 33	Critical	Job {0} {1} {2} {3} for Cycle {4} JobId {5} below Assay Range.	
503.000.00052	1F7 00 34	Critical	Job {0} {1} {2} {3} for Cycle {4} JobId {5} above Normal Range.	
503.000.00053	1F7 00 35	Critical	Job {0} {1} {2} {3} for Cycle {4} JobId {5} below Normal Range.	
503.000.00054	1F7 00 36	Critical	Job {0} {1} {2} {3} for Cycle {4} JobId {5} manipulated.	
503.000.00055	1F7 00 37	Critical	Job {0} {1} {2} {3} for Cycle {4} JobId {5} recalculated.	
503.000.00056	1F7 00 38	Critical	Job {0} {1} {2} {3} for Cycle {4} JobId {5} flagged with Control out of Manufacturer Range.	
503.000.00057	1F7 00 39	Critical	Job {0} {1} {2} {3} for Cycle {4} JobId {5} flagged with No Calibration.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.000.00058	1F7 00 3A	Warning	Manipulation aborted because no applicable integral found in database.	
503.000.00060	1F7 00 3C	Critical	Dark Count Average exceeds envelop.	
503.000.00061	1F7 00 3D	Critical	Dark Count was out of Range.	
503.000.00062	1F7 00 3E	Fatal	Three times in a row DarkCount Exceeds Envelop.	
503.000.00063	1F7 00 3F	Fatal	Three times in a row DarkCount out of Range.	
503.000.00064	1F7 00 40	Warning	Spikes Detected for Job {0} {1} {2} {3} for Cycle {4} Jobld {5}.	
503.000.00067	1F7 00 43	Message	Rerun Rule {0} did not create test {1} for {2} because dilution factor {3} is not defined in the assay.	
503.000.00068	1F7 00 44	Message	Rerun Rule {0} did not create test {1} because sample {2} has been unloaded.	
503.000.00069	1F7 00 45	Message	Flagged Result {0} {1} with 'caused rerun'.	
503.000.00070	1F7 00 46	Critical	Result {0} {1} flagged with QC Above Manufacturer Range.	
503.000.00071	1F7 00 47	Critical	Result {0} {1} flagged with QC Below Manufacturer Range.	
503.000.00072	1F7 00 48	Critical	Result {0} {1} flagged with QC Above User Range.	
503.000.00073	1F7 00 49	Critical	Result {0} {1} flagged with QC Below User Range.	
503.000.00074	1F7 00 4A	Critical	Result {0} {1} flagged with Above Assay Range.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.000.00075	1F7 00 4B	Critical	Result {0} {1} flagged with Below Assay Range.	
503.000.00076	1F7 00 4C	Critical	Result {0} {1} flagged with Above Normal Range.	
503.000.00077	1F7 00 4D	Critical	Result {0} {1} flagged with Below Normal Range.	
503.000.00078	1F7 00 4E	Critical	Result {0} {1} flagged with Control out of Manufacturer Range.	
503.000.00079	1F7 00 4F	Critical	Result {0} {1} flagged with Divided By Zero.	
503.000.00081	1F7 00 51	Message	Recalculated Result {0} {1}.	
503.000.00082	1F7 00 52	Message	Could not recalculated Result {0} {1}, undo changes.	
503.000.00090	1F7 00 5A	Critical	Job {0} {1} {2} {3} for Cycle {4} Jobld {5} flagged with Control Expired.	
503.001.00001	1F7 01 01	Critical	Abort Job {0} {1} {2} {3} for Cycle {4} Jobld {5} on SW Start or during Initialization.	Start again the aborted job; If the problems persists call Service.
503.001.00002	1F7 01 02	Warning	An internal error occurred	Call Service.
503.001.00003	1F7 01 03	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to darkcount subtraction.	Start again the aborted job; If the problems persists call Service.
503.001.00004	1F7 01 04	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to unexpected Result Cycle.	Start again the aborted job; If the problems persists call Service.
503.001.00005	1F7 01 05	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to Overdilution.	Start again the aborted job; If the problems persists call Service.

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.001.00006	1F7 01 06	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to not plausible Aspiration.	Start again the aborted job; If the problems persists call Service.
503.001.00007	1F7 01 07	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to starter prime not performed.	Start again the aborted job; If the problems persists call Service.
503.001.00008	1F7 01 08	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to mitigation not performed.	Start again the aborted job; If the problems persists call Service.
503.001.00009	1F7 01 09	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to divided by zero.	Start again the aborted job; If the problems persists call Service.
503.001.00010	1F7 01 0A	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to math error.	Start again the aborted job; If the problem persists call Service.
503.001.00011	1F7 01 0B	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to sent too late.	Start again the aborted job; If the problem persists call Service.
503.001.00020	1F7 01 14	Critical	Aborted Job {0} {1} {2} {3} due to starter lot switch.	Start again the aborted job; If the problem persists call Service.
503.001.00022	1F7 01 16	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to Sample Integrity Error.	Ensure the tube contains the needed amount of Sample; Check Sample integrity; Load the Sample tube in the appropriate rack; Start again the aborted job; If the problem persists call Service.
503.001.00023	1F7 01 17	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to clot detected in the sample probe.	Ensure the tube contains the needed amount of Sample; Check Sample integrity; Load the Sample tube in the appropriate rack; Start again the aborted job; If the problem persists call Service.
503.001.00024	1F7 01 18	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing System Liquid.	Ensure that the System Liquid tank is loaded, recognized and primed; Restart the job; If the problem persists call Service.

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.001.00025	1F7 01 19	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing Cuvette.	Ensure the needed Cuvettes are available and assigned on the system; Start again the aborted job; If the problem persists call Service.
503.001.00026	1F7 01 1A	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing Disposable Tip.	Ensure the needed Tips are available and assigned on the system; Start again the aborted job; If the problem persists call Service.
503.001.00027	1F7 01 1B	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to Tip Pickup Failure.	Check the assignment of the Tips on the SW; Restart the job; If the problem persists call Service.
503.001.00028	1F7 01 1C	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing Starter.	Ensure the Starter bottles are loaded and primed; Check liquid integrity; In case the reagent was set off line reset the LLD; Start again the aborted job; If the problem persists call Service.
503.001.00029	1F7 01 1D	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing Wash Buffer.	Ensure that the Wash Buffer tank is loaded, recognized and primed; Restart the job; If the problem persists call Service.
503.001.00030	1F7 01 1E	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to Washer Aspiration Error at Port {6}.	Start again the aborted job; If the problem persists call Service.
503.001.00031	1F7 01 1F	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to not enough or no liquid found in sample tube.	Start again the aborted job; If the problem persists call Service.
503.001.00032	1F7 01 20	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing Sample.	Start again the aborted job; If the problem persists call Service.
503.001.00033	1F7 01 21	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to not enough or no reagent found in the Integral vial {6}.	Start again the aborted job; If the problem persists call Service.

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.001.00034	1F7 01 22	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing Reagent {6}.	Start again the aborted job; If the problem persists call Service.
503.001.00035	1F7 01 23	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to not enough or no reagent found in the Ancillary vial {6}.	Start again the aborted job; If the problem persists call Service.
503.001.00036	1F7 01 24	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to missing Ancillary {6}.	Start again the aborted job; If the problem persists call Service.
503.001.00037	1F7 01 25	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to starter reagent temperature out of range.	Start again the aborted job; If the problem persists call Service.
503.001.00039	1F7 01 27	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to OLV Integrity Error during dispense.	Start again the aborted job; If the problem persists call Service.
503.001.00040	1F7 01 28	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to Starter Dispense Verification Check.	Start again the aborted job; If the problem persists call Service.
503.001.00041	1F7 01 29	Critical	Aborted Job {0} {1} {2} because calibrator can not be scheduled and its calibration was already started.	Start again the aborted job; If the problem persists call Service.
503.001.00042	1F7 01 2A	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to user request.	Start again the aborted job; If the problem persists call Service.
503.001.00043	1F7 01 2B	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} Jobld {5} due to OLV Integrity Error at Integral Lane {6} Position {7}.	Start again the aborted job; If the problem persists call Service.

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
503.001.00044	1F7 01 2C	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} JobId {5} due to OLV Integrity Error at Ancillary Position {6}.	Start again the aborted job; If the problem persists call Service.
503.001.00045	1F7 01 2D	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} JobId {5} due to DarkCountOutOfRange.	Start again the aborted job; If the problem persists call Service.
503.001.00046	1F7 01 2E	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} JobId {5} due to DarkCount Average Invalid.	Start again the aborted job; If the problem persists call Service.
503.001.00047	1F7 01 2F	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} JobId {5} due to DarkCount Average Exceeds Envelop.	Start again the aborted job; If the problem persists call Service.
503.001.00048	1F7 01 30	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} JobId {5} due to too many Spikes detected.	Start again the aborted job; If the problem persists call Service.
503.001.00050	1F7 01 32	Critical	Aborted Job {0} {1} {2} {3} for Cycle {4} JobId {5} due to overpressure during Sample Pipetting.	Ensure the tube contains the needed amount of Sample; Check Sample integrity; Load the Sample tube in the appropriate rack; Start again the aborted job; If the problem persists call Service.
503.001.00100	1F7 01 64	Message	Aborted Job {0} {1} {2} {3} for Cycle {4} JobId {5}, due to event {6} {7}.	Start again the aborted job; If the problem persists call Service.
503.001.00101	1F7 01 65	Message	Job with JobId {0} should be aborted due to critical event {1} {2} with Flag 'Nothing'.	

Table 49–3: Message list, Process Control for external-tank configuration

50 Infrastructure

Device ID: 0x1F8

50.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	VigilantOnlineFieldGatewayServiceConnected
1	0x01	EventCreationException
2	0x02	DeskShieldLanguageFileNotFoundException
3	0x03	WindowsTimeChanged

Table 50–1: Sub assemblies, Infrastructure

50.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
504.000.00000	1F8 00 00	Message	Connected to FieldGateway Service.	
504.000.00001	1F8 00 01	Warning	Failed to connect to the DoT system: The connection to the FieldGateway Service is failed.	Disable and Enable the DoT module; Restart the PC; If the problem persists call Service.
504.000.00002	1F8 00 02	Message	The system has been disconnected from FieldGateway Service.	
504.000.00003	1F8 00 03	Message	Connected to DoT system.	
504.000.00004	1F8 00 04	Message	The system has been disconnected from DoT system.	
504.000.00005	1F8 00 05	Message	DoT asked for the files: {0}.	
504.000.00006	1F8 00 06	Message	DoT asked for the files: {0} from {1} to {2}.	
504.000.00007	1F8 00 07	Message	Uploaded file {0} to DoT.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
504.000.00008	1F8 00 08	Message	Connected to DoT FastTrack service.	
504.000.00009	1F8 00 09	Message	The connection to the DoT system is temporarily interrupted.	If the problem persists, call Service.
504.000.00010	1F8 00 0A	Warning	Failed to connect to the DoT system: the authentication failed.	If the problem persists, call Service.
504.000.00011	1F8 00 0B	Warning	Failed to connect to the DoT system: an interfacing error occurred.	If the problem persists, call Service.
504.000.00012	1F8 00 0C	Warning	Failed to connect to the DoT system: connection was denied by the DoT provisioning service.	If the problem persists, call Service.
504.000.00013	1F8 00 0D	Warning	The internet connection to DoT is currently interrupted.	Disconnect and reconnect the instrument to DoT from the System Setting IoT dedicated page; If the problem persists, call Service.
504.000.00014	1F8 00 0E	Warning	Failed to connect to DoT system: an error occurred at the DoT Hub.	If the problem persists, call Service.
504.000.00015	1F8 00 0F	Warning	Failed to connect to the DoT FastTrack service.	If the problem persists, call Service.
504.000.00016	1F8 00 10	Warning	An error occurred during a file transfer from or to Xsense.	
504.000.00017	1F8 00 11	Warning	An error occurred during the connection to DoT.	Try to connect to DoT again; If the problem persists, call Service.
504.000.00018	1F8 00 12	Warning	Failed to connect to DoT system: The connection to the FieldGateway Service is interrupted.	If the problem persists, call Service.
504.000.00019	1F8 00 13	Message	The connection to the DoT is available.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
504.000.00020	1F8 00 14	Message	The connection to the DoT is available. Certificate Serial Number: {0}; Certificate Thumbprint: {1}.	If the problem persists, call Service.
504.000.00021	1F8 00 15	Message	The system has been disconnected from DoT Fast Track.	
504.000.00022	1F8 00 16	Message	The connection to the DoT Fast Track is temporarily interrupted.	
504.000.00030	1F8 00 1E	Message	Registration to DoT succeeded.	
504.000.00031	1F8 00 1F	Warning	Registration to DT failed.	
504.001.00000	1F8 01 00	Warning	An error occurred creating Event Log Entry for {0}.	
504.002.00000	1F8 02 00	Warning	The system could not find the desktop shield language file.	
504.002.00001	1F8 02 01	Warning	The system could not update the desktop shield language file.	
504.003.00000	1F8 03 00	Warning	The Windows Time was changed while LiaisonXS was running.	
504.004.00000	1F8 04 00	Message	Old unneeded trace files deleted.	
504.004.00001	1F8 04 01	Message	Trace files up to {0} deleted.	

Table 50–2: Message list, Infrastructure

51 Ancillary Loading

Device ID: 0x1F9

51.1 Sub assemblies

ID (dec)	ID (hex)	Name
2	0x02	AncillaryShelfExpired

Table 51–1: Sub assemblies, Ancillary Loading

51.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
505.002.00000	1F9 02 00	Warning	Ancillary {0} in position {1} is shelf expired.	
505.002.00001	1F9 02 01	Warning	Ancillary {0} in position {1} is onboard expired.	
505.002.00002	1F9 02 02	Problem	Ancillary in position {0} with Article Number '{1}', Vial Number '{2}' and Lot Number '{3}' is already loaded in another position.	
505.002.00003	1F9 02 03	Warning	Ancillary in position {0} has wrong identification: '{1}', expected: '{2}'	
505.002.00004	1F9 02 04	Warning	Ancillary in position {0} has wrong CRC checksum.	
505.002.00005	1F9 02 05	Warning	Ancillary in position {0} has wrong RFID data block size.	
505.002.00006	1F9 02 06	Warning	Ancillary {0} in position {1} disabled.	
505.002.00007	1F9 02 07	Warning	Ancillary {0} in position {1} enabled.	

51. Ancillary Loading

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
505.002.00010	1F9 02 0A	Message	Ancillary {0} in position {1} inserted.	
505.002.00011	1F9 02 0B	Message	Ancillary {0} from position {1} removed.	
505.002.00012	1F9 02 0C	Warning	Needed Ancillary {0} from position {1} removed.	
505.002.00013	1F9 02 0D	Warning	Onboard stability expired for ancillary {0} in position {1}.	
505.002.00014	1F9 02 0E	Warning	Shelf expired for ancillary {0} in position {1}.	
505.002.00015	1F9 02 0F	Warning	Ancillary {0} in position {1} taken offline.	
505.002.00016	1F9 02 10	Message	Ancillary rack released.	
505.002.00017	1F9 02 11	Warning	Ancillary rack removed without release.	
505.002.00018	1F9 02 12	Message	Needed Ancillary {0} at position {1} inserted.	
505.002.00019	1F9 02 13	Warning	Needed Ancillary {0} from position {1} removed and not reinserted.	

Table 51–2: Message list, Ancillary Loading

52 Consumables

Device ID: 0x1F9

52.1 Sub assemblies

ID (dec)	ID (hex)	Name
4	0x04	DrawerWithdrawRequest

Table 52–1: Sub assemblies, Consumables

52.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
505.004.00000	1F9 04 00	Message	Consumables Drawer was requested to be removed	
505.004.00001	1F9 04 01	Critical	Consumables Drawer was removed while In Use.	
505.004.00002	1F9 04 02	Warning	Not enough cuvettes available for priming.	
505.004.00003	1F9 04 03	Warning	Warn Level for Cuvettes has been reached	
505.004.00004	1F9 04 04	Message	Warn Level for Cuvettes has been left	
505.004.00005	1F9 04 05	Warning	Warn Level for Disposable Tips has been reached	
505.004.00006	1F9 04 06	Message	Warn Level for Disposable Tips has been left	
505.004.00098	1F9 04 62	Message	Consumables Drawer pushed in.	
505.004.00099	1F9 04 63	Message	Consumables Drawer pulled out.	

Table 52–2: Message list, Consumables

53 Liquid Containers

Device ID: 0x1F9

53.1 Sub assemblies

ID (dec)	ID (hex)	Name
6	0x06	WashBufferRemovedWhileIn Use

Table 53–1: Sub assemblies, Liquid Containers

53.2 Message list for internal-tanks configuration

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
505.006.00000	1F9 06 00	Fatal	Wash buffer was removed while in use.	
505.006.00001	1F9 06 01	Warning	Not enough wash buffer available for priming.	Refill the Wash Buffer.
505.006.00002	1F9 06 02	Warning	Not enough DI water available for priming.	Refill the DI water.
505.006.00003	1F9 06 03	Warning	Not enough DI water available for self diagnostics.	Refill the DI water.

Table 53–2: Message list, Liquid Containers for internal-tanks configuration

53.3 Message list for external-tanks configuration

The following message list applies from SW version 1.3.7.

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
505.006.00000	1F9 06 00	Fatal	Wash buffer was removed while in use.	
505.006.00001	1F9 06 01	Warning	Not enough wash buffer available for priming.	Refill the Wash Buffer.

53. Liquid Containers

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
505.006.00002	1F9 06 02	Warning	Not enough system liquid available for priming.	Refill the System Liquid.
505.006.00003	1F9 06 03	Warning	Not enough System Liquid available for self diagnostics.	Refill the System Liquid.
505.006.00004	1F9 06 08	Fatal	System Liquid available was removed while in use.	

Table 53–3: Message list, Liquid Containers, for external-tanks configuration

54 Reagent Loading

Device ID: 0x1F9

54.1 Sub assemblies

ID (dec)	ID (hex)	Name
1	0x01	IntegralShelfExpired

Table 54–1: Sub assemblies, Reagent Loading

54.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
505.001.00000	1F9 01 00	Message	Integral in lane {0} for assay {1} is shelf expired.	
505.001.00001	1F9 01 01	Message	Integral in lane {0} for assay {1} is onboard expired.	
505.001.00002	1F9 01 02	Problem	Integral in lane {0} with Article Number '{1}', Kit Number '{2}' and Lot Number '{3}' is already loaded in the instrument.	
505.001.00003	1F9 01 03	Message	Integral in lane {0} has wrong identification: '{1}', expected: '{2}'	
505.001.00004	1F9 01 04	Warning	Integral in lane {0} has wrong CRC checksum.	
505.001.00005	1F9 01 05	Warning	Integral in lane {0} has wrong RFID data block size.	
505.001.00006	1F9 01 06	Warning	Integral in lane {0} set offline as LLD is 0 while remaining volume is >0.	
505.001.00010	1F9 01 0A	Message	Integral {0} in lane {1} inserted.	

54. Reagent Loading

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
505.001.00011	1F9 01 0B	Message	Integral {0} from lane {1} removed.	
505.001.00012	1F9 01 0C	Warning	Onboard stability expired for integral {0} in lane {1}.	
505.001.00013	1F9 01 0D	Message	Integral {0} from lane {1} removed while in use.	
505.001.00020	1F9 01 14	Warning	Integral in lane {0} has wrong calibration block CRC checksum.	
505.001.00021	1F9 01 15	Warning	Integral in lane {0} has wrong combi block CRC checksum.	
505.001.00022	1F9 01 16	Warning	Assay {0} has Version {1}, but integrals needs {2}.	
505.001.00023	1F9 01 17	Warning	Integral {0} in lane {1} taken offline (error in position {2}).	
505.001.00024	1F9 01 18	Warning	Integral {0} in lane {1} disabled.	
505.001.00025	1F9 01 19	Warning	Integral {0} in lane {1} enabled.	
505.001.00026	1F9 01 1A	Warning	No Assay loaded for the Integral in lane {0} with the Article Number {1}.	
505.001.00030	1F9 01 1E	Message	Reset Liquid Levels on Integral {0} in lane {1}. Please unload and reload that Integral	
505.001.00031	1F9 01 1F	Message	Refilled Integral {0} in lane {1}. Please unload and reload that Integral	
505.001.00032	1F9 01 20	Message	Rack file '{0}' not found.	
505.001.00098	1F9 01 62	Message	Reagent Flap closed.	
505.001.00099	1F9 01 63	Message	Reagent Flap opened.	

Table 54–2: Message list, Reagent Loading

55 Sample Loading

Device ID: 0x1F9

55.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	SampleRackInserted

Table 55–1: Sub assemblies, Sample Loading

55.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
505.000.00000	1F9 00 00	Message	Sample rack in lane {0} inserted.	
505.000.00001	1F9 00 01	Message	Sample rack from lane {0} removed.	
505.000.00002	1F9 00 02	Message	Sample rack from lane {0} removed while in use.	
505.000.00003	1F9 00 03	Warning	Sample {0} in lane {1} taken offline.	
505.000.00004	1F9 00 04	Warning	Invalid barcode found in lane {0} position {1}: '{2}'.	
505.000.00005	1F9 00 05	Warning	Duplicate barcode found in lane {0} position {1}: '{2}'. The same sample is already loaded in lane {3} position {4}.	
505.000.00006	1F9 00 06	Problem	Invalid rack inserted into lane {0}.	
505.000.00007	1F9 00 07	Message	RackID '{0}' manually entered.	
505.000.00008	1F9 00 08	Message	Rack file '{0}' not found.	

55. Sample Loading

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
505.000.00009	1F9 00 09	Problem	Last read position barcode does not match the position defined in the RAC file in lane {0}.	
505.000.00010	1F9 00 0A	Problem	The Barcode {0} was denied because Barcode {1} is known in the database. Case conflict	
505.000.00011	1F9 00 0B	Warning	An aspiration error occurred for sample {0} in lane {1}.	
505.000.00012	1F9 00 0C	Problem	At least one position of the barcode could not be read in lane {0}.	
505.000.00098	1F9 00 62	Message	Sample Flap closed.	
505.000.00099	1F9 00 63	Message	Sample Flap opened.	

Table 55–2: Message list, Sample Loading

56 Starter Loading

Device ID: 0x1F9

56.1 Sub assemblies

ID (dec)	ID (hex)	Name
3	0x03	StarterRFIDDataBlockError

Table 56–1: Sub assemblies, Starter Loading

56.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
505.003.00000	1F9 03 00	Warning	Starter {0} has wrong RFID data block size.	
505.003.00001	1F9 03 01	Warning	Starter {0} has wrong identification: '{1}', expected: '{2}'.	
505.003.00002	1F9 03 02	Warning	Starter {0} has wrong CRC checksum.	
505.003.00003	1F9 03 03	Message	Starter {0} inserted.	
505.003.00004	1F9 03 04	Message	Starter from position {0} removed.	
505.003.00005	1F9 03 05	Warning	Starter in position {0} shelf expired.	
505.003.00006	1F9 03 06	Message	Starter in use from position {0} removed.	
505.003.00007	1F9 03 07	Warning	Starter {0} lot has changed and will invalidate all calibrations on prime.	
505.003.00008	1F9 03 08	Warning	Starter {0} lot switched from {1} to {2}.	
505.003.00009	1F9 03 09	Warning	Starter {0} taken offline.	
505.003.00010	1F9 03 0A	Warning	Not enough Starter {0} available for prime.	

56. Starter Loading

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
505.003.00098	1F9 03 62	Message	Starter Flap closed.	
505.003.00099	1F9 03 63	Message	Starter Flap opened.	

Table 56–2: Message list, Starter Loading

57 Top Cover

Device ID: 0x1F9

57.1 Sub assemblies

ID (dec)	ID (hex)	Name
7	0x07	TopCoverOpenForPriming

Table 57–1: Sub assemblies, Top Cover

57.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
505.007.00000	1F9 07 00	Warning	Prime not possible because top cover is open.	Close the top cover.
505.007.00098	1F9 07 62	Message	Top Cover closed.	
505.007.00099	1F9 07 63	Message	Top Cover opened.	

Table 57–2: Message list, Top Cover

58 Waste

Device ID: 0x1F9

58.1 Sub assemblies

ID (dec)	ID (hex)	Name
5	0x05	SolidWasteEmptied

Table 58–1: Sub assemblies, Waste

58.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
505.005.00000	1F9 05 00	Message	Solid waste was emptied.	
505.005.00001	1F9 05 01	Critical	Solid waste is full.	
505.005.00002	1F9 05 02	Message	Solid waste was removed, current level: {0}.	
505.005.00003	1F9 05 03	Message	Solid waste was inserted, current level: {0}.	
505.005.00004	1F9 05 04	Warning	Solid waste is nearly full.	
505.005.00005	1F9 05 05	Warning	Not enough solid waste available for priming.	
505.005.00006	1F9 05 06	Warning	Not enough liquid waste available for priming.	
505.005.00007	1F9 05 07	Warning	Not enough liquid waste available for self diagnostics.	
505.005.00008	1F9 05 08	Warning	Prime not possible because solid waste is not loaded.	

Table 58–2: Message list, Waste

59 Maintenance

Device ID: 0x1FA

59.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	ImportSuccessful

Table 59–1: Sub assemblies, Maintenance

59.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
506.000.00003	1FA 00 03	Message	Maintenance import finished successfully.	
506.000.00004	1FA 00 04	Problem	An error occurred while importing the maintenance file	
506.000.00007	1FA 00 07	Message	{0} maintenance task completed successfully.	
506.000.00011	1FA 00 0B	Message	Maintenance task {0} started.	
506.000.00012	1FA 00 0C	Problem	Insufficient {0} for starting this maintenance task.	
506.000.00013	1FA 00 0D	Problem	Maintenance task cannot be started. Starter temperature is out of range.	
506.000.00014	1FA 00 0E	Problem	Maintenance task cannot be started. Incubator temperature is out of range.	
506.000.00017	1FA 00 11	Problem	This maintenance task cannot be started because the instrument is not in the correct state	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
506.000.00018	1FA 00 12	Problem	There is no maintenance task running to abort.	
506.000.00019	1FA 00 13	Problem	{0} maintenance task aborted manually.	
506.000.00020	1FA 00 14	Problem	{0} maintenance task failed.	
506.000.00022	1FA 00 16	Problem	A maintenance task already exists with the same name. Import failed.	
506.000.00024	1FA 00 18	Problem	A maintenance step is not defined or is not available.	
506.000.00025	1FA 00 19	Problem	A maintenance step parameter was not found.	
506.000.00026	1FA 00 1A	Problem	All the parameters of a maintenance task are not correctly defined.	
506.000.00027	1FA 00 1B	Problem	Error in maintenance task definition.	
506.000.00029	1FA 00 1D	Problem	Maintenance import files are not valid xml files.	
506.000.00030	1FA 00 1E	Problem	Maintenance import file version incompatible with current Software.	
506.000.00031	1FA 00 1F	Problem	Maintenance import file has no XML schema version specified.	
506.000.00033	1FA 00 21	Problem	Maintenance procedure not available for maintenance task. Import failed.	
506.000.00034	1FA 00 22	Problem	Maintenance Task not startable because instrument in {0} state.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
506.000.00040	1FA 00 28	Problem	Light check left is not loaded.	
506.000.00041	1FA 00 29	Problem	Light check right is not loaded.	

Table 59–2: Message list, Maintenance

60 UI

Device ID: 0x1FB

60.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	ImportingAssayFile

Table 60–1: Sub assemblies, UI

60.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
507.000.00000	1FB 00 00	Message	Importing Assay File {0}.	
507.000.00001	1FB 00 01	Message	Importing Assay with Abbreviation {0}.	
507.000.00002	1FB 00 02	Message	Overwriting Assay with Abbreviation {0} version {1} with version {2}.	
507.000.00003	1FB 00 03	Message	Importing assay group file '{0}'.	
507.000.00004	1FB 00 04	Message	Importing assay profile file '{0}'.	
507.000.00005	1FB 00 05	Message	Exporting selected assay group(s) to file '{0}'.	
507.000.00006	1FB 00 06	Message	Exporting selected assay profile(s) to file '{0}'.	
507.000.00007	1FB 00 07	Message	Importing dilution sequences file '{0}'.	
507.000.00008	1FB 00 08	Message	Importing rerun rules file '{0}'.	
507.000.00009	1FB 00 09	Message	Exporting rerun rules file '{0}'.	
507.000.00010	1FB 00 0A	Message	Importing control definitions file '{0}'.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
507.000.00011	1FB 00 0B	Message	Exporting control definitions file '{0}'.	
507.000.00012	1FB 00 0C	Message	Assay '{0}' with article number '{1}', name '{2}' and LIS alias '{3}' deleted.	
507.000.00013	1FB 00 0D	Message	Importing scanned control definition '{0}', lot number '{1}', barcode '{2}'.	
507.000.00014	1FB 00 0E	Message	Importing scanned control definition(s) failed.	
507.000.00015	1FB 00 0F	Message	Importing control definitions file '{0}' failed.	
507.000.00020	1FB 00 14	Message	Resetting User Unit, Normal Range and Qualitative Labels for Assay {0}.	
507.000.00021	1FB 00 15	Message	Overwriting Assay Master Curve Unit from {0} to {1}.	
507.000.00022	1FB 00 16	Message	Overwriting Assay Sample Replicates Manufacturer from {0} to {1}.	
507.000.00023	1FB 00 17	Message	Overwriting Assay Control Replicates Manufacturer from {0} to {1}.	
507.000.00024	1FB 00 18	Message	Overwriting Assay Assay Range from {0} to {1}.	
507.000.00025	1FB 00 19	Message	Overwriting Assay Share Calibrations within Kit Lot Manufacturer from {0} to {1}.	
507.000.00030	1FB 00 1E	Message	The Assay {0} is missing for the Control Analyte Value.	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
507.000.00031	1FB 00 1F	Message	The Control Definition {0} with Lot {1} was not imported as no matching assays was found in the database	
507.000.00032	1FB 00 20	Message	Overwrite Control Definition {0} with Lot {1}.	
507.000.00033	1FB 00 21	Message	Importing generic sequences file '{0}'.	
507.000.00040	1FB 00 28	Message	Access rights imported.	

Table 60–2: Message list, UI

61 Reporting

Device ID: 0x1FC

61.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	PrintError

Table 61–1: Sub assemblies, Reporting

61.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
508.000.00001	1FC 00 01	Problem	An error occurred while trying to print the report.	
508.000.00002	1FC 00 02	Problem	The printer settings are not valid.	
508.000.00003	1FC 00 03	Message	The printer status has changed to online.	
508.000.00004	1FC 00 04	Message	The printer status has changed to offline.	

Table 61–2: Message list, Reporting

62 LIS

Device ID: 0x1FD

62.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	NoResponseToQuery

Table 62–1: Sub assemblies, LIS

62.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
509.000.00001	1FD 00 01	Message	No response to query for workorder for SID {0} within {1} seconds.	
509.000.00002	1FD 00 02	Warning	Structure error in received LIS message.	
509.000.00004	1FD 00 04	Warning	Error while transmitting data to LIS.	
509.000.00006	1FD 00 06	Message	Liaison XS LIS module running.	
509.000.00007	1FD 00 07	Message	Liaison XS LIS module stopped.	
509.000.00008	1FD 00 08	Warning	Liaison XS LIS module could not be started.	
509.000.00009	1FD 00 09	Warning	Liaison XS LIS module could not be stopped.	
509.000.00010	1FD 00 0A	Warning	Liaison XS LIS module received an invalid message (unsupported message type or structure error).	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
509.000.00011	1FD 00 0B	Warning	Liaison XS LIS module received an order message for a not supported assay.	
509.000.00012	1FD 00 0C	Warning	Liaison XS LIS module received an order message with an invalid sample id.	
509.000.00013	1FD 00 0D	Warning	The sample identifier received by the Liaison XS LIS module is a control identifier.	
509.000.00014	1FD 00 0E	Warning	The sample identifier received by the Liaison XS LIS module is a calibrator identifier.	
509.000.00015	1FD 00 0F	Warning	Liaison XS LIS module received an order message with an invalid dilution.	
509.000.00016	1FD 00 10	Message	Liaison XS LIS module received an order message.	

Table 62–2: Message list, LIS

63 ICoL

Device ID: 0x1FE

63.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	Unknown Message

Table 63–1: Sub assemblies, ICoL

63.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
510.000.00000	1FE 00 00	Problem	A communication issue between PC and instrument e occurred	
510.000.00001	1FE 00 01	Warning	A communication issue between PC and instrument e occurred	
510.000.00002	1FE 00 02	Critical	A communication issue between PC and instrument is occurred	
510.000.00003	1FE 00 03	Fatal	A communication issue between PC and instrument e occurred	
510.000.00004	1FE 00 04	Problem	A communication issue between PC and instrument e occurred	
510.000.00005	1FE 00 05	Problem	A communication issue between PC and instrument e occurred	
510.000.00006	1FE 00 06	Problem	A communication issue between PC and instrument e occurred	

Table 63–2: Message list, ICoL

64 TDef

Device ID: 0x1FF

64.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	ProcessInputDescriptionException

Table 64–1: Sub assemblies, TDef

64.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
511.000.00001	1FF 00 01	Problem	A communication issue between PC and instrument e occurred	
511.000.00002	1FF 00 02	Problem	It was not possible to correctly handle the Assay protocol	
511.000.00003	1FF 00 03	Problem	It was not possible to correctly handle the Assay protocol	
511.000.00004	1FF 00 04	Problem	It was not possible to correctly handle the Assay protocol	
511.000.00005	1FF 00 05	Problem	It was not possible to correctly handle the Assay protocol	
511.000.00006	1FF 00 06	Problem	It was not possible to correctly handle the Assay protocol	
511.000.00010	1FF 00 0A	Problem	It was not possible to correctly handle the Assay protocol	
511.000.00011	1FF 00 0B	Problem	It was not possible to correctly handle the Assay protocol	
511.000.00025	1FF 00 19	Problem	It was not possible to correctly handle the Assay protocol	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
511.000.00026	1FF 00 1A	Problem	It was not possible to correctly handle the Assay protocol	
511.000.00027	1FF 00 1B	Problem	It was not possible to correctly handle the Assay protocol	
511.000.00033	1FF 00 21	Problem	It was not possible to correctly handle the Assay Protocol	
511.000.00034	1FF 00 22	Problem	It was not possible to correctly handle the Assay protocol	
511.000.00035	1FF 00 23	Problem	It was not possible to correctly handle the Assay protocol	
511.000.00037	1FF 00 25	Warning	It was not possible to correctly handle the Assay protocol	
511.000.00041	1FF 00 29	Problem	It was not possible to correctly handle the Assay protocol	

Table 64–2: Message list, TDef

65 Data Reduction

Device ID: 0x200

65.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	CalibrationSetToInvalidDueToKulovic

Table 65–1: Sub assemblies, Data Reduction

65.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
512.000.00000	200 00 00	Problem	Calibration {0} set invalid due to internal error	
512.000.00001	200 00 01	Problem	Invalidated Calibration {0} due to max RLU CV.	
512.000.00002	200 00 02	Problem	Invalidated Calibration {0} due to RLU Tolerances.	
512.000.00003	200 00 03	Problem	Invalidated Calibration {0} due to Geometric Curve Check.	
512.000.00004	200 00 04	Problem	Invalidated Calibration {0} due to deleted calibrator result.	
512.000.00020	200 00 14	Problem	Set Calibration {0} to NotUsed due to Calibration {1}.	
512.000.00021	200 00 15	Problem	Set Calibration {0} to NotUsedLot due to Starter Lot Switch.	
512.000.00022	200 00 16	Message	Set Calibration {0} to Expired.	
512.000.00023	200 00 17	Message	Set Calibration {0} to Disabled.	

Table 65–2: Message list, Data Reduction

66 Auto Backup

Device ID: 0x201

66.1 Sub assemblies

ID (dec)	ID (hex)	Name
1	0x01	SecondPathDoesNotExist

Table 66–1: Sub assemblies, Auto Backup

66.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
513.001.00001	201 01 01	Problem	Could not copy autobackup file {0}. Target directory {1} does not exist.	
513.001.00002	201 01 02	Problem	Could not copy autobackup file {0}.	
513.001.00003	201 01 03	Problem	Autobackup operation failed.	

Table 66–2: Message list, Auto Backup

67 Backup

Device ID: 0x201

67.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	DefinitionFileNotFound

Table 67–1: Sub assemblies, Backup

67.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
513.000.00001	201 00 01	Problem	The configuration file {0} containing the backup definitions was not found.	
513.000.00002	201 00 02	Problem	The backup operation failed.	
513.000.00003	201 00 03	Problem	The backup type {0} is not found in the backup definitions file.	
513.000.00004	201 00 04	Problem	The backup of the database {0} failed.	
513.000.00005	201 00 05	Problem	Access denied to the directory {0}	
513.000.00006	201 00 06	Problem	Could not create backup of database {0}.	
513.000.00007	201 00 07	Problem	Could not copy backup file(s). Error: {0}.	
513.000.00008	201 00 08	Problem	Could not copy backup file(s). The system doesn't have permissions to write in the destination directory {0}	

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
513.000.00009	201 00 09	Problem	There is not enough space on {0} to copy the file(s).	
513.000.00010	201 00 0A	Problem	Could not delete the backup file(s). Error: {0}.	
515.000.00001	203 00 01	Problem	A Low Level Message was sent.	

Table 67–2: Message list, Backup

68 QC Stat

Device ID: 0x202

68.1 Sub assemblies

ID (dec)	ID (hex)	Name
0	0x00	RulePassed

Table 68–1: Sub assemblies, QCStat

68.2 Message list

Event ID (dec)	Event ID (hex)	Severity	Message text	Corrective action
514.000.00001	202 00 01	Message	Quality Control {0} for analyzer '{1}', assay '{2}', control name '{3}' at {4}. The rule type was '{5}' with the comment: '{6}'.	
514.000.00002	202 00 02	Problem	Quality Control {0} for analyzer '{1}', assay '{2}', control name '{3}' at {4}. The rule type was '{5}' with the comment: '{6}'.	
514.000.00003	202 00 03	Message	Quality Control gets informed for analyzer '{0}', assay '{1}', control name '{2}'.	
514.000.00004	202 00 04	Critical	Communication error with Quality Control occurred. Please check if all controls are in range.	

Table 68–2: Message list, QCStat

69 Command Codes for Service Manual

Command code is shown in hexadecimal code.

69.1 Command Codes COP

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
1Ah	Reset	Command resets CAN-Kernel
6Dh	Num Links	Command returning total number of defined linkages between two devices on different CAN buses.
6Eh	Num Nodes	Command returning total number of defined CAN devices.
6Fh	Num Connections	Command returning total number of defined CAN connections.
72h	Get Connection Info	Command returning information about connection of two CAN devices.

ID. Hex code	Name	Description
73h	Get Node Info	Command returning address information about defined CAN device.
74h	Get Node Id	Command returning identifier for defined CAN device.
75h	Get Node Mode	Command returning current operation mode of defined CAN device.
76h	Define Connection	Command to define connection between two CAN devices.
77h	Define Routing	Command to define routing information for CAN device connected to master.
78h	Set Headless Com	Command to enable port of master for headless communication by omitting DevCom device address while communication.
79h	Establish	Command to complete and activate defined CAN configuration.
7Ah	Define Unused Port	Command to define a port of CAN device as unused.
7Bh	Define Link	Command to define linkage between two devices connected to different CAN buses
7Ch	Get Link Info	Command returning information about defined linkage between two devices on different CAN buses.
7Dh	Set Pn P Login	Command to define CAN device as plug-and-play device.

Table 69–1: Command Code List: COP

69.2 Command Codes PwrCAN distribution

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
08h	Set Actor	Command sets the selected output.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
19h	Get Max Power	Command returns the maximum power consumption of the module.
1Ah	Reset	Command resets the CU and restarts with the CAN-Kernel.

Table 69–2: Command Code List: PwrCAN distribution

69.3 Command Codes User Access Panel

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
08h	Set Actuator	Command sets the selected output.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM
0Eh	Get Version	Command returns the version of the application firmware.
12h	Get Parameter Type	Command reads information about the selected parameter
13h	Get Parameter	Command reads a formatted parameter from module RAM.

ID. Hex code	Name	Description
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
19h	Get Max Power	Command returns the maximum power consumption of the module.
1Ah	Reset	Command resets CAN-Kernel.
20h	Self Diagnostic	Command starts the self diagnostic tests
21h	Enable Buck Converter	Command enables the 24V to 12V buck converter.
22h	Get PC State	Command returns the state of the panel PC.
B0h	Service Memory Read	Service Memory read selection.
B1h	Service Memory Reset Sd	Service Memory reset SD selection -> protected.
B2h	Service Memory Reset All Errors	Service Memory reset all errors -> protected.

Table 69–3: Command Code List: User access panel

69.4 Command Codes Incubator

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
03h	Move	Command moves the drive to its target position.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
08h	Set Actuator	Command sets the selected output.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
11h	Move Relative	Command moves the drive relative to its current position.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
17h	Define Ramp	Command sets drive parameter of a selected drive.
18h	Stop	Command stops the motor.
19h	Get Max Power	Command returns the maximum power consumption of the module.
1Ah	Reset	Command resets CAN-Kernel.

ID. Hex code	Name	Description
1Bh	Get Position	Command returns the current position of a selected drive.
1Fh	Get Command Format	Command returns the format of a selected command.
20h	Hardware Init	Initialize the hardware assemblies.
30h	Self Diagnostic	Starts the self diagnostic tests.
40h	Move Position	Move cuvette to position.
41h	Check Cuvette	Move cuvette to sensor position and check presence.
42h	Calibrate Sensor	Calibrate Cuvette Sensor.
43h	Un Calibrate Init Pos	Delete saved Microstep Position of Init.
44h	Calibrate Init Pos	Init and save Microstep Position of Init.
45h	Get Cuvette State	Get presence of all cuvettes.
50h	Heater Start	Start temperature control.
51h	Heater Stop	Stop temperature control.
B0h	Service Memory Read	Service Memory read selection.
B1h	Service Memory Reset Sd	Service Memory reset SD selection -> protected.
B2h	Service Memory Reset All Errors	Service Memory reset all errors -> protected.

Table 69–4: Command Code List: Incubator

69.5 Command Codes Washer

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
03h	Move	Command moves the drive to its target position.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
08h	Set Actor	Command sets the selected output.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
11h	Move Relative	Command moves the drive relative to its current position.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
17h	Define Ramp	Command sets drive parameter of a selected drive.
18h	Stop	Command stops a selected drive.
19h	Get Max Power	Command returns the maximum power consumption of the module.
1Ah	Reset	Command resets CAN-Kernel.

ID. Hex code	Name	Description
1Bh	Get Position	Command returns the current position of a selected drive.
1Ch	Get Process Data	Command sends the process data of a selected sensor.
20h	Move Lift To Logical Position	Command moves the wash lift to the position index.
25h	Move Ring One Position	Command moves the wash ring one position.
26h	Auto Teach Lift Bottom	Command starts the auto teach routine for the bottom detection.
2Fh	Self Diagnostic	Command starts the self diagnostic tests.
30h	Define Actual Position Only For Internal Use	Internal command for exception messages. External request not possible!
31h	Start Running For Time Only For Internal Use	Internal command for exception messages. External request not possible!
32h	Move To Position Only For Internal Use	Internal command for exception messages. External request not possible!
33h	Start Running Only For Internal Use	Internal command for exception messages. External request not possible!
34h	Get Motor Position Only For Internal Use	Internal command for exception messages. External request not possible!
40h	Calibrate Bubble Sensor	Command calibrates the aspiration sensors.
41h	Start Bubble Measure	Command starts measure bubble volume check.
42h	Stop Bubble Measure	Command stops measure bubble volume check and returns result.
50h	Service Memory Read	Service Memory read selection.
51h	Service Memory Reset Sd	Service Memory reset SD selection -> protected.
52h	Service Memory Reset All Errors	Service Memory reset all errors -> protected.

Table 69–5: Command Code List: Wash module

69.6 Command Codes Reader

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
03h	Move	Command moves the drive to its target position.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
08h	Set Actor	Command sets the selected output.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
11h	Move Relative	Command moves the drive relative to its current position.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
19h	Get Max Power	Command returns the maximum power consumption of the module.
1Ah	Reset	Command resets CAN-Kernel.
1Bh	Get Position	Command returns the current position of a selected drive.

ID. Hex code	Name	Description
1Ch	Get Process Data	Command sends the process data of a selected sensor.
20h	Self Diagnostic	Command starts the self diagnostic tests.
25h	Move Rotor To Position	Command moves the rotor to target position.
26h	Move Lift To Position	Command moves the lift to target position.
27h	Start Dark Count Measurement	Command starts one dark count measurement.
28h	Start RLU Measurement	Command starts one complete measurement.
29h	Get Flash Data	Command transmits all data.
2Bh	High Voltage On	Command switches the high voltage of the PMT on.
2Ch	High Voltage Off	Command switches the high voltage of the PMT off.
2Dh	Set Rotor Stop Current	Command sets new rotor stop current.
2Eh	Prepare For Transfer	Command prepares rotor for transfer.
2Fh	Transfer Completed	Command resets transfer modifications for rotor.
30h	Calibrate Bubble Sensor	Command calibrates the aspiration sensors.
31h	Start Bubble Measure	Command starts measure bubble volume check.
32h	Stop Bubble Measure	Command stops measure bubble volume check and returns result.
40h	Auto Teach Lift Bottom	Command starts the auto teach routine for the bottom detection.
50h	Fan On	Command switches the fan on.
51h	Fan Off	Command switches the fan off.
55h	Move Rotor To Position Backwards	Command moves the rotor backwards to target position.
80h	Test Pin Set Mode	Set mode of Test Pin function.
81h	Test Pin Get Mode	Get mode of Test Pin function.
B0h	Service Memory Read	Service Memory read selection.
B1h	Service Memory Reset Sd	Service Memory reset SD selection -> protected.
B2h	Service Memory Reset All Errors	Service Memory reset all errors -> protected.

Table 69–6: Command Code List: Reader

69.7 Command Codes Loading Bay

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the selected subassembly.
11h	Move Relative	Command moves the drive relative to its current position.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
19h	Get Max Power	Command returns the maximum power consumption of the module.
1Ah	Reset	Command resets CAN-Kernel.
22h	Scanner Send Receive	Provides direct communication with the scanner.
23h	Scanner Custom Default	Sets the scanner configuration to custom default values.
2Fh	Self Diagnostic	Command starts the self diagnostic tests.
40h	Loading Activate	Activates the loading process.

ID. Hex code	Name	Description
41h	Loading Deactivate	Deactivates the loading process.
4Ah	Cooling Activate	Activates the cooling control.
48h	Cooling Deactivate	Deactivates the cooling control.
50h	ISO 1569 3_Inventory	Command returns the VICC Unique Identifiers (UID), found in the reader field.
51h	ISO 1569 3_Read Blocks	Command reads the specified number of blocks from VICC.
52h	ISO 1569 3_Write Blocks	Command writes the specified number of blocks to VICC.
53h	ISO 1569 3_Read Blocks With Uid	Command reads the unique identifier (UID) and the specified number of blocks from VICC.
54h	ISO 1569 3_Write Blocks With Uid	Command reads the unique identifier (UID) and writes the specified number of blocks to VICC.
55h	ISO 1569 3_Write Verify Blocks	Command writes and verifies the specified number of blocks.
56h	ISO 1569 3_Write Verify Blocks With Uid	Command reads the unique identifier (UID), writes and verifies the specified number of blocks.
57h	ISO 1569 3_Get Systeminformation	Command returns the system informations of the VICC.
58h	ISO 1569 3_Get Multiple Block Security Status	Command reads the security status of the requested blocks from the VICC.
59h	ISO 1569 3_Lock Block	Command locks permanently the requested blocks from the VICC.
5Ah	Select Internal Antenna	Command selects the requested internal antenna output of the RFID CU PCB or switches off the antenna signal.
5Bh	Select External Antenna	Command selects the requested antenna output of the external antenna CU PCB (expander) via I2C interface.
5Dh	Read Single Reader Register	Reads one single reader IC register.
5Eh	Write Single Reader Register	Writes one single reader IC register.
5Fh	Read All Reader Registers	Reads all reader IC registers.
B0h	Service Memory Read	Service Memory read selection.
B1h	Service Memory Reset Sd	Service Memory reset SD selection -> protected.
B2h	Service Memory Reset All Errors	Service Memory reset all errors -> protected.

Table 69–7: Command Code List: Loading bay

69.8 Command Codes Liquid Support

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
08h	Set Actor	Command sets the selected output.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
19h	Get Max Power	Command returns the maximum power consumption of the module.
1Ah	Reset	Command resets CAN-Kernel.
24h	Tube Heating Set Limits	Set target temperature and limits for the selected heater.
25h	Tube Heating Regulate	Start or stop heat control for the selected heater.
26h	Get Flap Status	Returns the door state of selected flap.
27h	Get Liquid Level Status	Gets the liquid level of a selected bottle.

ID. Hex code	Name	Description
28h	Get Presence	Command gets the presence state of a selected bottle.
32h	Start Bubble Measure	Command starts measure bubble volume check.
33h	Stop Bubble Measure	Command stops measure bubble volume check and returns result.
34h	Get Process Data	Returns the last measured volume.
36h	Calibrate Bubble Sensor	Command calibrates the selected sensor.
37h	Check Bubble Sensor	Checks the bubble sensor voltage for liquid and air.
37h	Code Coverage Reset	Command to reset all instrumentation variables to zero.
38h	Code Coverage Save	Command to save all instrumentation values.
50h	ISO15693_Inventory	Command returns the VICC Unique Identifiers (UID), found in the reader field.
51h	ISO15693_ReadBlocks	Command read the specified number of blocks from VICC.
52h	ISO15693_WriteBlocks	Command write the specified number of blocks (4 Bytes) to VICC.
53h	ISO 1569 3_Read Blocks With Uid	Command reads the unique identifier (UID) and the specified number of blocks from VICC.
54h	ISO 1569 3_Write Blocks With Uid	Command reads the unique identifier (UID) and writes the specified number of blocks to VICC.
55h	ISO 1569 3_Write Verify Blocks	Command writes and verifies the specified number of blocks.
56h	ISO 1569 3_Write Verify Blocks With Uid	Command reads the unique identifier (UID), writes and verifies the specified number of blocks.
57h	ISO 1569 3_Get System Information	Command Returns the system information of the VICC.
58h	ISO15693_GetMultipleBlockSecurityStatus	Command read the block security status from VICC.
59h	ISO15693_LockBlock	Command lock permanently the requested block.
5Ah	Select Internal Antenna	Command selects one of the antenna output, connected on the RFID CU PCB or switches off the antenna signal.

Table 69–8: Command Code List: Liquid support

69.9 Command Code Wash Dispense Pump

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
03h	Move	Command moves the drive to its target position.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
08h	Set Actuator	Command sets the selected output.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
17h	Define Ramp	Command sets drive parameter of a selected drive.
18h	Stop	Command stops the pump.
19h	Get Max Power	Command returns the maximum power consumption of the module.
1Ah	Reset	Command resets CAN-Kernel.
1Bh	Get Position	Command returns the current position of a selected drive.

ID. Hex code	Name	Description
20h	Aspirate	Command aspirates liquid.
21h	Dispense	Command dispenses liquid.
22h	Aspirate Back	Command aspirates liquid in backward direction.
23h	Dispense Back	Command dispenses liquid in backward direction.
24h	Empty	Command dispenses all liquid.
25h	Calibrate Init	Command calibrates the init position.
26h	Fill	Command aspirates liquid until at least target volume is present.
27h	Fill Back	Command aspirates liquid in backward direction.
28h	Empty Back	Command dispenses all liquid.
29h	Purge	Command dispenses all air.
30h	Self Diagnostic	Command starts the self diagnostic tests.
31h	Purge Back	Command dispenses all air in backward direction.
B0h	Service Memory Read	Service Memory read selection.
B1h	Service Memory Reset Sd	Service Memory reset SD selection -> protected.
B2h	Service Memory Reset All Errors	Service Memory reset all errors -> protected.

Table 69–9: Command Code List: Wash dispense pump

69.10 Command Code WAP

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
03h	Move	Command moves the drive to its target position.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
08h	Set Actuator	Command sets the selected output.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
11h	Move Relative	Command moves the drive relative to its current position.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.

ID. Hex code	Name	Description
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
17h	Define Ramp	Command sets drive parameter of a selected drive.
18h	Stop	Command stops a selected drive.
19h	Get Max Power	Command returns the maximum power consumption of the module.
1Ah	Reset	Command resets CAN-Kernel.
1Fh	Get Command Format	Command returns the format of a selected command.
20h	Set Speed	Command set speed for next move and changes speed on the fly.
30h	Self Diagnostic	Command starts the self diagnostic tests.
B0h	Service Memory Read	Service Memory read selection.
B1h	Service Memory Reset Sd	Service Memory reset SD selection -> protected.
B2h	Service Memory Reset All Errors	Service Memory reset all errors -> protected.

Table 69–10: Command Code List: WAP

69.11 Command Code Front Cover

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
09h	Read Serial Number	Command returns the serial number
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.

ID. Hex code	Name	Description
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
1Ah	Reset	Command resets CAN-Kernel.
20h	Activate Buzzer	Activates the Buzzer.
21h	Deactivate Buzzer	Deactivates the Buzzer.
22h	Set Display Mode	--
23h	Get Display Mode	--
26h	Get Flap Status	Returns the door state of selected flap.
40h	Self Diagnostic	Starts the self diagnostic tests.
B0h	Service Memory Read	Service Memory read selection.
B1h	Service Memory Reset Sd	Service Memory reset SD selection -> protected.
B2h	Service Memory Reset All Errors	Service Memory reset all errors -> protected.

Table 69–11: Command Code List: Front cover

69.12 Command Code Job Manager

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
0Eh	Get Version	Command returns the version of the application firmware.
20h	Load Job	Loads a job.
21h	Delete Job	Deletes a job.
22h	Cancel Job	Cancels a job
24h	Get Job State	Returns the state of a job.
25h	Cancel Test	Cancels a test.
26h	Cancel Test Due Suspend Resource	Cancels a test due a resource is suspended.
30h	Add Special Event	Add an event to the list of special events
31h	Remove Special Event	Remove an event from the list of special events.

Table 69–12: Command Code List: Job manager

69.13 Command Code Assay Manager

ID. Hex code	Name	Description
01h	State Request	Command returns the application's state.
02h	Init	Command initializes the application. This commands reset all application internal data structures.
0Eh	Get Version	Command returns the version of the application. COMMAND DESCRIPTION DOCUMENT 2290_AssayMan.docx Version: 2.0 Page 29 of 58 ID Name Description
1Ah	Reset	Resets the application.
20h	Load T Def	Download of a Test Definition.
21h	Load Pip Seq	Download of a Pipetting Sequence.
22h	Get Test List	Provides a list of stored test definitions.
23h	Get Pip Seq List	Provides a list of stored pipetting sequences.
24h	Delete T Def	Delete a Test Definition.
25h	Delete Pip Seq	Delete a Pipetting Sequence.
26h	Delete All T Defs	Delete all Test Definitions.
31h	Get Queue List	Gets the work list queues of the application.
40h	Get Assay Basics	Get the overall assay details including resources to be used.
41h	Get Assay Step	Get details of an assay step including resources to be used.
42h	Get Pip Seq	Gets the details of a pipetting sequence.
43h	Update Assay Link Count	Increments or decrements the assay usage link count. When the link count is not zero the assay cannot be deleted as it is in use.
44h	Check Assay Seq	Check a list of Test Definitions are suitable for processing.
45h	Get Assay Seq Basics	Get the overall assay details including resources to be used for one or more test definitions combined.
50h	Get Step Struct Sizes	Gets the struct sizes for each assay process step type definition. Provided for diagnostics.

Table 69–13: Command Code List: Assay manager

69.14 Command Code Dispatcher

ID. Hex code	Name	Description
01h	State Request	Command returns the application's state.
02h	Init	Command initializes the application. This commands reset all application internal data structures.
0Eh	Get Version	Command returns the version of the application.
1Ah	Reset	Resets the application.
20h	IPC_Set Task	Command to provide all information necessary to perform those tests that change from one assay step to another.
31h	Get Queue List	Gets the worklist queues of the application.
40h	Maint System Test	Command to perform maintenance system test.

Table 69–14: Command Code List: Dispatcher

69.15 Command Code Cuvette Handler

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
0Eh	Get Version	Command returns the version of the application firmware.
20h	Occupation Status	Command returns current occupation for all positions.
30h	Seek Initial Status	Command to determine initial occupation status for all positions.
31h	Get Queue List	Gets the worklist queues of this process.
40h	Set Incubator Status	Set cuvette occupation for all positions of incubator ring.
60h	Position Incubator	Command for moving incubator position to target.
61h	Position Washer	Command for moving washer position to target.
62h	Position Muc Rotor	Command for positioning measurement rotor to target.
63h	Transfer Cuv	Command for cuvette transfer between positions.
64h	Insert Cuv	Command for loading new cuvette to position.
90h	Com Gen Sync To	Worklist command generation for synchronizing subsequent actions for specific area.
91h	Com Gen Move Incubator	Worklist command generation for moving incubator position to specific target.
92h	Com Gen Step Washer	Worklist command generation for moving forward washer.
93h	Com Gen Transfer Cuv	Worklist command generation for cuvette transfer between areas.
94h	Com Gen Insert Cuv	Worklist command generation for loading new cuvette to area.
95h	Com Gen Pos Rotor Muc	Worklist command generation for positioning measurement rotor to position.
A0h	Get Cuv Status	Command to get (latest) status information for given position.

ID. Hex code	Name	Description
A1h	Set Cuv Status	Command to update status information for given position.
A2h	Com Gen Set Cuv Status	Worklist command generation for updating status information for given position.
A7h	Get Distance Target	Command to get distance of cuvette position to target position.
A8h	Get Pos Target	Command to get cuvette position at given target position.
A9h	Get Next Pos	Command to get cuvette position with requested status next to given position.
AAh	Get Next Target	Command to get cuvette position with requested status next to given target.
AFh	Start Iteration	Command to enable/disable iteration for searching cuvette positions in series.
B0h	Get Context Pos	Command to get context data assigned to cuvette position.
B1h	Com Gen Set Context Pos	Worklist command generation to assign context data to cuvette position.
B2h	Com Gen Release Context Pos	Worklist command generation to clear context data and release from cuvette position.
BFh	Finalize Context Pos Internal	Command to finalize oldest queued context change for position.

Table 69–15: Command Code List: Cuvette handler

69.16 Command Code Loading Manager

ID. Hex code	Name	Description
01h	State Request	Command returns the application's state.
02h	Init	Command initializes the application. This commands reset all application internal data structures.
0Eh	Get Version	Command returns the version of the application.
1Ah	Reset	Resets the application.
2Bh	Set Display Mode	Command to display state of LEDs.
2Eh	Buzzer On	Command to activate buzzer.
2Fh	Buzzer Off	Command to switch off buzzer.
30h	IPC_Store Occupation Change	Command to update area status and to process further information for change e.g. for identification.
31h	Get Queue List	Gets the worklist queues of the application.
32h	Start Barcode Reading	Command to start barcode reading at given lane.
33h	Confirm Segment	Command to confirm segment of inserted resource for job processing.
40h	Level Status	Command to request current range states of containers.
41h	Set Level Status	Command to set/clear counters for consumables and waste.
42h	Fill Levels	Command to request current filling levels of containers.
48h	Temp Status	Command to request current temperature range state of container.
50h	Get Fill Height	Command to get information on fill height for resource.
51h	Update Fill Height	Command to set/update information on fill height for resource.
60h	RFID Read Data	Command to read RFID data.
61h	RFID Write Data	Command to write RFID data.
80h	IPC_Get Resource State	Command to query current availability of resource.
82h	IPC_Book Resource	Command to book amount of resource becoming reserved for a test.
83h	IPC_Taken Resource	Command to notify about usage of resource.

ID. Hex code	Name	Description
84h	IPC_Cancel Resource	Command to cancel booked amount of resource now becoming available again for new tests.
85h	IPC_Set Resource Offline	Command to set resource into offline state.

Table 69–16: Command Code List: Loading manager

69.17 Command Code Diti Manager

ID. Hex code	Name	Description
25h	Consumable Area	Define area and groups for consumables.
28h	Consumable Presence	Report consumables presence.
29h	Get Number Of Available Consumables	Report the number of available consumables (cuvettes/tips).
2Ah	Lock Area	Lock / Unlock a consumables area.
2Ch	Refill Group	Refill group in a consumables area.

Table 69–17: Command Code List: Diti manager

69.18 Command Code Service Driver

ID. Hex code	Name	Description
01h	State Request	Command returns the driver's state.
02h	Init	Command initializes the driver. This commands reset all driver internal data structures.
0Eh	Get Version	Command returns the version of the driver.
1Ah	Reset	Resets the driver.
20h	Init Instrument	Initializes the whole instrument.
21h	Emergency Stop Instrument	Starts an emergency stop for the whole instrument.
22h	Stop Instrument	Stops the whole instrument.
30h	Load Cuv	Fills up positions in incubator ring.
31h	Get Queue List	Gets the worklist queues of the driver.
32h	Clear Ring	Removes all cuvettes from incubator ring.
33h	Clear Washer	Removes all cuvettes from washer.
34h	Clear Measure	Removes all cuvettes from measurement.
38h	Prime Pipettor	Performs priming of the pipettor.
39h	Prime	Performs priming of the washer and/or measurement component.
40h	Self Diagnostic	Performs the self diagnostic process of all modules.

Table 69–18: Command Code List: Service driver

69.19 Command Code Incubator Driver

ID. Hex code	Name	Description
01h	State Request	Command returns the driver's state.
02h	Init	Command initializes the driver. This commands reset all driver internal data structures.
04h	Emergency Stop	Command stops the module immediately.
07h	Read Sensor	Command reads a single sensor.
0Eh	Get Version	Command returns the version of the application firmware.
18h	Stop	Command stops the Motor.
20h	Hardware Init	Initialize the hardware assemblies.
30h	Self Diagnostic	Starts the self diagnostic tests.
31h	Get Queue List	Gets the work list queues of the driver.
40h	Move Position	Move cuvette to position.
41h	Check Cuvette	Move cuvette to sensor position and check presence.
42h	Calibrate Sensor	Calibrate cuvette Sensor.
45h	Get Cuvette State	Get presence of all cuvettes.
50h	Heater Start	Start temperature control.
51h	Heater Stop	Stop temperature control.

Table 69–19: Command Code List: Incubator driver

69.20 Command Code Washer Driver

ID. Hex code	Name	Description
01h	State Request	Command returns the driver's state.
02h	Init	Command initializes the driver. This commands reset all driver internal data structures.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
08h	Set Actor	Command sets the selected output.
0Eh	Get Version	Command returns the version of the driver.
18h	Stop	Command stops a selected drive.
1Ah	Reset	Resets the driver.
1Bh		
20h	Hardware Init	Initialize the hardware assemblies.
21h	Move Wash Lift Index	Move to index position for the selected wash lift.
22h	Inc Ring	Increment one ring position.
23h	Aspirate	Set aspiration status.
24h	Load	Prepare coming wash buffer dispense.
25h	Dispense	Set dispense sequence.
26h	Wap Set Fast Speed	Set fast WAP speed.
2Dh	Wdp Purge	Set piston pump purge sequence.
2Fh	Self Diagnostic	Starts the self diagnostic tests.
31h	Get Queue List	Gets the worklist queues of the driver.

Table 69–20: Command Code List: Washer Driver

69.21 Command Code Reader Driver

ID. Hex code	Name	Description
01h	State Request	Command returns the driver's state.
02h	Init	Command initializes the driver. This commands reset all driver internal data structures.
04h	Emergency Stop	Command immediatly stops all assemblies using their emergency stop command.
0Eh	Get Version	Command returns the version of the driver.
18h	Stop	Command stops a selected drive.
1Ah	Reset	Resets the driver.
20h	Self Diagnostic	Starts the self diagnostic tests.
31h	Get Queue List	Gets the work list queues of the driver.
40h	Hardware Init	Initialize the hardware assemblies.
41h	Set Cuvette State	Command sets the cuvette state.
42h	Get Cuvette State	Gets the current state of the cuvette.
45h	Move Rotor To Position	Command moves rotor to position.
46h	Move Rotor To Position Backwards	Command moves rotor backwards to position.
52h	Measure	Command starts measurement.
53h	Lift Down	Command moves rotor to position and moves lift down.
54h	Lift Up	Command moves lift up.
56h	Inject Pump	Command injects pump.
57h	Fill Pump	Command fills pump.
60h	Bubble Measure Start	Command starts measure bubble volume check.
61h	Bubble Measure Get Data	Command starts the transfer of the bubble measure data.
64h	Aspirate	Command moves lift down and starts the bubble measure volume check.
71h	Prepare For Transfer	Command prepares rotor for transfer.

Table 69–21: Command Code List: Reader (measurement) Driver

69.22 Command Code Reagent Sample Bay Driver

ID. Hex code	Name	Description
01h	State Request	Command returns the driver's state.
02h	Init	Command initializes the driver. This commands reset all driver internal data structures.
04h	Emergency Stop	Command immediately stops the module in case of any emergency.
0Eh	Get Version	Command returns the version of the driver.
18h	Stop	Command stops all active module sub assemblies in a controlled manner.
1Ah	Reset	Resets the driver.
20h	Hardware Init	Initialize the hardware assemblies.
31h	Get Queue List	Gets the worklist queues of the driver.
36h	Activate Auto Stirrer	Command to activate the stirrer.
37h	Deactivate Stirrer	Command to deactivate the stirrer.
40h	Loading Activate	Command to activate the sample barcode scanner in order to start the loading process.
41h	Loading Deactivate	Command to deactivate the sample barcode scanner in order to stop the loading process.
4Ah	Cooling Activate	Command to activate the cooling control.
48h	Cooling Deactivate	Command to deactivate the cooling control.
51h	Read RFID Data	Command to read RFID data.
52h	Write RFID Data	Command to write RFID data.
60h	Set Volume And Fill Height	Command to write volume and fill height to defined RFID tag.
A0h	Self Diagnostic	Command starts the self diagnostic tests.

Table 69–22: Command Code List: Reagent Sample Bay driver

69.23 Command Code Liquid Support Drive

ID. Hex code	Name	Description
01h	State Request	Command returns the driver's state.
02h	Init	Command initializes the driver. This commands reset all driver internal data structures.
04h	Emergency Stop	Command stops the module immediately.
0Eh	Get Version	Command returns the version of the driver.
18h	Stop	Command stops a selected drive.
1Ah	Reset	Resets the driver.
25h	Tube Heating	Regulate Start or stop heat control for the selected heater.
26h	Get Flap Status	Gets flap status.
27h	Get Liquid Level Status	Gets the liquid level of a selected bottle.
28h	Get Presence	Command gets the presence state of a selected bottle.
30h	Self Diagnostic	Starts the self diagnostic tests.
31h	Get Queue List	Gets the worklist queues of the driver.
32h	Bubble Measure Start	Command starts measure bubble volume check.
33h	Stop Bubble Measure	Command stops measure bubble volume check and returns result.
40h	Init Hardware	Initialize all Hardware.
50h	Inventory	Command to perform a complete liquid support inventory.
51h	Read RFID Data	Command to read RFID data.
52h	Write RFID Data	Command to write RFID data.
53h	Decrement Shots	Decrements the remaining shots in RFID tag of starter 1 or starter 2.

Table 69–23: Command Code List: Liquid support drive

69.24 Command Code Front Cover Driver

ID. Hex code	Name	Description
01h	State Request	Command returns the driver's state.
02h	Init	Command initializes the driver. This commands reset all driver internal data structures.
04h	Emergency Stop	Command stops the module immediately.
0Eh	Get Version	Command returns the version of the driver.
18h	Stop	Commands stops the driver. Current active command will be executed.
1Ah	Reset	Resets the driver.
20h	Hardware Init	Initialize the hardware assemblies
21h	Buzzer ON	Activates the buzzer.
22h	Buzzer OFF	Deactivates the buzzer.
31h	Get Queue List	Gets the work list queues of the driver.

Table 69–24: Command Code List: Front cover driver

69.25 Command Code COM GEN Driver

ID. Hex code	Name	Description
01h	State Request	Command returns the driver's state.
02h	Init	Command initializes the driver. This commands reset all driver internal data structures.
04h	Emergency Stop	Emergency stops both pipettor arms including all pumps.
0Eh	Get Version	Command returns the version of the driver.
1Ah	Reset	Resets the driver.
20h	Hardware Init	Initialize the hardware assemblies.
21h	Arm Init Z	Initialize the Z axis.
30h	Self Diagnostic	Starts the self diagnostic tests.
31h	Get Queue List	Gets the worklist queues of the driver.
3Ah	Set Job Id	Set job id to be reported with pressure/olv data.
40h	Pump Move	Pump move, aspirate or dispense, fluid or airgap.
41h	Airgap Via Bypass	Airgap aspirate via bypass valve.
42h	Wash	Wash steel probe.
45h	Di Ti Pickup New	Pickup a new tip.
46h	Di Ti Eject	DiTi eject.
47h	Di Ti Pickup	Pickup a tip.
50h	XY Move Abs	Horizontal move of arm.
51h	Z Move	Vertical move of pipettor axis.
52h	LLD	Liquid level detection.
5Eh	Waste Pump Off	Switch off the waste pump.
5Fh	Set Pres Thres	Set pressure thresholds.
60h	Gripper Pick New	Pick new cuvette.
61h	Gripper Pick	Pick cuvette.
62h	Gripper Place	Place cuvette.

Table 69–25: Command Code List: COM GEN driver

69.26 Command Code Pipettor

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes a subassembly.
03h	Move	Command moves the drive to its target position.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
08h	Set Actuator	Command sets the selected output.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
11h	Move Relative	Command moves the drive relative to its current position.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
17h	Define Ramp	Command sets drive parameter of a selected drive.
18h	Stop	Command stops a selected drive.
1Ah	Reset	Command resets CAN-Kernel.
1Bh	Get Position	Command returns the current position of a selected drive.

ID. Hex code	Name	Description
1Ch	Get Process Data	Command sends the process data of a selected sensor.
20h	Self diagnostic	Start the modules self diagnostic process.
44h	Report Coordinate	Report the values of a coordinate.
45h	Expand Flash Coor	Expand coordinate data from flash to RAM.
70h	Define Target Array	Define a regular grid of positions.
71h	Define Bottle Shape	Define the shape of a bottle.
7Eh	Eject Pickup	Eject/Pickup of disposable tip.
82h	Define Coordinate Set	Define a coordinate set.
83h	XY Move	XY move to specified coordinate.
96h	Z Relative Move	Z relative move, slow but defined speed.
98h	XY Relative Move	XY relative move.
9Ah	Z Move	Z absolute move.
9Bh	Enhanced Z Move	Enhanced Z absolute move.
A1h	Waste Pump	Triggers the drain pump.
A2h	Enable Channel	Enable Z channels.
A6h	Define Actual Position	Sets the horizontal arm position.
AAh	OLV Dead Volume Calibration	Olv dead volume calibration.
ACh	OLV Sensor Calibration	Olv sensor calibration.
AEh	Wash Steel Probe	Wash steel probe.
B0h	Detect Liquid Level	Detect liquid level.
B2h	Get Pip Position	Report actual pipettor position.
BEh	Cap Clot Detection	Start cap. clot detection.
C0h	Gripper Pickup	Gripper picks a cuvette.
C1h	Gripper Eject	Gripper ejects a cuvette.
C2h	Gripper Cmd Buffer	Buffer a gripper command.
C3h	Gripper Cmd	Sends a command to the gripper.
D0h	Pump Cmd	Pump command.
D3h	Define Z Travel	Set reference of Z-coordinates.
D5h	Pump String Cmd	Pump string command.
D7h	Pump String Cmd Buffer	Buffer a pump string command.
DEh	Aspirate Dispense	Aspirate / Dispense specified number of pump steps.
DFh	Set Aspirate Dispense Parameter	Load channel specific pump move data.
E0h	Synchronize	Send response for synchronisation.
F2h	Analyze OLV Trace	Analyze OLV signal trace.

Table 69–26: Command Code List: Pipettor

69.27 Command Code Gripper

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
03h	Move	Command moves the drive to its target position.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
08h	Set Actuator	Command sets the selected output.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
11h	Move Relative	Command moves the drive relative to its current position.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.
16h	Unlock Parameter	Command sets the access authorization to locked parameters.

ID. Hex code	Name	Description
17h	Define Ramp	Command sets drive parameter of a selected drive.
18h	Stop	Command stops a selected drive.
19h	Get Max Power	Command returns the maximum power consumption of the module.
1Ah	Reset	Command resets CAN-Kernel.
1Bh	Get Position	Command returns the current position of a selected drive.
1Fh	Get Command Format	Command returns the format of a selected command.
20h	Hold	Command closes the gripper to hold the cuvette.
21h	Release	Command opens the gripper to release the cuvette.
22h	Close	Command closes the gripper, no cuvette check.
23h	Open	Command opens the gripper, no cuvette check.
30h	Self Diagnostic	Starts the self diagnostic tests.

Table 69–27: Command Code List: Gripper

69.28 Command Code Air Pipettor Left Arm

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
03h	Move	Command moves the drive to its target position.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read	Sensor Command reads a single sensor.
08h	Set Actuator	Command sets the selected output.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
1Ah	Reset	Command resets the CU and restarts with the CAN-Kernel.
1Bh	Get Position	Command returns the current position of a selected drive.
1Ch	Get Process Data	Command sends the process data of a selected sensor.
1Eh	Read Write SCTL	Command reads/writes a SCTL register.

ID. Hex code	Name	Description
20h	String	Command VP9101 compatible Pump Command.
21h	Define Ramp	Set Velocity, Acceleration and Stop Velocity.
22h	Stop	Stops the selected drive.
24h	LLD	Starts the LLD.
25h	AP Head	Command Direct command to APHead device.
2Bh	Analyse Pressure Trace	Analyses the Pressure Trace.
2Ch	Set Reference Pressure	Saves actual pressure as Reference pressure.
2Eh	Get Level Oscillator Buffer	Read the actual content of the level oscillator buffer: samples before and after the last LLD trigger.
2Fh	Pressure Recording	Starts or stops pressure recording.
30h	Pump Move	Complex move command with real time pressure monitoring.
31h	Verify Tip Type	Verify attached tip type.
32h	Get Submodule Version	Returns the version string of selected submodule.
33h	Start Move	Start a plunger move and send response immediately, movement can be stopped any time.
40h	Self Diagnostic	Command starts the self-diagnostic tests.
41h	Encoder	Calibration Executes a calibration move.

Table 69–28: Command Code List: Air pipettor left arm

69.29 Command Code SPOLV Right Arm

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
03h	Move	Command moves the drive to its target position.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
08h	Set Actuator	Command sets the selected output.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
1Ah	Reset	Command resets the CU and restarts with the CAN-Kernel.
1Bh	Get Position	Command returns the current position of a selected drive.
1Ch	Get Process Data	Command sends the process data of a selected sensor.
1Eh	Read Write SCTL	Command reads/writes a SCTL register.

ID. Hex code	Name	Description
1Eh	String	Command VP9101 compatible Pump Command.
21h	Define Ramp	Set Velocity, Acceleration and Stop Velocity.
22h	Stop	Stops the selected drive.
24h	LLD	Starts the LLD.
25h	AP Head	Command Direct command to APHead device.
2Bh	Analyse Pressure Trace	Analyses the Pressure Trace.
2Ch	Set Reference Pressure	Saves actual pressure as Reference pressure.
2Eh	Get Level Oscillator Buffer	Read the actual content of the level oscillator buffer: samples before and after the last LLD trigger.
2Fh	Pressure Recording	Starts or stops pressure recording.
30h	Pump Move	Complex move command with real time pressure monitoring.
32h	Get Submodule Version	Returns the version string of selected submodule
33h	Start Move	Start a plunger move and send response immediately, movement can be stopped any time.
34h	Set OLV Reference Values	Read OLV Signal and set Fluid / Air thresholds.
35h	Start Stop OLV	Start or stop the OLV.
40h	Self Diagnostic	Command starts the self-diagnostic tests.
41h	Encoder Calibration	Executes a calibration move.

Table 69–29: Command Code List: SPOLV right arm

69.30 Command Code Sledge Y

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
03h	Move	Command moves the drive to its target position.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
11h	Move Relative	Command moves the drive relative to its current position.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
17h	Define Ramp	Command sets drive parameter of a selected drive.
1Ah	Reset	Command resets CAN-Kernel.
1Bh	Get Position	Command returns the current position of a selected drive.
20h	Self diagnostic	Start the modules self diagnostic process.
22h	Set Position	Command sets a new position.

Table 69–30: Command Code List: Sledge Y

69.31 Command Code Sledge Z

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
03h	Move	Command moves the drive to its target position.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
08h	Set Actuator	Command sets the selected output.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
11h	Move Relative	Command moves the drive relative to its current position.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.

ID. Hex code	Name	Description
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
1Ah	Reset	Command resets CAN-Kernel.
1Bh	Get Position	Command returns the current position of a selected drive.
1Eh	Read Write Sctl	Command reads/writes a Sctl register.
20h	Set Move Par	Sets speed and acceleration of the selected drive.
22h	Define Actual Pos	Defines the actual position of the selected drive.
23h	Get Actual Pos	Returns the encoder position of the drive.
24h	LLD	Starts a LLD move and configures the LLD functionality for the selected mode.
27h	Eject Pickup	Eject/Pickup of disposable tip.
28h	Set Command Pos	Sets the command position of the selected drive.
30h	Self diagnostic	Start the modules self diagnostic process.

Table 69–31: Command Code List: Sledge Z

69.32 Command Code Pump Starter 1 and 2

ID. Hex code	Name	Description
01h	State Request	Command returns the module state.
02h	Init	Command initializes the module.
03h	Move	Command moves the drive to its target position.
04h	Emergency Stop	Command stops the module immediately.
05h	Standby	Command set the module in standby mode.
06h	Resume	Command resume the module from standby mode.
07h	Read Sensor	Command reads a single sensor.
08h	Set Actor	Command sets the selected output.
09h	Read Serial Number	Command returns the serial number.
0Ah	Read Article Number	Command returns the article number.
0Bh	Set Default	Command sets the parameter to default values.
0Ch	Read EE Prom	Command reloads the saved parameter to module RAM.
0Dh	Write EE Prom	Command saves the parameter to EEPROM.
0Eh	Get Version	Command returns the version of the application firmware.
0Fh	Read Error	Return the stored errors.
10h	Reset Error	Reset the stored errors.
12h	Get Parameter Type	Command reads information about the selected parameter.
13h	Get Parameter	Command reads a formatted parameter from module RAM.
14h	Set Parameter	Command writes a formatted parameter from module RAM.
15h	Get Parameter Index	Command returns the minimum and the maximum index of the parameters.
16h	Unlock Parameter	Command sets the access authorization to locked parameters.
17h	Define Ramp	Command sets drive parameter of a selected drive.
18h	Stop	Command stops the pump.
19h	Get Max Power	Command returns the maximum power consumption of the module.
1Ah	Reset	Command resets CAN-Kernel.

ID. Hex code	Name	Description
1Bh	Get Position	Command returns the current position of a selected drive.
20h	Aspirate	Command aspirates liquid.
21h	Dispense	Command dispenses liquid.
22h	Aspirate Back	Command aspirates liquid in backward direction.
23h	Dispense Back	Command dispenses liquid in backward direction.
24h	Empty	Command dispenses all liquid.
25h	Calibrate Init	Command calibrates the init position.
26h	Fill	Command aspirates liquid until at least target volume is present.
27h	Fill Back	Command aspirates liquid in backward direction.
28h	Empty Back	Command dispenses all liquid.
29h	Purge	Command dispenses all air
30h	Self Diagnostic	Command starts the self diagnostic tests.
31h	Purge Back	Command dispenses all air in backward direction.

Table 69–32: Command Code List: Starter Pump

70 Glossary

00h; 0x00	Shows hexadecimal (hex) code, used in trouble shooting error messages.
APH	Air Pipettor Head Module:
APM	Aspirate pressure monitoring system
APP	Air Pipettor Pump Module:
ASCII	American Standard Code for Information Interchange (ASCII), pronounced is a character encoding based on the English alphabet.
ASTM	ASTM International, is an international standards organization that develops and publishes voluntary consensus technical standards. ASTM 1381 and ASTM 1394 are determinations for the communication between computers.
Bisects	Halved every step (To cut or divide into two parts, especially two equal parts.)
Bleach	Hydrogen peroxide (bleach, 3%)
CCD	Charge Coupled Device
CCW	Counter clock wise
CE	Conformité Européenne (European conformity)
CFR	Code of Federal Regulations (Codes of FCC, US government)
COM	COM is the original, yet still common, name of the serial port interface on PCs. It might not only refer to physical ports, but also to virtual ports, such as ports created by RS232 or USB adapters.
COP	Central operating processor
CSA	Canadian Standards Association
CU	Control Unit
CV	The abbreviation CV stands for 'coefficient of variation' and is the relative standard deviation, which is the quotient of the absolute standard deviation and the mean of a measured value. %CV: The CV multiplied with 100%
DITI	Disposable Tip
EEPROM	An Electrically Erasable Programmable Read-Only Memory, is a type of non-volatile memory used in computers and other electronic devices to store small amounts of data that must be saved when power is removed, e.g., calibration tables or device configuration.
EIA	Enzyme Immuno Assay

ELISA	Enzyme Linked Immuno Sorbent Assay
Fatal Error	An unexpected and non-recoverable termination of instrument operation including, but not limited to assay processing, maintenance, service, start-up, and shut-down.
FCC	Federal Communications Commission, US government
Host	In computer networking a host is a main computer (e. g. server, central computer).
IC	Integrated circuit
ID	Identification (Number).
IFU	Instructions for use manual
IP	Integrated pipettor
IVD	In-vitro Diagnostic
LAN	A Local Area Network is a computer network.
LAS	Laboratory Automation System
LED	Light Emitting Diode
LIMS	A Laboratory Information Management System (LIMS) is computer software that is used in laboratories for the management of samples, laboratory users, instruments, standards and other laboratory functions such as invoicing, plate management, and work flow automation.
LIS	A Laboratory Information System, is a class of software which handles receiving, processing and storing information generated by medical laboratory processes.
LIS01-A2	Specification for Low-Level Protocol to Transfer Messages Between Clinical Laboratory Instruments and Computer Systems
LIS02-A2	Specification for Transferring Information Between Clinical Laboratory Instruments and Information Systems
LLD	Liquid Level Detection B-LLD - Barometric Liquid Level Detection C-LLD - Capacitive Liquid Level Detection
LWP	Liquid waste pump
ms	Milli seconds (1000 ms = 1 s)
MSDS	Material safety data sheet
MTP	Microtiter Plate, microplate
nm	A nanometer is a unit of length in the metric system. (1 nm = 0.000001 mm = $1 \cdot 10^{-9}$ m = $39.37 \cdot 10^{-9}$ inch)
NVM	None volatile memory is computer memory that can retain the stored information even when not powered.

OD	Optical density
OLV	Optical Liquid Verification
PC	Personal Computer
PCB	Printed Circuit Board
PMT	photomultiplier tube, sensitive photo detector that detects individual photons of light. Also known as photo electron multiplier.
PS2	The PS/2 (PS2) connector allows the connection of a keyboard or a mouse to a computer.
QA	Quality control Analysis
RAM	Random access memory When the computer is shut down, anything contained in RAM is lost.
RFID	Radio frequency identification
RLU	Relative luminescence unit
SCTL	Servo Motor Controller (off the shelf)
SM	Service Manual
SSW	Service Software
TAB	Tabulator
TDS	Technical data sheet
USB	The Universal Serial Bus is a serial bus standard to connect devices to a computer.
V	Voltage
VC	Validation criteria
VCD	Proximity coupling device (reader IC)
VGA	The Video Graphics Array is a standard interface to connect a screen to a computer.
VICC	Proximity integrated circuit card (tag or smart label)
ZSPIP	Gear Rack Pipettor
µl	A microliter is a unit of volume in the metric system. (1 µl = 0.001 ml = 1*10 ⁻⁶ l)

